TRANSMISSION CORPORATION OF ANDHRA PRADESH LIMITED

SPECIFICATION NO: E-proc No.04/2026-27-1st Call

NAME OF THE WORK: Replacement of Old/Damaged 220 KV & 132 KV CTMBs, CVTMBs, PTMBs at 220/132/33 KV Pendurthi SS Under O&M.

EXECUTIVE ENGINEER, O&M DIVISION, PENDURTHI
APTRANSCO, VISAKHAPATNAM.

Phone & Fax: 9177555090, 9490155067;

Website: www.apeprocurement.gov.in
www.aptransco.gov.in

NOTICE INVITING BIDS

1.	Department Name	:	APTRANSCO-EE-O&M Division-VISAKHAPATNAM
2.	Circle/Division	:	SE/O&M/Visakhapatnam **/** EE/O&M/PDT
3.	Bid Number	:	e-Proc No.04/2026-27- 1st Call of EE/O&M/Pendurthy
4.1	Bid Subject	:	
4.2	Name of the Project.	:	Replacement of Old/Damaged 220 KV & 132 KV CTMBs, CVTMBs, PTMBs at 220/132/33 KV Pendurthi SS Under O&M
5.	Package No.	:	**e.Proc.No:**04/2026-27 – 1st Call
6.	Estimated Contract Value	:	Rs **6,04,120.00**
7.	Period of Contract	:	3 months
8.	Form of Contract	:	L.S
9.	Bid Type	:	**Open Competitive Bidding**
10.	Bid Category	:	**Works**
11.	Bid Validity	:	120 days from Scheduled Bid submission closing date
12.	Bid Security	:	2% of ECV, Rs.12,082.00
13.	Bid Security Payable to	:	In the shape of Demand Draft in favour of The Superintending Engineer, O&M Circle, A.P.TRANSCO, Visakhapatnam payable at Visakhapatnam (or) Bank Guarantee issued by a Nationalized Bank/ Scheduled Bank (as per the list enclosed) in favour of the Superintending Engineer, O&M Circle, APTRANSCO, Visakhapatnam and shall cover a period of 45 days over the bid validity .
14.	Transaction Fee	:	**Refer e-procurement web-site**
15.	Transaction Fee Payable to	:	**Refer e-procurement web-site**
16.	Schedule Sale opening Date	:	**23.10.2026 at 6.00PM (IST)**
17.	Schedule sale Closing Date	:	**30.10.2026 at 6.00PM (IST)**
18..	Bid Submission Closing Date	:	**31.10.2026 at 3.00PM (IST)**
19.	Bid Submission	:	Online
20.	Price Bid Opening Date		**31.10.2026 at 4.00PM (IST)**
21.	Place of Bid Opening	:	O/o Superintending Engineer, O&M Circle, APTRANSCO, Visakhapatnam
22.	Officer Inviting Bids	:	Executive Engineer /O&M Division / Pendurthy/ A.P.TRANSCO/ Visakhapatnam
23.	Address	:	O/o Executive Engineer, O&M Division, Pendurthy, APTRANSCO, Viratnagar, Simhachalam temple road, Visakhapatnam-530 029.
24.	Contact Details/Telephone/Fax	:	Phone & Fax :9177555090, 9490155067
25.	Eligibility Criteria	:	Refer Clause No. 3 'Qualification Requirements' of General Terms and conditions of contract of specification Volume-1
26.	Procedure for Bid Submission	:	Refer Instructions to the bidder of specification Volume-1
27	General Terms & Conditions	:	Refer General Terms and conditions of contract of specification Volume-1 Note: Bidders are requested to go through all the clauses and contents of specification thoroughly and quote the bid. Ignorance of clauses will not be entertained by the purchaser at later date.

I N D E X

Sl. No.	Description
	Invitation for Bids
1)	**Instructions To Bidders**
	1.0 General
	2.0 Bidders / Firms eligible to Bid
	3.0 Documents to be furnished
	4.0 Submission of bids
	5.0 Contents of the specification
2)	**General Terms & Conditions (Section-I)**
3)	**Financial Terms & Conditions (Section-II)**
4)	**Schedules (I – X)**
5)	**Check List (XI)**
6)	**Forms (A- D)**
7)	**List of approved Manufacturers**
8)	**List of approved Banks**
9)	**Schedules**
	Price Schedules for Works
	Financial Bid/Price Bid
	List of makes of the materials proposed to supply if the contract is awarded
10)	Project Data
11)	Schedule-A
12)	Schedule-D(Summary of Price)
13)	Schedule-E(Completion Period)

INSTRUCTIONS TO THE BIDDERS

1.0 **General:** The APTRANSCO invites bids on e-procurement platform for the works mentioned in the NIT and Project Data sheet for which date and time is specified in the NIT and will be opened online by either Bid inviting authority or his nominee based on the date and time mentioned in the NIT. The details of Bid inviting authority are mentioned in the Project Data Sheet.

1.1 The Bids should be in the prescribed form as per this specification and as per the guidelines of e-procurement and as specified in NIT and Bidding documents which can be downloaded free of cost from the website www.eprocurement.gov.in

1.2 The dates stipulated in the NIT are firm and under any circumstances they will not be relaxed unless officially extended by APTRANSCO in e-procurement plat-form.

1.3 The e-Procurement application is PKI (Public Key Infrastructure) enabled and supports the digital certificates issued by APTS Ltd.,(Andhra Pradesh Technology Services Limited) for signing the bids at the time of submission by bidder. The bidder has to procure the digital certificates issued by APTS Ltd., Vijayawada by following the procedure laid for the same. Digitally signed bids are to be submitted electronically through e-procurement without which the Bid will not be considered for opening.

1.4 The intending Bidders would be required to enroll themselves on the 'e'-procurement market place at www.eprocurement.gov.in. If the bidders intend to know the procedure of bid submission on e-procurement platform, suitable training will be given by M/s **Vupadhi Techno Services Pvt. Ltd.** 1st Floor, Ramky Grandiose, Sy.No: 136/2&4, Gachibowli, Hyderabad-500032/Near PWD Grounds, Vijayawada.. Bidders are required to submit their bids in two parts:

1. Part-I : Eligibility and Technical bid.
2. Part-II: Financial bid.

1.5 The Bidder should upload scanned copies as specified in checklist and all enclosures required for the schedules and Appendices. The bidder should produce the originals of all uploaded documents for verification if asked for by the Competent Authority within 3 (Three) days.

1.6 The uploading of bids on-line through e-procurement cannot be possible after expiry of due date and time and it is sole discretion of APTRANSCO to consider any requests for extension.

1.7 Transaction fee: The Transaction fee at 0.03% on Estimate Contract value (ECV) of work plus 18 % GST on transaction fees shall be payable by electronic Payment on e-Procurement platform, which provides a facility to participating suppliers/ contractors to electronically pay the transaction fee from the platform, using their credit cards. The Payment Gateway accepts all Master and VISA credit cards issued by any bank. The payment of transaction fee online is mandatory for participating in the bidding.

1.8 Successful Bidder shall pay the Corpus fund to APTS @ 0.04% on ECV in the form of DD payable at Vijayawada and drawn in favour of MD, APTS, Hyderabad / Vijayawada to sustain 'e' procurement initiatives, research and development of software application for automation for process in user departments.

1.9 **The successful Bidder is expected to complete the work within the time period specified in the Schedule – D and as per PERT/ Bar Chart to be furnished by the successful bidder at the time of kickoff meeting.**

1.10 APTRANSCO reserves the right to cancel/extend/alter the bid conditions at any time. In the process, if the tender process / works are stalled due to legal intervention or due to natural calamities, no compensation will be paid.

2.0 **Bidders / Firms eligible to Bid:**

The Bidders / Firms who

i) Possess the valid registration and satisfies all the conditions therein.
ii) Are not blacklisted or debarred or suspended by the Government /any power utility for what ever the reason, prohibiting them not to continue in the contracting business.
iii) Have complied with the eligibility criteria specified in the Clause 3 of General Terms & Conditions in Section -1 of this Specification are the eligible Bidders /Firms.

2.1 Bidders / Firms ineligible to Bid:

(i) A retired officer of the Govt. of AP or Govt. of India or APTRANSCO or their subsidiaries' is disqualified from tendering for a period of two years from the date of retirement without the prior permission of the Parent Department. The contractor shall intimate the list of his employees who were retired from Government/APTRANSCO along with the permission possessed from parent department. The contractor shall intimate the list of his employees who were retired from Government/APTRANSCO along with the permission, they posses from parent department.

(ii) The Bidder who has employed any retired officer as mentioned above shall be considered as an ineligible Bidder.

(iii) The contractor himself or any of his employee is found to be Gazetted Officer who retired from Government Service and had not obtained permission from the parent department for accepting the contractor's employment within a period of 2 years from the date of his retirement.

(iv) The Contractor or any of his employees is found at any time after award of contract, to be such a person who had not obtained the permission of the Government as aforesaid before submission of the Bid or engagement in the Contractor's service.

(v) Contractor shall not be eligible to Bid for works in APTRANSCO where any of his near relatives are employed in the rank of Assistant Engineer or Assistant Executive Engineers and above on the Engineering side and Assistant Accounts Officer and above on the accounts side. The Contractor shall intimate the names of persons who are working with him in any capacity or are subsequently employed. He shall also furnish a list of Gazetted / Non-Gazetted State Government Employees related to him. Failure to furnish such information Bidder is liable to be removed from the list of approved contractors and his contract is liable for cancellation.

Note: <u>Near relatives include</u>

a) Sons, step sons, daughters, and step-daughters.
b) Son-in-law and daughter-in-law.
c) Brother-in-law and sister-in-law.
d) Brothers and Sisters.
e) Father and Mother.
f) Wife / Husband.
g) Father-in-law and Mother-in-law
h) Nephews, nieces, uncles and aunts
i) Cousins and any person residing with or dependent on the contractor.

vi) The Companies which are blacklisted by any other utility and whose promoters firms or companies were already blacklisted /debarred by any other utility shall be considered as ineligible bidder. Further, the bidder shall give an undertaking(as per Schedule V- (E)that, if the information furnished in this regard is found to be false at a later date the bidder will be made ineligible or the contract will be terminated.

3.0 Documents to be furnished for Qualification requirements of the Bidders:

3.1 The Bidder shall furnish the following particulars in the formats provided online and supported documentary evidence shall be uploaded.
Attested copies of documents relating to the Registration of the firm, Registration as Electrical Licensed Contractor in A.P, Partnership deed, Articles of Association, GSTIN Provisional Registration, Latest Income Tax Clearance certificate / PAN number from IT Department, Labour department registration, work experience of similar type of works, self declaration, any other documents etc.
<u>Note</u>: The Partnership firms, which are registered as Contractors shall intimate the change in partnership deed, if any, as per GO Ms No.58, I & CAD, dt.23.4.2002 within one month of such change. Failure to notify the change to the registration authority in time will entail the firms to forfeit their registration and their Bid will be rejected. The intimation of change of partners if any and the acceptance by the Registration authority may be enclosed. Further the change if any in the official address, phone/fax numbers etc., shall be intimated.

3.2 Availability of key personnel for administration /site management and execution viz., technical personnel required for the work.

3.3 Qualification criteria for opening of the price bid: Bidders qualified in the Technical / Pre-qualification Bid if they meet the requirements as per clause no. 3 'Qualification Requirements' of Section -1

3.4 Even though the Bidders are meeting the above qualification requirements, they are liable to be disqualified / debarred / suspended / blacklisted if they have

a) Furnished false / fabricated particulars in the forms, statements and / annexure submitted in proof of the qualification requirements and/or
b) Not turned up for entering into agreement, when called upon with in the time specified in the letter of acceptance
c) Record of poor progress such as abandoning the work, not properly completing the contract, inordinate delays in completion, litigation history or financial failures etc., and
d) A history of criminal record in which the Bidder is involved if any.
e) History of litigation with Government/APTRANSCO during the last 5 years in which the Bidder is involved.
f) Any wrong declaration or hiding of information on clause (vi) above shall be viewed seriously and such firms will be blacklisted /debarred.

3.5 Even while execution of the work, if found that the work was awarded to the Contractor based on false /fake certificates of experience, the Contractor will be blacklisted and work will be taken over.

3.6 The bidders shall submit a written Power of Attorney authorizing the signatory of the bid to commit on behalf of the bidder.

3.7 Bidders are advised to find out the latest tax structure, applicability and rates before quoting the plus or minus tender percentage on ECV. Further to reduce the billing difficulties, bidders are advised to quote the percentage with **not more than two decimal places.**

3.8 **SITE VISIT:** The bidders are advised to visit and examine the site of proposed Substations and the route of transmission line and obtain all information that may be necessary for preparing the bid such as access to site, communication, transport, leads and availability for water, sand, metal etc., right of way, the quantity of various items of work, the availability of local labour, availability and rates of materials, local working conditions, uncertainties of weather and period of crops and working seasons, involvement of river crossings, railway crossings etc. The cost of visiting the site shall be at bidder's own expense. Ignorance of site conditions shall not be accepted by the Purchaser as basis for any extra claim or additional rates or any compensation at a later date. The submission of a bid by the bidder will be construed as evidence that such an examination was made and later claims/disputes in this regard shall not be entertained.

3.9 It is the responsibility of the bidder to satisfy himself that sufficient quantities of construction materials required for the works shall exist in the borrow areas or quarries sites. The APTRANSCO does not accept any responsibility either in identifying the quarries or procuring the material or any other facilities in this regard. The bidder will not be entitled for any extra cost /rate or claim for the misjudgment on his part for quantity and quality of materials available in the quarries.

3.9.1 If Bidders have any clarifications/ doubts on any portion of the specification or enclosures in writing thereto, they shall obtain the required clarifications from the officer to whom the bids are to be addressed. The purchaser will issue clarification in writing, if necessary.

4.0 BID SUBMISSION PROCEDURE :

The bidder shall submit his response through Bid submission to the tender on eProcurement platform at www.apeprocurement.gov.in by following the procedure given below. The bidder would be required to

register on the e-procurement market place www.apeprocurement.gov.in or https://tender.apeprocurement.gov.in and submit their bids online. Offline bids shall not be entertained by the Tender Inviting Authority for the tenders published in e-procurement platform.

4.1.1 The bidders shall submit their eligibility and qualification details, Technical bid, Financial bid etc., in the online standard formats displayed in e-Procurement web site. The bidders shall upload the scanned copies of all the relevant certificates, documents etc., in support of their eligibility criteria/technical bids and other certificate/documents in the e-Procurement web site. The bidder shall sign on the statements, documents, certificates, uploaded by him, owning responsibility for their correctness/authenticity. The bidder shall attach all the required documents for the specific tender after uploading the same during the bid submission as per the tender notice and bid document.

4.1.2 Registration with e Procurement platform: For registration and online bid submission bidders may contact HELP DESK.
www.apeprocurement.gov.in or https://tender.apeprocurement.gov.in.
Digital Certificate authentication:

The bidder shall authenticate the bid with his Digital Certificate for submitting the bid electronically on e Procurement platform and the bids not authenticated by digital certificate of the bidder will not be accepted on the e Procurement platform.

You may please Contact Registration Authorities of any Certifying Authorities in India. The list of CAs are available by clicking the link https://tender.apeprocurement.gov.in/digital-signature.html#

4.1.3. Hard copies:

i) Vide ref. G.O.Ms.No.174, I&CAD dept dated:1-9-2008, submission of original hard copies of the uploaded scanned copies of Proof of online Payment (Remittance)/BG towards Bid security by participating bidders to the tender inviting authority before the opening of the price bid is dispensed forthwith.

ii) All the bidders shall invariably upload the scanned copies of Proof of online Payment (Remittance)/BG in e Procurement system and this will be the primary requirement to consider the bid responsive.

iii) The department shall carry out the technical evaluation solely based on the uploaded certificates/documents, Proof of online Payment (Remittance)/BG towards Bid security in the e Procurement system and open the price bids of the responsive bidders.

iv) The department will notify the successful bidder for submission of original hardcopies of all the uploaded documents Proof of online Payment (Remittance)/BG towards Bid security prior to entering into agreement.

v) The successful bidder shall invariably furnish the original Proof of online Payment (Remittance)/BG towards Bid security, Certificates/Documents of the uploaded scan copies to the Tender Inviting Authority before entering into agreement, either personally or through courier or post and the receipt of the same within the stipulated date shall be the responsibility of the successful bidder. The department will not take any responsibility for any delay in receipt/non-receipt of original Proof of online Payment (Remittance)/BG towards Bid security, Certificates/Documents from the successful bidder before the stipulated time. On receipt of documents, the department shall ensure the genuinity of the Proof of online Payment (Remittance)/BG towards EMD and all other certificates/documents uploaded by the bidder in e Procurement system. In support of the qualification criteria before concluding the agreement.

4.1.4. The GO. Ms. No. 174 -I&CAD dated: 1-9-2008

Deactivation of Bidders If any successful bidder fails to submit the original hard copies of uploaded certificates/documents, Proof of online Payment (Remittance)/BG towards Bid security within stipulated time or if any variation is noticed between the uploaded documents and the hardcopies submitted by the bidder, as the successful bidder will be suspended from participating in the tenders on e Procurement

platform for a period of 3 years. The e Procurement system would deactivate the user ID of such defaulting bidder based on the trigger/recommendation by the Tender Inviting Authority in the system. Besides this, the department shall invoke all processes of law including criminal prosecution of such defaulting bidder as an act of extreme deterrence to avoid delays in the tender process for execution of the development schemes taken up by the government. Other conditions as per tender document are applicable.

The bidder is requested to get a confirmed acknowledgement from the Tender Inviting Authority a proof of Hardcopies submission to avoid any discrepancy.

4.1.5. Payment of Bid security:

It is mandatory for all participant bidders to electronically pay Bid security Online by utilizing the "Payment Gateway Service on E-Procurement platform". The Electronic Payment Gateway accepts all Master and Visa Credit Cards issued by any bank and Direct Debit facility/Net Banking/NEFT payment modes through ICICI Bank and/or Axis Bank Payment Gateways to facilitate the transaction. This is in compliance as per G.O.Ms. 08 dated 08.05.2016. A GST of 18.00% + Bank charges on the transaction amount payable to APTS shall be applicable. In addition to this, Bidders can also pay the Bid security through Download of PDF format of RTGS Challan for respective Payment gateway and pay the EMD through their Parent Bank account. Once the Bid security is received by the e Procurement application, Bidders can automatically continue with their Bid Submission online.

IMPORTANT NOTE REGARDING Bid security PAYMENT:

1. Bidders are encouraged to use only Net banking facility for payment of EMDs as far as possible for faster refunds in case of unsuccessful Bids for the Tender.

2. Bidders are advised not to use RTGS Challan downloads at the penultimate hour of Bid submission closing as any delay by their banker would not enable Bid submission on the platform. Please allow a minimum of 60 minutes for enabling "Continuation of Bid Submission" from the time the Pool Account receives credit of the EMD from the Bidder's Bank for both NEFT and RTGS Transfers. For RTGS Transfers, the Pool Account can get immediate credit whereas NEFT transfers would follow RBI Payment Cycle time.

3. Bidders are advised to pay Bid security Online at least T-1 or T-2 days before Bid submission closing date (T= Bid submission closing date) to avoid last minute delays and denials of successful Bid submission and to take care of any delays in Banking procedures.

Un Successful Bidder Bid security Refund process: -

The bid is declared unsuccessful, under the following circumstances.
Bid submitted by the bidder is not the lowest bid.
Upon Finalization of the L1 Bid.
Technical Disqualification of the Bid in case of 2 cover system.
Bid security paid but bid not submitted

Bid security refund will be initiated by the Tender Inviting Authority directly and through Online only and through the same payment channels as Bid security received by the Department. (RTGS / NEFT /Credit Card /Debit Card refund), within 30 days from the date of publishing the Decision / Result. However, Vupadhi /GOAP will not be held responsible for the delays occurring due to banking channels/procedures/processes of the respective vendor.

IMPORTANT NOTE REGARDING Bid security REFUNDS:

Bidders are requested to use discretion in their choice of payment channel for remittance of Bid security.
Time taken for Refunds under Ideal conditions:
1. Net Banking / NEFT / RTGS Challan: One (1) Banking Business Day from time of initiation of refund by Tender Inviting Authority subject to RTGS/NEFT timings of RBI.
2. Credit card/ Debit card: 7-10 working days from time of initiation of refund by the Tender Inviting Authority. However, this may be longer in case of certain bank cards. In case of delays, bidders are requested to contact the Card issuing Bank for faster resolution.

4.1.6. Payment of Transaction Fee:

It is mandatory for all the participant bidders from 1st January 2006 to electronically pay a Non-refundable Transaction fee to M/s. APTS, the service provider through "Payment Gateway Service on e-Procurement platform". The Electronic Payment Gateway accepts all Master and Visa Credit Cards issued by any bank and Direct Debit facility/Net Banking of ICICI Bank, Axis Bank to facilitate the transaction. This is in compliance as per G.O.Ms. 13 dated 07.05.2006. A GST of 18% + Bank charges on the transaction amount payable to APTS shall be applicable.

4.1.7. Corpus Fund:

As per GO MS No.4 User departments shall collect 0.04% of ECV (estimated contract value) with a cap of Rs. 10,000 (Rupees ten thousand only) for all works with ECV upto Rs.50 Crores, and Rs. 25,000/- (Rupees twenty-five thousand only) for works with ECV above Rs.50 Crores, from successful bidders on e Procurement platform before entering into agreement / issue of purchase orders, towards e procurement fund in favour of Managing Director, APTS. There shall not be any charge towards e Procurement fund in case of works, goods and services with ECV less than and upto Rs. 10 lakhs

4.1.8. Tender Document:

The bidder is requested to download the tender document and read all the terms and conditions mentioned in the tender Document and seek clarification if any from the Tender Inviting Authority. Any offline bid submission clause in the tender document could be neglected.
The bidder has to keep track of any changes by viewing the Addendum/Corrigenda issued by the Tender Inviting Authority on time-to- time basis in the E-Procurement platform. The Department calling for tenders shall not be responsible for any claims/problems arising out of this.

4.2. Bid Submission Acknowledgement:

The bidder shall complete all the processes and steps required for Bid submission. The system will generate an acknowledgement with a unique bid submission number after completing all the prescribed steps and processes by the bidder. Users may also note that the bids for which an acknowledgement is not generated by the e-procurement system are treated as invalid or not saved in the system. Such invalid bids are not made available to the Tender Inviting Authority for processing the bids. The Government of AP is not responsible for incomplete bid submission by users.

4.3 THE BIDS RECEIVED BY, OTHER THAN E-PROCUREMENT PLATFORM SHALL NOT BE ENTERTAINED UNDER ANY CIRCUMSTANCES.

4.4 The bidders are therefore advised to ensure that their bids are uploaded in complete shape in the first instance itself. Post bid rebates, revisions or deviation in quoted price and/or conditions or any such offers which will give benefit to the bidder over others will not only be rejected straight but the original bid itself will get disqualified on this account and the bidder's bid security amount will be forfeited.

4.5 COMPLETENESS OF BID:

a) The bid shall be quoted for all the works & all the items as per the bid specification.
b) Part bids or incomplete bids will not be accepted and will be rejected.

5.0 Contents of the specification:

i) **Volume –I**:

a)	Instructions To Bidders	In the word file
b)	General Terms & Conditions (Section-I)	
c)	Financial Terms & Conditions (Section-II)	
d)	Schedules (I – X)	
e)	Check List (XI)	
f)	Forms (A- F)	
g)	List of approved Manufacturers	
h)	List of approved Banks	
i)	Price Schedules for Equipment/Material(Schedule)	(As separate XL file)
j)	Price Schedules for Works (Schedule –A)	
k)	Financial Bid/Price Bid (Schedule – D)	
l)	Completion period (Schedule – E) and Bar Chart	
m)	List of makes of the materials proposed to be supplied if the contract is awarded (Schedule-F)	
n)	Notice Inviting Bid -	(As separate PDF file)
o)	Project Data sheet -	
p)	Quality Assurance Plan -	
q)	Qualification Requirements -	

i) Formats

A. Bank Guarantee proforma for Bid Security

B. Bank Guarantee proforma for performance security.

C. Bid form

D. Format for Bank's Certificate for credit facilities

E. Format for joint venture agreement

F. Format for power of attorney for joint venture

iii) Volume-II: Technical Specifications:

a) General
b) Technical specifications of all Material

iv) Schedules: To be filled up and furnished along with the bid. In case of joint venture, schedules are to be filled up by both the partners

a)	Particulars of Registration	:	Schedule-I
b)	Details of Technical personnel	:	Schedule-II
c)	Details regarding Financial standing	:	Schedule-III(A)
d)	Financial Turnover	:	Schedule-III(B)
e)	Details of Works /Contracts executed	:	Schedule-IV(A)
f)	Details of Fabricated Towers and structures	:	Schedule-IV(B)
g)	Details of works under execution (APTRANSCO)	:	Schedule - V (A)
h)	Details of works under execution other than APTRANSCO	:	Schedule – V (B)
i)	Details of all ongoing works and balance Commitment:		Schedule - V (C)
j)	Under taking with regards to the Q.R	:	Schedule - V (D)
k)	Details of gangs available	:	Schedule-VI
l)	Details of vehicles and T & P available	:	Schedule-VII (A&B)
m)	Undertaking regarding relatives	:	Schedule-VIII

n)	Declaration for taking up awarded works	:	Schedule-IX
o)	Bank A/c. details for RTGS	:	Schedule-X
p)	Check List	:	Schedule-XI
q)	Price Schedules of Material	:	Schedule-A
r)	Price Schedules of Works	:	Schedule-B&C
s)	Price Bid	:	Schedule-D
t)	Completion Period & Bar Chart	:	Schedule-E
u)	Makes of Materials proposed to supply	:	Schedule-F

NOTE:

1. ***APTRANSCO is not responsible for any omissions in the bid specification downloaded by the bidders from the web site.***
2. ***The bidders can verify the accuracy of the downloaded bid specification with the master copy available at the office mentioned in the Project Data Sheet.***
3. ***The bidders are requested to check / verify the web site before scheduled closing date for any changes/ amendments in the tender.***

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TABLE OF CLAUSES - SECTION – I :: GENERAL

Clause. No.	Description
1.	Definition of terms
2.	Scope of the contract
3.	Qualification requirements
3.1	Physical experience
3.4	Financial turnover
4.	General
4.1	Performance based experience
4.2	Awarding contract for more than one work
5.	Bid validity
6.	Bid security
7.	Discrepancies
8.	Evaluation of bids
9.	Notification of award and acknowledging:
10.	Service conditions
11.	Sub-contractor and sub-orders
12.	Power to vary or omit work
13.	Execution of works relating to power line crossings and railway crossings:
14.	Electricity rules:
15.	Regulation of local authorities and statutes
16.	Co-ordination with statutory bodies and other agencies
17.	Co-operation with other contractors
18.	Contractor's cooperation with the purchaser
19.	Employment of technical staff for supervision, skilled and unskilled labour
20.	Liability for accidents to persons:
21.	Liability for damage to works and plants:
22.	Approval of sub-vendors for equipment / materials
23.	Quality assurance plan
24.	Quality control & inspections
25.	Time control
26.	Management /co-ordination meetings
27.	Site stores
28.	Contractor's site office
29.	Clearing up the site
30.	Change orders
31.	Water & power
32.	Operation and maintenance manuals:
33.	Safety plans
34.	Bill of materials:
35.	Photographs and progress report
36.	Measurement of work and progress payments
37.	Protection of work
38.	Fire protection
39.	Security (watch & ward)
40.	Testing and commissioning
41.	Indemnity bond for all materials
42.	Accounting of surplus materials/ settlement of material account
43.	Completion of Balance works and Handing Over
44.	Final account
45.	Taking over
46.	Technical Audit

SECTION – I: GENERAL

The Transmission Corporation of Andhra Pradesh Limited, hereinafter referred to as the APTRANSCO, invites bids for the Works as mentioned in the **NIT** and **PROJECT DATA SHEET** of this Specification.

1. DEFINITION OF TERMS:

In the contract, the following expression shall, unless the contract otherwise requires, have the meanings hereby respectively, assigned to them.

I) The 'Purchaser/Employer' shall mean the Transmission Corporation of Andhra Pradesh Limited and shall include its successors and permitted assigns.

II) The 'Bidder' shall mean the firm/company who are uploading their bids online to APTRANSCO.

III) The 'Contractor' shall mean the bidder whose bid has been accepted by the purchaser and shall include the bidder's heirs, legal representatives, and successors and permitted assigns.

IV) The 'Manufacturer' shall mean the firm/company named in the contract, to whom the supply of material/ equipment and services of erection is entrusted by the contractor.

V) The 'Engineer' shall mean the officer placing the order for the work with the Contractor and such other officer as may be authorized and appointed in writing by the Purchaser to act as Engineer for the purpose of supervising the Contract.

VI) The 'Supervising Engineer' shall mean such officer as may be duly appointed from time to time by the Purchaser or his Engineer to take general control and supervision of work.

VII) The 'Inspector' shall mean any person or persons nominated by the Purchaser and/or the Engineer to inspect stores or works under the agreement and/or his duly authorized representative appointed to act as the Inspector.

VIII) The 'Contract' shall mean and include the bid and acceptance thereof, the general conditions, special conditions, Specifications, Schedules, drawings, Form of Bid, covering letter, schedules of prices, the final General conditions and the format Agreement executed on the stamped paper.

IX) The 'Specification' shall mean collectively all the terms, stipulations, and conditions of contract, technical provisions and annexure thereto and list of corrections and amendments mutually agreed upon from time to time in writing.

X) 'Tests on completion' shall mean such tests, as are prescribed by specification, to be made by the Contractor to the satisfaction of the purchaser before the work is taken over by the Purchaser.

XI) 'Commissioning' shall mean the satisfactory commissioning and operation of the equipment/work as specified in the project, after all necessary initial tests, checks and adjustments required at site, if any, have been satisfactorily completed and the equipment / works have been successfully charged.

XII) 'Commercial use' shall mean the use of the work, which the contract contemplates or of which it is commercially capable.

XIII) 'Approval' shall mean the written approval of the Engineer and of the statutory authorities wherever such authorities are specified by any codes or otherwise.

XIV) 'Month' shall mean calendar month.

XV) 'Writing' shall include any manuscript, typewritten or printed, statements, under or over signature or seal as the case may be.

XVI) 'Letter of Intent' shall mean the purchaser's letter conveying his acceptance of the bid, subject to such reservations as may have been stated therein.

XVII) 'ECV' shall mean Estimated Contract Value.

XVIII) 'RTGS' shall mean Real Time Gross Settlement.

XIX) GST means Goods and Service Tax

2. **SCOPE OF THE CONTRACT: The brief description of work/ project is mentioned in the NIT and Project Data Sheet. The detailed scope of work/ project is as per the schedules enclosed and as per the technical specifications given in Volume –II.**

(A) Transmission Lines/ Substations:

a) The APTRANSCO shall provide approved profiles, approved layouts, tower schedules

b) The contractor shall procure the materials / equipment with technical parameters as per the technical specification volume-II, from the approved list of manufacturers / suppliers only and execute the works as per the approved profiles / approved layouts. It is the responsibility of the contractor to complete the works in full shape and handover to APTRANSCO in fully operational condition within the time specified in the contract.

c) Being the turnkey work, if any petty/ minor items required for erection of the substations & lines, even though they have not been specifically included in this specification but are necessary for successful erection / operation of the substations & lines are deemed to be included in the specification.

d) All the approved GTP'S &drawings of equipment/material are available at APTRANSCO website i.e www.aptransco.gov.in.

3 QUALIFICATION REQUIREMENTS:

3.1 Physical Experience Requirements: For awarding of notified work/ project the bidder should meet the following qualifying criteria.

3.2 The bidder should have valid 'A" Class electrical license issued for executing the electrical works related to Extra High voltages.

3.3 The bidder in his name should have executed the following quantity of works mentioned below against each type of notified work.

Sl. No.	Notified Work	Technical Qualification Criteria	
		The bidder should have executed at least the following works in the last 10 years as a prime contractor / as a lead partner for qualifying the notified work.	The following works done by the bidder should be in successful operation for at least one year. The bidder shall enclose the proof of successful operation of 1 year.
Sub station			
1.	220kV SS	50% of the work (refer schedule) of 220kV or higher voltage line (or) 100% of the work of 132kV or higher voltage line.	50% of the work (refer schedule) of 220kV or higher voltage line (or) 100% of the work of 132kV or higher voltage line.

Note: The joint Venture clause shall be applicable for the works with ECV greater than Rs. 10.00 Crores only. The lead Partner of the Joint Venture shall meet at least either one of the Q.R mentioned against Part-1 i.e. Line and 2nd ckt or Part-II i.e. Sub-Station and bays above otherwise the bid will be disqualified.

If the notified work is combination of the above items, the bidder shall meet the qualification requirements mentioned against each notified item separately. "The bidder should invariably furnish the Performance certificates in support of their Qualifying requirements and the **Performance certificates shall be issued by an officer not below the rank of Executive Engineer or equivalent cadre of state power utility / Government organizations / PSU only otherwise the bid will become non-responsive**".

3.4 FINANCIAL TURNOVER:

The Bidder should have the bid capacity not less than the estimated contract value or value of work put to tender. The bid capacity will be calculated as per formula **(2AN-B)> ECV**. The bidder/Firm/ Company shall be a profit making Company.

where

A = Max. Value of Works executed in any one financial year during the preceding 5 financial years (updated to the current price level with a simple weightage of 10% per each financial year) taking into account the completed as well as works in progress. With regard to execution of the works during the last 5 years, the bidder shall produce the Chartered Accountant's certificate with audited and approved profit and loss statement shall be produced.

N = Number of years prescribed for completion of the works for which tenders are invited. If N is less than 6 months it will be taken as 6 months

B = Value of existing commitments and ongoing works to be completed during the period of completion of work for which tenders are invited.

ECV = Estimated Contract Value (or) The Value of work put to tender

The bidder shall produce the statement showing the **value of existing commitment** and ongoing works as well as the stipulated period of completion remaining for each of the works. The statement must be supported by certificates issued by officers of the State/Central Government Department / Under takings not below the rank of Executive Engineer or the officer of equivalent rank and counter signed by the Superintending Engineer or the officer of equivalent rank (or) The bidder shall produce a declaration as mentioned below on a non-judicial stamp paper of Rs.100/- duly notarized.

"I, Sri/ Smt. /M/s..... .... do hereby solemnly affirm and declare that the balance works on hand as per the details furnished in tender schedule are true and also declare that in case it is found to be false fake/incorrect at a later date, I/We will abide by any action such as disqualification or termination of contract under clause 60(a) PS to APSS or black listing or any other action deemed fit by APTRANSCO ."

Signature of the Bidder

3.5. JOINT VENTURES:

Joint Venture Clause:

The bidders can also participate as a **Joint-Venture** containing not more than two firms/companies. The Physical / Technical experience and financial turnover of the partners of the Joint venture shall be added together to determine the JV bidder's compliance with the minimum qualifying criteria setout in the specification. **However, in order for a Joint Venture to qualify, the partners of Joint Venture must meet the following minimum criteria.**

a) The joint venture here in mean, Firms/companies not more than two, which are having **Joint Venture agreement** between them to participate and to execute the notified transmission project of APTRNASCO and shall clearly mention in the JV agreement, that both firms jointly and severally shall have responsibility in performing the project.

b) The lead partner shall have at least 40% of the bid capacity which will calculated as per formula (2AN-B)> ECV. However both the JV partners put together shall meet the required bid capacity.

c) The lead partner shall have the following minimum technical experience.

(i) In case of tendered project involves multiple works like Sub-Station and Transmission line, then the lead partner shall meet technical qualification mentioned in the Specification either for one Sub-Station or for Transmission line.

(ii) In case the tendered project involves only Transmission line, then the lead partner shall meet at least 50% of the technical qualification mentioned in the Specification.

(iii) In case the tendered project involves only Sub-Station / Switching station / bay extensions, then the lead partner shall meet 100% of the technical qualification mentioned in the Specification.

However both the JV partners put together shall meet the technical qualification and financial bid capacity as required / as mentioned in the Specification.

d) Any of the JV partners participating in the present tender, shall not participate in association with other bidders or participate individually in the same tender. If any one of the JV partners participates / submits more than one bid either individually or as JV partner, all such bids will be made disqualified and their bids will not be evaluated further.

e) However, in order to qualify the Bid submitted by a JV partners, they shall comply with the following requirements. The joint venture bid with non-compliance of the following provisions will be disqualified.

a. Joint venture shall be represented by technical member as lead partners for the purpose of execution of the Bid. All the Payments shall be made through RTGS to the lead partner account as per the bank details mentioned in the Bid.

b. The uploaded scanned bid shall have authorised signatures of both the partners so as to be legally binding upon both members.

c. The lead partner shall be authorized to upload the bid in e-procurement and to incur liabilities and receive instructions for and on behalf of the partners of JV, and the entire execution of the Contract shall be done with the Lead Partner on behalf of the Joint Venture as per the power conferred to him in the Power of Attorney. The authorization shall be evidenced by submitting a power Attorney by legally authorized signatory of all partners as per proforma provided in the bidding documents.

d. **The performance security shall be submitted by both the Joint Venture partners in the ratio of 75:25(Lead Partner : Associate Partner)**

e. In case of any breach of the contract, **both partners will be held fully responsible for the fulfilment of the contract.** In case of any default in execution of the contract, it is the sole discretion of APTRANSCO to transfer the order on any one of the partner to complete the work individually in such case the default partner will not be considered for participating in future tenders individually / JV or to

terminate the total contract and get the balance works executed through alternate agency at the cost and risk of both the partners of the Joint Venture duly en-cashing the Performance Security and any other securities and withholding pending payments if any in the present contract or in the other contracts of the JV partners individually or jointly. In such cases, both the JV partners are liable for blacklisting. It is mandatory for both the partners to attend the review meetings, failure of which will be viewed seriously.

f. The Joint Venture partners shall be profit making Companies / Firms.

g. **The lead partner of the joint venture shall upload the following documents pertains to both the JV Partners. (a)Authority of the partner/company to enter into joint venture business in memorandum of association (b) A certified copy of the board resolution according permission to entering into the joint venture and to take up the notified work. (c) A certified copy of the board resolution according permission for affixing the company seal on the joint venture deed**.

h. APTRANSCO shall in no way be responsible or connected or liable or suffer for the disputes arise between the Joint venture partners.

i. **While calculating the bid capacities at the time of evaluation, the existing works with JV Partners will be taken as 75:25 ratio between lead partner and associated partner (for calculating value of the existing commitment i.e. B in 2AN-B). The lead partner of the joint venture shall upload the Schedule-V(C) pertains to both the JV Partner for calculating the balance commitments.**

4 GENERAL:

i) Bidders shall submit audited financial reports (audited Profit & Loss statement, audited Income & Expenditure statement and audited Balance Sheet) counter signed by Charted Accountant for last **Five Financial years** preceding the date of bid submission.

ii) **Sub-Contractors' experience may be considered if it is officially recognized by** Order Placing authority of the State Power Utility / Government organizations / PSU only.

iii) Responsibility for correctness of the information submitted in the bid lies with bidder. If any information furnished in the bid is proved to be false at a later date, the bid will not only be rejected but the bidder will be BLACKLISTED.

4.1 PERFORMANCE BASED EXPERIENCE:

In case of any bidders who are already having one or more works awarded to them by APTRANSCO which are under execution at the time of bidding and who fails to execute the works already awarded as per the stipulated time schedule of completion, bids of such bidders will not be considered.

4.2 AWARDING CONTRACT FOR MORE THAN ONE WORK

In case of contractors quoting more than one work / tender at similar time and becomes lowest in more than one work, the aggregate requirement of turnover will be examined at the time of awarding based on the works already awarded / to be awarded / under consideration by APTRANSCO.

5. BID VALIDITY:

120 days from the Bid submission closing date. Bids having validity less than **120 days** are liable for rejection at the discretion of Transmission Corporation of Andhra Pradesh Limited.

6. BID SECURITY

The bidder has to upload the scanned copy of bid security for the amounts as indicated in the 'Notice Inviting Bids'. In the absence of indicating the same in Bid notice, the bidder has to furnish the amount specified in the NIT as bid security. The bid security shall be in the form of a Demand Draft or a Bank Guarantee.

The amount towards bid security will not be accepted by cheque, cash or money order. It shall be paid by way of Demand Draft on any approved Bank as per the list enclosed, drawn in favour of **Superintending Engineer, O&M Circle, A.P.TRANSCO, Visakhapatnam , payable at Visakhapatnam**.

In case the bid security is paid by way of a Bank Guarantee, the BG shall be issued by any approved Bank as per the list enclosed, and the validity of the BG shall cover a period of **45 days** over and above the period of bid validity. The BG shall be executed on Rs.100/- non-judicial stamp paper in favour of **the Superintending Engineer, O&M Circle, APTRANSCO, Visakhapatnam**. In case of extension of bid validity by the bidder at the request of APTRANSCO, the validity of the bid security Bank Guarantee shall be suitable extended along with confirmation of bid validity. The Bank Guarantee shall be issued by the Bank on behalf of the bidder or either of the Joint Venture partners only.

BIDS ON E-PROCUREMENT PLATFORM, WHICH DO NOT CONTAIN THE SCANNED COPY OF BID SECURITY, WILL BE REJECTED AND THE BIDDER WILL BE BLACKLISTED WITHOUT ANY CONSIDERATION WHATSOEVER. Bids with insufficient value or validity of bid security will be rejected without consideration whatsoever. The APTRANSCO will not be responsible for any omissions in this regard on the part of the bidders, which result in the rejection of the bids and the bidders will be blacklisted by the APTRANSCO.

The Bid security of the successful Bidder on his request can also be adjusted as part of Performance Security and the balance security deposit shall be paid at the time of concluding Agreement by the successful Bidder provided the bidder extends the validity of the bid security bank guarantee suitably.
The Bid Security will be forfeited in the following cases.

i) When the successful bidder does not accept the order after issue of preliminary acceptance letter or the Letter of Intent.

ii) When the successful bidder fails to furnish the Performance Security within 15 days from the date of issue of preliminary acceptance letter or Letter of Intent.

iii) When a bidder withdraws his bid or alters the prices after schedule bid submission closing date and before the expiry of the bid validity period or does not accept the correction of errors made in his bid or Offers post Bid rebates, revisions or deviations in quoted prices and / or conditions or any such offers.

7. DISCREPANCIES: In case of difference in e-procurement lump sum amount and the amount arrived from the schedules the following will be adopted:

i) Where there is a discrepancy between the total lump sum rate quoted in the e-procurement platform web page and in the schedules uploaded, the total lump sum rate quoted in the e-procurement platform will govern for deciding L1.

ii) If any discrepancies arises after applying the tender percentage on the schedules:-
   (a) If the amount arrived is less than the e-procurement lump sum amount, then the lesser amount shall be taken for award of contract.
   (b) If the amount arrived is more than the e-procurement lump sum amount, the e-procurement lump sum amount shall be taken for award of contract. The unit rates shall be adjusted to match the e-procurement lump sum amount.

8.0 EVALUATION OF BIDS:

8.1 The bids will be opened on-line by bid inviting authority or assignee at the time and date specified in the Bid documents.

Physical and Financial capacities of all the bidders will be evaluated based on the documentary evidence furnished along with the bid. Evaluation of Technical/ Commercial and Price bids will be made based on the available documentary evidence submitted along with the bid. However, the Tender opening Authority may seek clarification with the bidder as per clause 8.2, if required. If any information furnished in the bid is proved to be false later, the bid will not only be rejected but the bidder will be black listed.

If any significant discrepancies are observed in the Technical/Price schedules of any bid, due to which the bidder becomes lowest and withdraws the bid, the bid will be liable for rejection and forfeiture of Bid security and also liable for black listing of the bidder.

They are liable for rejection for any of the deviations mentioned below.

a) No Bid Security or Insufficient Value or Validity of Bid Security
b) Insufficient Validity of the Bid.
c) Non-uploading of the required documents, schedules, forms, supporting documents for Qualifying Requirement. Insufficient or illegible supporting documents, schedules, forms.
d) Deviation from the Delivery schedule of the Specification.
e) Deviation from the Payment terms of the Specification.
f) Deviation from the General terms and conditions
g) Deviation from the Technical Specification.
h) Non-Compliance with the Qualifying Requirements.
i) Record of Poor Performance in the previous / on-going works. Record of poor progress i.e. abandoning the work, not properly completing the contract, inordinate delays etc.
j) Record of Black listing in any other Power Utilities / Government Department.
k) Record of financial failures if any.
l) History of litigation with Government/APTRANSCO during the last 5 years.
m) History of criminal record in which the Bidder is involved.
n) Non-uploading of the Price schedules. Insufficient or illegible Price Schedules. Change of Quantities / Description of the work / materials items in the Price Schedule uploaded by APTRANSCO.
o) Furnishing of false / fabricated particulars in the forms, statements and annexure, submitted in proof of the qualification requirements.
p) Not turned up for entering into agreement, when called upon with in the time specified in the letter of acceptance.
q) Any other deviations of clause mentioned in this specification.
The price bids of qualified bidders in the pre-qualification analysis will only be considered.

While being equal or compatible in other aspects preference will be given to those bidders who have good experience / proven performance, past experience is superior in terms of quality and timely completion in similar works.

It is the sole discretion of APTRANSCO regarding finalization of the bids. Mere quoting and becoming of L1 (Lowest quoted) bidder shall not automatically entitle any rights for any bidder for award of contract. The past performance of the contractor and performance of the contractor in the ongoing works shall be taken in to account while finalizing the bids. The APTRANSCO reserves its right to reject any or all the bids and not to accept the lowest or any other without assigning reasons. The APTRANSCO reserves its rights to cancel the tender at any stage without assigning the reasons.

8.2 Clarification on the Bid: "The Tender opening authority may call upon any Tenderer for clarification on the statements, documentary proof relating to the bid. The request for

clarification and response thereto shall be in writing and it shall be only on the qualification information uploaded online by the Tenderer. The clarification called for from the Tenderers shall be furnished within the stipulated time, which shall not be more than a week.

The Tenderer if so desirous, shall agree in writing to furnish the clarification called for within the stipulated time and, for disqualification and rejection of his Tender in the event of failure to do so."

No Clarification shall be asked/entertained regarding qualification requirements (Supply, Performance and Financial Turnover). Tenders shall be evaluated with uploaded documents in the e-procurement website only.

9. NOTIFICATION OF AWARD AND ACKNOWLEDGING:

a) The Bidder who's Bid has been accepted will be notified, the award of the work by Bid inviting authority prior to expiration of the Bid validity period by registered letter / Fax. This letter (hereinafter and in the Conditions of Contract called "Letter of Intent") will indicate the sum that the employer will pay the Contractor in consideration of the execution, completion, and maintenance of the Works by the Contractor as prescribed by the Contract (hereinafter and in the Contract called the "Contract Amount").

b) Upon intimation being given by the Bid inviting authority, of acceptance of his Bid, the Bidder shall make payment of the balance Security Deposit and additional security deposit wherever needed by way of Demand Draft or unconditional and irrevocable Bank Guarantee obtained from approved Bank with required validity period and sign an agreement in the form prescribed by the department for the due fulfillment of the contract. Failure to attend enter into the required agreement or failure to submit the Performance and Additional Securities as intimated in Detailed contract Order shall entail forfeiture of the Bid Security deposited. The written agreement to be entered into between the Contractor and the **Executive Engineer / O&M** shall be the foundation of the rights and obligations of both the parties and the contract shall not be deemed to be complete until the agreement has first been signed by the contractor and then by the proper officer authorized to enter into contract on behalf of the Transmission Corporation of Andhra Pradesh Limited.

c) The successful bidder shall submit the performance Bank Guarantee within 15 days of the issuance of the detailed contract.

d) The successful Bidder has to sign and return a copy of the detailed contract within a period of 7 days from the date of issue of contract award letter.

e) Any technical clarifications shall be done by the concerned **Executive Engineer /O&M**. **The Superintending Engineer / O&M** shall arrange to hand over the approved layout, Final Ground Level for substation and approved profile for line on the day of kickoff meeting.

f) The contractor or his representative shall invariably attend the meetings whenever called. The representative of the contractor who attends the kick-off meetings and review meetings shall be qualified and shall be authorized by the contractor with the details of qualification and responsibility and designation of the authorized person.

g) The concerned **Executive Engineer / O&M** shall be the agreement authority and the contractor shall attend the office of the **EE/ O&M** within 15 days of the detailed order to enter in to an agreement. The **Executive Engineer / O&M** shall enter an agreement with the contractor duly verifying the following.

i. The PAN card, GST Provisional registration, Labor license, valid A-grade electrical license to execute EHT works in A.P.
ii. Valid Performance Bank Guarantee.
iii. Valid Project insurance
iv. Authorization / power of attorney of the contractor representative to sign the agreement.

The following documents shall also be made part of the agreement.

i. The copy of the letter of intent.
ii. The copy of the Purchase order.
iii. The copy of the volume-1 of the specification.

v. Copy of the Performance Bank Guarantee and its approval.
vi. Authorization / power of attorney of the contractor representative to sign the agreement.
vii. Copies of the statutory registrations of the contractor etc.,

10. SERVICE CONDITIONS:

The Substation / Transmission Line will be operating under the following tropical conditions.

i)	Humidity	-	60 to 80%
ii)	Max. isoceraunic level	-	40 thunder storm days per year.
iii)	Max. temperature in shade	-	45 Deg.C
iv)	Min. temperature in shade	-	10 Deg.C
v)	Max. temperature in sun	-	55 Deg.C
vi)	Altitude above sea level	-	550 meters to 1000 meters.

Due consideration will be given to any special devices or attachments, put forward by the bidder which are calculated to enhance the general utility and the safe and efficient operation of the line.

11. SUB CONTRACTOR AND SUB ORDERS:

The successful Bidder shall carry out all the works mentioned in the schedules by their company only. **Sub contracting or sub ordering is not allowed by APTRANSCO.**

12. POWER TO VARY OR OMIT WORK:

No alterations, amendments, omissions or variations of the works under the contract shall be made by the contractor except as directed in writing by the APTRANSCO, but the APTRANSCO shall have full powers to instruct the contractor by notice in writing to make such variations without prejudice to the contract. The contractor shall carry out such variations and be bound by the same conditions as far as applicable as though the said variations occurred in the contract documents. If any suggested variations would, in the opinion of the contractor, if carried out, prevent him from fulfilling any of his obligations under the contract, he shall notify the APTRANSCO thereof in writing and the APTRANSCO shall decide forthwith whether or not, the same shall be carried out and if the APTRANSCO confirms its instructions, contractor's obligations shall be modified to such an extent as may be mutually agreed. Any agreed difference in cost occasioned by any such variation shall be added to or deducted from the contract price as the case may be.

APTRANSCO reserves the right to allot any surplus materials from APTRANSCO stores to this project (even though the scope of specification covers supply of materials) duly omitting that part of materials from Schedule-A. In such cases transportation shall be carried out by the Contractor and transporting charges shall be paid as per clause (3) Section-II Financial 'Supplemental Quantities / Items'. The bidders shall supply only the balance materials required after allotment of such materials.

13. EXECUTION OF WORKS RELATING TO POWER LINE CROSSINGS AND RAILWAY CROSSINGS:

Works such as erection of towers underneath an existing power line and paving out of conductors and earth wire and stringing the power line crossing span or a railway crossing span, will have to be done only after receipt of approval from the concerned officer, which, sometimes, may not match with the program of the contractor. In such cases, the contractor shall execute such works as and when approvals are received. His rates for tower erection and stringing shall take into consideration, such contingencies also.

14. ELECTRICITY RULES:

All works shall be carried out in accordance with the relevant clauses of the Indian Electricity Act and Rules unless modified by this specification. In case of variation between the two, the latter shall be binding on the contractor.

Unless otherwise specified, works shall be carried out in accordance with the Indian Electricity Act, Electricity Rules or any revisions thereof, which may be issued during the currency of the contract and the requirements of any other Regulations and Acts in India to which the APTRANSCO may be subjected to.

All railway tracks, communication lines or other important track crossings and routing the line through Air field region shall conform to the relevant rules and procedures as may be laid down by Railway, Communications, Aviation or other concerned authorities from time to time.

Aviation signals will have to be provided at the top of the transmission line towers in the vicinity of civil and military aerodromes or airfield regions, if any. Similar provisions will have to be made on the special river crossing towers. The rates for tower erection shall include such works also.

15 REGULATION OF LOCAL AUTHORITIES AND STATUTES

i) The CONTRACTOR shall ensure compliance with all statutes laws of India, rules and regulations of the Central or State Government or any other authority such as the Workmen's Compensation Act 1923, Payment of Wages Act, Minimum Wages Act 1948, Employees State Insurance Act, Employees Provident Fund Act, etc., and any statutory modifications thereof in connection with employees engaged by him in the work. The contractor's offer shall be presumed to include financial liabilities arising from the above and the purchaser shall not be liable for any extra costs on this account.

ii) The CONTRACTOR shall conform to the provisions of Indian Boiler Regulation, Factory Laws, Indian Electricity Act and rules made there under, and any other acts of legislature relating to the work and to the regulations and bye-laws of any authority and of any matter, lighting and other companies and/or authorities with whose systems the plant/structure proposed to be connected and shall, before making variations from the drawings or specifications may be necessitated by so conforming, give to PURCHASER/ENGINEER written notice, specifying the variations proposed to be made and the reason for making it and apply for instructions thereof.

iii) The CONTRACTOR shall arrange to give all notices required by the said Acts, Regulations or Bye-laws to be given to any Authority or to any Public Officer and pay all fees that may be properly chargeable in respect of the 'Works' and lodge the receipts with the PURCHASER/ ENGINEER, unless otherwise specified. Obtaining all permits and licenses required thereupon shall be the responsibility of the CONTRACTOR.

iv) All registration and statutory inspection fees, if any, in respect of his work pursuant to this 'Contract' shall be to the account of the CONTRACTOR. However, any registration, statutory inspection fees lawfully payable under the provisions of the Indian Boiler Regulations and any other statutory laws and its amendments from time to time during erection in respect of the plant and equipment ultimately to be owned by the PURCHASER shall be to the account of the PURCHASER. Should any such inspection or registration need to be re-arranged due to the fault of the CONTRACTOR or his VENDOR, the additional fees for such inspection and/or registration shall be borne by the CONTRACTOR.

16 CO-ORDINATION WITH STATUTORY BODIES AND OTHER AGENCIES

The Contractor is fully responsible for carrying out all co-ordination and liaison work to be required with Electrical inspectors, Factory inspector and other statutory bodies for implementation of the work.

The application on behalf of the Purchaser for submission to the electrical inspector and other statutory bodies along with copies of drawings complete in all respects shall be done by the Contractor and approval/ certificates taken well ahead of time so that the actual commissioning of equipment/ materials is not delayed for want of inspection and approval by the inspector and statutory bodies. The actual inspection work by the electrical inspector shall be arranged by the Contractor. However, any fees paid to electrical inspector/ statutory bodies, etc., in this regard shall be borne by the Purchaser. Any modification on the installation suggested by the electrical inspector or any other statutory authority shall also be carried out by the contractor at no extra cost by the Purchaser.

17 CO-OPERATION WITH OTHER CONTRACTORS

The Contractor shall co-operate with all other contractors or tradesmen of the Purchaser, who may be performing other Works on behalf of the Purchaser and the workmen who may be employed by the Purchaser and doing work in the vicinity of the Works under the contract. The Contractors shall also so arrange to perform his work as to minimize, to the maximum extent possible, interference with the work of other contractors and their workmen. Any injury or damage that may be sustained by the employees of the other contractors and the Purchaser, due to the Contractor's work shall promptly be made good at his own expense.

The Purchaser / Engineer shall determine the resolution of any difference or conflict that may arise between the Contractor and other contractors or between the Contractor and the workmen of the Purchaser in regard to their work. If the Works of the Contractor is delayed because of any acts or omissions of another contractor, the Contractor shall have no claim against the Purchaser on that account other than an extension of time for completing his Works. However extension of time will be considered after examination of its merits and at the discretion of the purchaser.

18 CONTRACTOR'S COOPERATION WITH THE PURCHASER

In cases where the performance of the erection work by the CONTRACTOR affects the operation of the system facilities of the PURCHASER/ENGINEER, such erection work of the CONTRACTOR shall be scheduled to be performed only in the manner stipulated to be performed by the PURCHASER/ ENGINEER and the same shall be acceptable at all times to the PURCHASER/ ENGINEER. It will be the responsibility of the CONTRACTOR to provide all necessary temporary instrumentation, measuring devices and all other material/ arrangements required during the work for proper functioning of the equipment.

The CONTRACTOR at all times shall work in coordination with the PURCHASER/Engineer's staff and offer them all reasonable facilities to become familiar with the erection, operation and maintenance of the equipment. In respect of observations of local rules, administrative orders, working hours etc., the CONTRACTOR and his personnel shall cooperate with the PURCHASER/ENGINEER.

19 EMPLOYMENT OF TECHNICAL STAFF FOR SUPERVISION, SKILLED AND UNSKILLED LABOUR:

The contractor shall provide experienced, technically qualified supervising Engineers for the supervision. The Chief Supervising Engineer of the contractor or his agent shall have full power as the representatives of the contractor who can negotiate at site in regard to execution of the contract. The minimum qualification of site Engineer is a degree in Engineering. The contractor must make his own arrangements for recruiting skilled, semi- skilled and un-skilled labour in sufficient numbers. The contractor shall engage only competent skilled workers. The Executive Engineer/ In-charge of the work will have the right to remove any skilled worker employed by the contractor, if found not suitable.
The contractor shall employ at least the following Technical Staff indicated against each work.

Work	Technical Personnel to be employed by the Contractor
220 kV or 132 kV Sub-Station or Switching Station	1 B.Tech / B.E Electrical +1 B.Tech / B.E Civil + 1 Diploma Electrical
220 KV or 132 kV Line	1 Diploma Electrical +1 Diploma Civil
220kV/132 kV bays	1 Diploma Electrical / Civil

The contractor shall have one of the above technical staff as a project manager with five year's experience in executing such contract of comparable nature.

In case of failure of the contractor to employ technical staff during execution as above, recovery shall be made from his bills at Rs. 10,000/- per month for each work, besides other penal action.
The contractor shall make his own arrangements for engagement of all labour, local or otherwise, their transportation, housing, feeding and payment thereof, in accordance with labour law, unless the contract otherwise provides. No idle labour payment will be made to the contractor.

19.1 FACILITIES TO WORKERS

Minimum facilities for workers as per Factory Act and other relevant acts shall be provided for by the CONTRACTOR. If the CONTRACTOR fails to do so, the same shall be provided by the Purchaser and deduct the same from the CONTRACTOR'S bills.

First-Aid

The Contractor shall provide necessary first-aid facilities for all his employees, representatives and workmen working at the Site. Enough number of Contractor's-personnel shall be trained in administering first-aid.

Cleanliness

The CONTRACTOR shall be responsible for keeping the entire area allotted to him clean and free from rubbish, debris etc., during the period of 'Contract'. The CONTRACTOR shall employ enough number of special personnel

to thoroughly clean his work area. All such rubbish and scrap material shall be stacked or disposed in a place to be identified by the PURCHASER/ENGINEER. Materials and stores shall be so arranged as to permit easy cleaning of the area. Similarly, the offices of the CONTRACTOR shall be kept clean and neat to the entire satisfaction of the PURCHASER/ ENGINEER. Proper sanitary arrangements shall be provided by the CONTRACTOR, in the work areas and office of the CONTRACTOR and residential colonies. If contractor fails to comply with these requirements in spite of instructions from the Purchaser, the areas will be cleaned by the purchaser at contractor's cost and the cost of the same will be deducted from the contractor's due amounts.

20. LIABILITY FOR ACCIDENTS TO PERSONS:

a. The contractor shall indemnify the purchaser against all actions, suits, claims, demands, cost or expenses arising in connection with injuries suffered by persons employed by him, on the works whether under the General Law or under the Workers Compensation Act.1923 or any other statute in force at the date of the contract dealing with question of liability of employer for injuries suffered by employees and have taken steps properly to insure against any claims there under. Contractor should take insurance cover to all his workmen for injuries, disablement and death. All compensation against workmen's compensation act should be settled by the Contractor.

b. On the occurrence of an accident which results in the death of workmen employed by the contractor which is due to the contract work and of so serious as to be likely to result in the death of any such workman, the contractor shall within 24 hours of happening of such accident intimate in writing to the Engineer and such other officers required by the provision of the workmen's compensation act, the fact of such accident. The contractor shall indemnify the APTRANSCO against all loss or damage sustained by the APTRANSCO resulting directly or indirectly from his failure to give information in the manner aforesaid including the penalties or fines if any payable by the APTRANSCO as a consequence of the APTRANSCO's failure to give notice under the workmen's compensation act or otherwise conform to the provisions of the said act in regard to such accident.

c. In the event of any claim being made, or action brought against the purchaser involving the contractor and arising out of the matters referred to and in respect of which the contractor is liable under the clause of the contract shall be immediately notified thereof, and he shall conduct all negotiations for the settlement of the same or any litigation that may arise there from.

d. In the event of an accident in respect of which compensation may become payable under workmen's compensation act VIII of 1923 and any subsequent amendment thereof, whether by the contractor or by the APTRANSCO, as principle it shall be lawful for the Engineer to retain out of moneys due and payable to the contractor such sum or such sums of money as may be in the opinion of the said Engineer be sufficient to meet such liability. The opinion of the Engineer shall be final in regard to all matters arising under this clause and will not be subject to any claim.

e. Liability for damage or loss to third party including inspection officers due to act of the contractor or by his representatives connected with the execution of this contract shall be fully borne by the contractor. The contractor shall maintain such detailed records to furnish information regarding engagement and discharge of all workmen employed under this contract as to be adequate for the timely and full settlement of claims under the workmen's compensation act. All cases of accidents or injuries shall be reported to the Engineer with full details required for the settlement under the workmen's compensation act.

f. The contractor should report about all accidents within 24 hours to the Engineer of the APTRANSCO in the preliminary accident form. He should furnish other particulars such as medical certificates, wage particulars, fitness certificates, and proof of having paid the compensation as per the rules in vogue in the due course without delay.

21. LIABILITY FOR DAMAGE TO WORKS AND PLANTS:

21.1 The contractor during the progress of the work shall take every reasonable, proper, timely and useful precaution against accident or injury to the workmen from any cause and shall remain answerable and liable for all the accidents or injuries thereto which until the same, or be occasioned by the acts of omissions of the contractor or his workmen or his representatives and all losses and damages to the works as aforesaid, shall be made in the most complete and substantial manner by and at the sole cost of the contractor and to the reasonable satisfaction of the Engineer. Should such loss or damage happen to units of works or plant or materials falling outside the scope of this contract and due to the lapses of the contractor, these shall be replaced or compensated for by the contractor to the satisfaction of the Engineer.

21.2 In the case of loss or damage to any portion of the work occasioned by the causes, the same shall if

required by the purchaser, be made good by the contractor in like manner but at the cost of the purchaser at a price to be agreed between the contractor and the purchaser and the purchaser shall pay to the contractor the contract value of the portion of the work so lost or damaged or any balance of such contract value remaining unpaid as the case may be.

21.3 Until the work shall be or deemed to be taken over as aforesaid, the contractor shall also be liable for and shall indemnify the purchaser in respect of all damage or injury to any person or to any property of the purchaser or other occasioned by the act of the contractor or his workmen or his representatives or by defective design, work or materials but not due to cause beyond his reasonable control.

21.4 Provided that the contractor shall not be eligible under the contract for any loss of profit or loss of contracts or any claims made against the purchaser not already provided for in the contract, not for any damage or injury caused by or arising from the acts of the purchaser or of others (save as to damage by fire, as hereinafter provided) due to the circumstances over which the contractor has not control nor shall his total liability for loss, damage or injury exceed the total value of the contract.

22. APPROVAL OF SUB-VENDORS FOR EQUIPMENT / MATERIALS AND APPROVAL OF GENERAL TECHNICAL PARTICULARS, DRAWINGS, BOMS ETC.

The materials / equipment required for this project shall be invariably purchased from the manufacturers listed in the specification who have already been supplied similar materials to APTRANSCO and have proven performance. The bidder shall furnish in Schedule-E, the list of manufacturers to a maximum of three companies per each material which were considered while quoting the prices and from whom they are going to purchase the materials if order is placed. All the equipment / material offered in the Bid shall be fully type tested as per the relevant standards. The type tests should have been conducted not earlier than 5 years from the date of bid opening. If the type tests conducted are earlier than 5 years the bidder shall arrange to conduct type tests at his own cost. The bidder shall submit all the type test certificates and General Technical Particulars of equipment / material along with the bid. APTRANSO has approved the General Technical Particulars and drawings of all equipment/ materials of certain manufacturers and are available on APTRANSCO website www.aptransco.gov.in. However, the successful bidder shall obtain the approval of General Technical Particulars and drawings of the equipment/materials which are not available in the APTRANSCO website from the Superintending Engineer/ O&M Circle//Visakhapatnam. Within 15 days of receipt of the Letter of Intent, the Contractor shall finalize the sub-vendors as per the list they have submitted in the bid and submit the finalized vendor list to APTRANSCO for approval along with the GTPs, Drawings, BOMs etc., Any delay in submitting the above list will lead to delay in execution of the project and responsibility lies with the Contractor.

23. QUALITY ASSURANCE PLAN:

The Quality Assurance Plan of APTRANSCO is appended to this specification. The bidders are advised to go through the same before quoting the bids. The successful bidder will have to adhere to the Quality Assurance Plan during execution of works.

Bidder shall maintain the quality standards as per specification and shall attend the remarks of APTRANSCO Quality Control wing on top most priority without any reminders from APTRANSCO.

Setting up of Quality Control Laboratory at Site:

(i) Appropriate grade of concrete and cement content for R.C.C. items may be scrupulously followed as envisaged in IS 456-2000 considering appropriate exposure conciliations.

(ii) Using of M30 grade shall be mandatory for R.C.C. members of all structures located in coastal areas.

(iii) A quality control laboratory shall be setup by the contractor at the construction site for all major works in which the value of the civil works costing more than Rs. 1.00 Crore. List of testing equipment to be kept in the field laboratory is listed below.

Sl.No.	Description of testing Apparatus/Equipment
1.	Core Cutter Apparatus.
2.	Standard Cone Penetration Test Equipment.
3.	Proctor Density test Apparatus.
4.	Compression Testing Machine.
5.	Concrete Test Hammer.
6.	Slump Cone Test Apparatus.
7.	Vicat Needle Apparatus.
8.	Sieve Analysis Apparatus for fine and coarse aggregates.
9.	Screw guage, Vernier calipers.
10.	Auto level, Theodalite, Chains and Ranging rods.

Note: In addition to the above field tests, the contractor has to be conduct necessary lab tests as requested by the Engineer-in-charge/QC wing at Govt. Labs or at any reputed Govt./Private Engineering College testing labs.

24. QUALITY CONTROL & INSPECTIONS

Standard: The goods supplied under this contract shall conform to the standards mentioned in the Technical Specifications and when no applicable standard is mentioned, the standard specified by the Institution of Central/State Government or internationally recognized Institutions shall be applicable and such standards shall be of latest version issued by the concerned institution.

Inspections and Tests:

i) The purchaser or his representative shall have access to the Contractor's or Manufacturer's work at any time during working hours for the purpose of inspecting and testing the materials during manufacturing of the materials / equipment and may select test samples from the materials going into plant and equipment.

ii) The inspections and tests may be conducted in the premises of the manufacturer/supplier, at the point of delivery and/or at the final destination stores i.e. at the site. Where tests are conducted in the premises of Manufacturer / supplier, all reasonable facilities and assistance including access to drawings and production data shall be furnished at no extra charge to the Purchaser.

iii) Should any inspected or tested materials fail to conform to specifications, the inspection officer may reject them and the Manufacturer / supplier shall either replace the rejected materials or make all alterations necessary to meet specification requirements free of cost to the Purchaser.

iv) The Purchaser's right to inspect, test and where necessary, reject the materials/equipment after their arrival at the site, shall in no way be limited or waived by reason of the materials/equipment having been previously inspected, tested and passed by the Purchaser or his representative prior to the dispatch

v) The cost of making any test shall be borne by the contractor, if such test is clearly intended by or provided for in the contract.

Cost of tests not provided for: APTRANSCO may decide to conduct certain other tests not covered in this specification on the materials supplied by the bidder by an independent person or agency at any place other than the site of the place of manufacture of the materials. The cost of such tests shall still to be borne by the contractor. If the tests show that the workmanship or quality of materials are not in accordance with the provisions of the contract the same may be replaced with new one conforming to specification at Contractor's cost.

Quality of Materials and Workmanship: All materials and workmanship shall be of the respective kinds described in the contract and in accordance with the Engineer's instructions and shall be subjected to change from time to time to such tests as the Engineer may direct at the place of manufacture or fabrication or on the site or at all or any such places. The contractor shall provide such assistance, instruments, machines, labour and materials that are normally required for examining, measuring and testing any work and the quality, weight or quantity of any materials used and shall supply samples of materials before incorporation in the works, for testing as may be selected and required by Engineer.

Cost of Samples: The contractor at his own cost shall supply all samples, if the supply thereof is clearly intended by or provided for in the contract.

Uncovering and Making Openings: The APTRANSCO reserves the right to uncover and examine any part of the works if it is found to be not according to specification. The contractor shall uncover any part of the works or make openings as the Engineer may from time to time direct and shall reinstate and make good such part or parts to the satisfaction of the Engineer.

Inspection of operation: The Engineer and any person authorized by him shall at all times have access to the works and to all workshops and places where work is being prepared or from where materials are being

obtained for the works and the contractor shall afford every facility and every assistance in or in obtaining the right to such access.

Removal of improper work and materials:

a) The APTRANSCO representative shall, during the progress of works, have power to order in writing from time to time the removal from the site within one week of receipt of notice, of any materials which in their opinion are not in accordance with the contract.

b) In case of default on the part of the Contractor in carrying out such an order, the APTRANSCO shall be entitled to employ and pay other persons to carry out the same and all expenses consequent therein or incidental thereto shall be recoverable from the contractor by APTRANSCO from any money due to or which may become due to the contractor.

INSPECTION, DURING ERECTION:
The provisions of the clauses entitled in the specification for other sections Inspection, Testing and Inspection Certificates shall also be applicable to the erection portion of the Works. The PURCHASER/ENGINEER shall have the right to re-inspect any equipment though previously inspected and approved by him, at the CONTRACTOR's works, before and after the same are erected at Site. If by the above inspection, the PURCHASER/ENGINEER rejects any equipment, the CONTRACTOR shall make good for such rejections either by replacement or modifications/repairs as may be necessary to the satisfaction of the PURCHASER/ENGINEER. Such replacements will also include the replacements or re-execution of those works of other CONTRACTORS and/or agencies, which might have got damaged or affected by the replacements or re-work done by the CONTRACTOR's work.

ACCESS TO SITE AND WORKS ON SITE: Suitable access to and possession of the 'Site' shall be afforded to the CONTRACTOR by the PURCHASER in reasonable time.

The work so far as it is carried out on the Purchaser's premises, shall be carried out at such time as the Purchaser may approve.

In the execution of the Works, no persons other than the Contractor or his duly appointed representative, Sub-contractor and workmen, shall be allowed to do work on the Site except by the special permission, in writing of the Engineer or his representative.

Access to the 'Site' at all times shall be accorded to the PURCHASER/ENGINEER and other authorized officials and statutory Public Authorities. Nevertheless, the CONTRACTOR shall not object to the execution of the work by other Contractors or tradesmen whose names shall have been previously communicated in writing to the CONTRACTOR by the PURCHASER/ ENGINEER and afford them every facility for the execution of their several functions simultaneously with his own.

Provision of Vehicle during the execution of line works – The contractor shall provide one vehicle (Jeep) till the transmission line works are completed for site / filed visits of the field AEEs/AEs of both civil and electrical wings and also Quality control wings from section office to site. Non-providing of vehicle as per the requests of the section officer shall attract a penalty of **Rs. 500/- per day.**

25. TIME CONTROL

Program: Time is the essence of the contract. The contractor shall organize his resources and perform his works so as to complete the contract in the stipulated completion period.

The contractor shall execute the project as per completion period given in Schedule-E. The timing of the tasks / activities given in the bar chart shall be followed for timely completion of the contract.

During the execution of the contract, if the Engineer opines that proper progress is not maintained, suitable changes shall be made in the balance program of works to ensure proper progress.

Progress of Work:
If it is found that the progress of work is not commensurate with the program of completion, APTRANSCO will be entitled to terminate the contract in part or full, giving 15 days notice and get the balance works / supplies

completed through other agencies at the contractor's cost and risk. This right of APTRANSCO is in addition to levying of "Penalty" for delayed works.

MANAGEMENT /CO-ORDINATION MEETINGS

Review of works shall be conducted by the Superintending Engineer/ O&M/ Visakhapatnam or Chief Engineer / Zone.

Early Warning: The contractor has to intimate the APTRANSCO at the earliest opportunity of specific likely future events or circumstances, which may adversely affect the Project Implementation. The contractor shall cooperate with the officers of APTRANSCO in making and considering proposals for how the effect of such an event or circumstance can be avoided or reduced.

26. SITE STORES:

The contractor shall establish temporary stores at his cost at the sub-station site for storing material/equipment such as cement, steel, panels etc. This store should be dismantled and site cleared after the completion of the work.

27. CONTRACTOR'S SITE OFFICE:

The CONTRACTOR shall establish a Site Office at the 'Site' and keep posted an authorized representative for the purpose of the 'Contract'. Any written order or instruction of the PURCHASER/ENGINEER or his duly authorized representative shall be communicated to the authorized representative of the CONTRACTOR at the Site Office and the same shall be deemed to have been communicated to the CONTRACTOR at his legal address. A complete set of specifications, drawings and a copy of the contract agreement shall be kept in the site office at all times.

a. The attested copies of the following shall be invariable kept by the Contractor at the site office and Sub-Station site / Line Site locations and shall have access to the Field Engineers and Inspecting officers.
   i) Agreement copy along with price schedules.
   ii) Volume-1, Volume-2 and volume-3 of the Specification.
   iii) Approved copies of Guaranteed Technical particulars of materials / equipment, BOMs, approved drawings for all structures, foundations, equipment and materials.
   iv) Applicable Technical Standards like IS and ISS which are mentioned in the Volume-2 of the specification.

b. In cases where 'New works' are taken up as 'supplemental items' under the 'Original Agreement' all the above documents related to the extent of these new works also shall be made available at Site / locations.

28. CLEARING UP THE SITE:

During construction, the contractor shall keep the work site and storage area used by him free from accumulation of waste materials or rubbish and before completing the works the contractor shall, at his own

cost, remove or dispose off in a manner satisfactory to the engineer in charge. All temporary structures waste and debris shall be cleaned, all holes in the ground shall be filled in and the land restored to its original state as far as is practicable and the entire premises should be in a neat and tidy condition as directed by the Engineer. Any damage done to the permanent or temporary work of the Department by the contractor shall be made good at contractor's expenses.

29. CHANGE ORDERS:

The Purchaser may at any time by a written order given to the Contractor, make changes within general scope of the contract in any one or more of the following:

(a) Drawings, designs or specifications for material/equipment.
(b) Method of packing,
(c) The services to be provided by the contractor.

If any such change causes an increase or decrease in the cost of, or the time required for the material or works, an adequate adjustment shall be made to the contract price or delivery schedule or both and the contract shall be accordingly amended.

30. WATER & POWER: The successful bidder has to make use of the water and power supply available if any at Substation site. Transport of water from the available source to the Sub-station site will be the bidder's responsibility. The contractor has to use power driven equipment at his own cost for pumping water. In case water is not available at site the bidder has to make his own arrangements and any extra claim on account of this will not be entertained. Temporary Power supply if required for construction purpose will be arranged by APTRANSCO at one point in the substation site from DISCOM. Further cabling up to the contractor's office/works is in the contractor's scope only.

The following are the responsibility of the contractor:

a) The power consumption charges including deposits if any for temporary supply will have to be borne by the contractor.

b) To settle all the bills raised by DISCOM regarding the consumption charges and to obtain & submit "No Due Certificate" from DISCOM on completion/closing of works.

c) If any mal-practice/theft of energy takes place during the execution of contract by the contractor, it is the responsibility of the contractor to settle such cases by DISCOM before completion of substation works.

The estimated cost of 11 kV or 33 kV line if any for getting temporary supply shall be borne by APTRANSCO.

31. OPERATION AND MAINTENANCE MANUALS:

Contractor shall supply 10 copies of operation and Maintenance manuals and approved drawings along with the equipment. If the contractor does not supply the operation and Maintenance manuals and approved drawings as stated above the equipment bills will be kept on hold.

32. SAFETY PLANS

32.1 <u>GENERAL SAFETY RULES FORMED FOR FOUNDATIONS, TOWER ERECTION AND STRINGING:</u>

1. Steel/ Aluminums /Wooden ladders of 3.5 meters length shall be used for movement of labour in excavation, stub setting and concreting works. Keep cement out of the eyes.
2. All polypropylene ropes and slings, shackles proposed to be used shall be tested and inspected thoroughly before use.
3. Walkie talkie sets & Cell Phones shall be used for effective communication.
4. Binoculars shall be used for closer view
5. Red and Green flags shall be used for illiterate communication.
6. Pilot wires and joints shall be tested and thoroughly inspected before use
7. Ropes and guy wires shall be tested and thoroughly inspected before use
8. Ensure travelers in frictionless and good condition.
9. Load rating of stringing, pulling lines, conductor grips, load bearing hardware and accessories, rigging and hoists shall not be exceeded.
10. Traveler grounds should be installed at the fist and last tower between tensioner and puller, when stringing in proximity to energized lines. Traveler ground at a maximum distance not exceeding 3 KM. Personnel shall never be in series with a ground lead. Traveler grounds shall have a suitable

grounding stub located in an accessible position to enable placing and removing the ground clamps with hot sticks when necessary.

11. The conductor reel stands shall have breaking device to prevent over running and backlash as the conductor is pulled out.
12. The location of the puller, tensioner and intermediate anchor sites shall be selected such that the structures are not over loaded.
13. Slack shall be removed from all anchor lines prior to loading to minimize the possibility of equipment movement or impact loads to the anchors.
14. All equipment, conductors, anchors and structures within the work area shall be bonded together and connected to the ground source.
15. A running ground shall be installed between the reel stand and tensioner for conductor, or puller for pulling line, and the first tower.
16. When clipping is being done, care must be exercised to ascertain that the conductor are grounded prior to clipping, placing a local ground upon the conductor at the location of work.
17. Care must be exercised to ensure that concentrated load of the men, car and equipment dose not increase the sag appreciably to cause a hazard from obstructions over which the car will pass while installing spacers in bundle conductors.
18. If two conductors are to be spliced, the conductors shall be bonded and grounded before being spliced.
19. To cross over of energized lines, the line clear permit system shall be followed without any ambiguity to de-energize the lines. The de-energized lines shall be earthed effectively. Men and material clearance on line shall be ensured before returning permit.

32.2 GENERAL SITE SAFETY RULES & REGULATIONS

1. All personals engaged for this job shall wear Safety Helmet & safety shoes.
2. Appropriate Hand Gloves shall be used for Materials handling, Welding, Grinding, Gas cutting, Chemical Handling & Electrical work.
3. Suitable goggles must be ensured for the personnel deployed for Welding, Gas Cutting Chipping etc Grinders shall use face shield.
4. The personnel engaged in the noise zone such as Compressor Operator, Pneumatic Vibrators & Breakers, DG Operator etc. shall be equipped with suitable ear protection.
5. All personal working over 2.0-meter height shall wear full body safety harness with double lanyard and scaffold hook, before climbing. Further he will have to anchor the safety harness to a fixed structure before starting the work at a height, Fall Arrest Devices will be used in conjunction with safety harness, wherever the requirement arises.
6. Smoking is prohibited in all restricted area except in authorized smoking areas/ shelters.
7. Horseplay is prohibited at workplace. Running at the site is prohibited, except in case of emergency.
8. Material shall not be thrown from the height. It should be lowered from height.
9. Other than the electricians, no one is allowed to carry out electrical connections, repairs on electrical equipment or other jobs related thereto.
10. Inserting of bare wires for tapping the power from electrical sockets is completely prohibited.
11. Electrical supply to portable equipments should be through Earth leakage circuit breaker (ELCB) of 30 milliamps sensitivity.
12. All major /minor accidents and near misses to be reported to Site in charge / Site Safety Engineer to enable the management to take necessary steps to avoid the recurrence.
13. All tools and tackles shall be inspected before use. Defects to be reported immediately. No lifting tackle to be used unless it is certified by a competent person/ Safety Engineer.
14. Workers should possess induction tester. Whenever a shut down is taken this device should be used to ensure no current is on the conductor.
15. Move carefully when walking or climbing.
16. Keep the working area as neat as possible.
17. Remove hazards as soon as they are noticed.

32.3 ENERGIZED ELECTRICAL WORKS, BASIC REQUIREMENTS ARE:

1. Face shields/ Voltage rated gloves/ Voltage rated rubber floor matting Shepherd's Hook/ Warning/ Safety signs/ barricades.

2. Pay close attention to what is being done.
3. Disconnect power tools before adjusting them.
4. Keep power tool guards in place.
5. Avoid interrupting another person who is using a power tool.

32.4 STORES

1. A layout demarcating areas for stacking, storing and disposing the materials is made.
2. The materials are stacked with passage to reach them. The aisle is marked. Materials should not protrude beyond the marked area posing tripping hazard.
3. Name bards shall be displayed to mention the place for every item.
4. The racks installed must be supported well to prevent from falling.
5. To reach the rack top, person should not climb on the rack shelf, ladder should be used.
6. Vertical stacking of materials should not exceed the prescribed norms, posing falling hazard because of imbalance.
7. Adequate lighting is provided.
8. Flammable, materials like Dissolved acetylene, paints etc. are stored under well- ventilated shed. Electrical connection in these locations should be proper and maintained well such that they do not cause short circuit. Smoking, carrying matchbox or any other fire causing materials is prohibited in these areas.
9. Sufficient fire extinguishers are kept at conspicuous places and the path to reach them shall not be blocked anytime.
10. Toxic materials are labeled and kept at secured place where only authorized persons shall handle.
11. Nail pullers shall be used whenever possible to remove nails from boxes and crates. Metal strapping should be cut with proper safety tool.
12. Barrels and drums shall preferably be placed on one end. If placed on their side, these shall be provided with racks or blocked so these cannot roll.
13. Oils, greases and paints shall not be openly stored at any time.
14. The scrap pile and junk materials shall be kept as orderly as conditions will permit. Extreme care shall be used in handling scrap material to prevent personal injury.
15. All T&P issued to the site to be ensured in safe working conditions by a competent person and damaged T&P to be identified and place at a marked place.
16. First aid box shall be available at work site.

32.5 TRANSPORTATION OF MATERIALS

1. Only identified crew does the loading and unloading of the materials.
2. The gang leader and the crew should be aware of the weight of the materials they are to handle.
3. Proper and reliable tackles like slings, D- shackles etc. shall be used for material handling.
4. Lashing to be done after the materials loaded on the trailer.
5. Red flag is tide on structures extending beyond the trailer body. At night reflectors to be fixed.
6. Structures extending beyond the cabin height are prohibited from transporting.
7. The drive should be aware of the path to the terminal he is to transport the material and ensure on blockaded are there.
8. The gang leader should know the specific spot for unloading the materials.
9. Two tag lines shall be tide to the end of the structures while handling by crane.
10. All structural steel shall be placed on wooden sleepers.
11. Stable and sufficient wooden sleepers to be ensured by the Foreman /gang leader.
12. Workmen should not travel along with the loaded material.
13. It is better to plan material shifting in sequence in such a manner that one gang is engaged for loading and another gang should be made available for unloading. This will reduce the traveling of workmen on transport vehicles.
14. The supplier/Manufacturer shall provide such packing for the material/equipment as is required to prevent their damage or deterioration during transit to their final destination as indicated in the Technical specification for material/equipment. The packing shall be sufficient to withstand, without limitation, rough handling during transit and exposure to extreme temperatures, humidity etc. during

transit and open storage.

32.6 SAFETY SIGNS

1. Safety warning signs shall be placed to provide adequate warning of hazards. Sign shall be removed or covered when the hazards no longer exist.
2. Warning signs shall indicate the cautions to be observed regarding the hazards concerned and the necessary information and, also, actions of employees required to initiate.
3. It is recommended that safety signs on the Project conform with the current national colour codes.
4. Signs shall be in English & Local language where located near public highways and facilities.
5. Sign required to be seen at night shall be illuminated. Kerosene and open flame pots shall not be used for illumination.

34 BILL OF MATERIALS:

Wherever the drawings and designs are provided by APTRANSCO, the suitability of these designs and drawings shall be verified by the bidder for design requirements as per standards, site and soil conditions based on the soil investigation and environmental conditions prevailing for that particular project before submitting BOMs for approval. Substitutions if any of steel sections of the tower parts of higher size due to non-availability or otherwise, shall be to the contractors account. The supplier shall raise the invoices as per the approved BOMs.

35. PHOTOGRAPHS AND PROGRESS REPORT

Contractor shall furnish three (3) prints to the PURCHASER/ENGINEER of progress photographs of the work done at 'Site'. Photographs shall be taken when and where indicated by the PURCHASER/ENGINEER or his representative. Photographs shall be adequate in size and number to indicate various stages of erection. Each photograph shall contain the date, the name of the CONTRACTOR and the title of the photograph. Normally photographs may be taken once every 15 days and for all special events.

The above photographs shall accompany the fortnight/ monthly progress report detailing out the progress achieved on all erection activities as compared to the Schedules. The report shall also indicate the reasons for the variance between the scheduled and actual progress and the action proposed for corrective measures wherever necessary. The contractor shall furnish progress reports as and when requested by the PURCHASER/ ENGINEER.

Photographs shall be taken by the contractor for each tower location of EHT lines in the following stages of

1. *After excavation is over in normal soils.*
2. *Before and after blasting in Hard rock soils.*
3. *During dewatering.*
4. *During concreting.*
5. *Shoring & strutting.*

For Substations:

At all important stages like leveling, foundations, laying of slab etc.

Note: The photographs should be clear and hard copy or soft copy (2 Nos) shall be submitted to the Engineer in charge immediately.

36. MEASUREMENT OF WORK AND PROGRESS PAYMENTS

The PURCHASER/ENGINEER may, from time to time, intimate the CONTRACTOR that he requires the works to be measured and the CONTRACTOR shall attend or send a qualified agent to assist the PURCHASER/ENGINEER or his representative in taking such measurements and calculations and to furnish all particulars as may be required by him.

Where the erection of equipment, vessels and structural steel is involved, the basis of such measurements and progress evaluation shall be weights specified in the bidding documents or invoices or drawings as decided by the PURCHASER/ ENGINEER.

Should the CONTRACTOR not attend or neglect or omit to send such agents, then, the measurement taken by the PURCHASER/ENGINEER or approved by him shall be taken to be the correct measurements of the work.

The CONTRACTOR or his agent may, at the time of measurement, take such notes of measurements as he may require.

When measurements are affected by conditions already established, the CONTRACTOR shall take field measurements notwithstanding scale or dimensions shown on the drawings.

The measurements so taken and certified correct by the PURCHASER/ENGINEER shall be the basis for the progress payment to the CONTRACTOR. Where the breakup of 'Contract on unit basis price' is difficult to arrive at, the PURCHASER/ENGINEER and the CONTRACTOR shall work out at the commencement of the 'Contract', the weightages or the cost break-ups to arrive at a mutually agreeable basis for computation of the progress estimates.

To the value so arrived at on the basis of the CONTRACTOR's monthly progress evaluated, shall be added the amounts earned by the CONTRACTOR under supplemental contracts and orders, if any, till date of progress estimate. From the total thus computed, all previous payments plus any amounts due to the PURCHASER in accordance with the terms of this contract shall be deducted. The remainder shall be paid by the PURCHASER to the CONTRACTOR under interim Certificates from the ENGINEER.

In case of work is nearly suspended, or in case only unimportant progress is being made, or in case it is apparent that the CONTRACTOR is about to forfeit his 'Contract' or that the money yet due to him shall not complete his 'Contract', the PURCHASER may at his discretion withhold any payment which may be due to the CONTRACTOR.

The PURCHASER may withhold part or whole of any payment for erection claimed by the CONTRACTOR, which in opinion of the PURCHASER, is necessary to protect himself from loss on account of:

a) Defective work not remedied or guarantees not met
b) Claims filed against the CONTRACTOR
c) Failure by the CONTRACTOR to make due payment for materials supplied or labour employed by him.
d) Damage to other Contractor's, PURCHASER's or OTHERS' property.
e) Failure to meet the mutually agreed schedules for completion of the contract.

When the grounds for withholding payments are removed, payments of the amount due to the CONTRACTOR shall be made by the PURCHASER without delay.

The CONTRACTOR shall not demand nor be entitled to receive payment for the work or portion thereof, except in the manner set forth in this 'Contract' and only after the PURCHASER/ ENGINEER shall have given a certificate for such payment.

37. PROTECTION OF WORK:

The CONTRACTOR shall have total responsibility for protecting his 'Works' till it is finally accepted by the PURCHASER/ ENGINEER. No claim will be entertained by the PURCHASER/ ENGINEER for any damage or loss to the CONTRACTOR's works and the CONTRACTOR shall be responsible for the complete restoration of the damaged 'Works' to its original condition to comply with the specifications and drawings. If any such damage occur to the CONTRACTOR's Works because of other party not under his claim directly with the party concerned the same shall be dealt by the contractor alone. During construction of the project, the PURCHASER will continue to operate the plant and equipment already put in service. The CONTRACTOR shall protect all such plant, structures, piping, conduits, equipment and facilities against damage during its operations.

38. FIRE PROTECTION

The Work procedures that are to be used during the erection shall be those, which minimize fire hazards to the extent practicable. Combustible materials, combustible waste and rubbish shall be collected and removed from the 'Site'. Fuels, oils and volatile or inflammable materials shall be stored away from the construction and equipment and materials, storage areas, in safe containers. Untreated canvas paper, plastic or other inflammable flexible materials shall not at all be used at 'Site' for any other purposes unless otherwise specified. If any such materials are received with the equipment at the 'Site', the same shall be removed and replaced

with acceptable material before moving into the construction area or storage.

Similarly, corrugated paper fabricated cartons, etc., will not be permitted in the construction area either for storage or for handling of materials. All such materials used shall be of waterproof and flame resistant type. All the other materials such as working drawings, plans, etc., which are combustible but are essential for the 'Works' to be executed shall be protected against fire resulting from welding sparks, cutting flames and other similar fire sources.

All the CONTRACTOR's supervisory personnel and sufficient number of workers shall be trained for fire fighting and shall be assigned specific fire protection duties. Adequate number of such trained personnel must be available at the 'Site' during the entire period of the 'Contract'.

The CONTRACTOR shall provide and maintain fire protection equipment, adequate in design and numbers for the warehouses, office, temporary structures etc. Access to such fire protection equipment shall be easy and kept open at all times. The compliance of the above requirements under fire protection shall in no way relieve the CONTRACTOR of any of his responsibilities and liabilities due to fire accidents occurring either to his materials and equipment or to those of others working in the area.

39. SECURITY (WATCH & WARD)

The CONTRACTOR shall have total responsibility for all equipment and materials in his custody stored, loose, semi-assembled and/or erected by him at 'Site'. The CONTRACTOR shall make suitable security arrangements including employment of security personnel to ensure the protection of all materials, equipment and 'Works' from theft, fire, pilferage and any other damage and loss.

40. **TESTING AND COMMISSIONING**

The MRT wing of APTRANSCO shall supervise and assists the Contractor and his sub-vendors in testing & commissioning of Equipment. However the 'Testing and Commissioning' of the equipment erected by the CONTRACTOR shall be the responsibility of the CONTRACTOR. The CONTRACTOR shall provide in addition, test instruments, calibrating devices, etc. which are not available with MRT wing for successful and speedy completion of testing and commissioning. The labour required for the successful performance of these inspection tests shall have to be arranged by the Contractor. If it is anticipated that the above tests may prolong for a long time, the CONTRACTOR's workmen required for the above inspection tests shall always be present at 'Site' during such inspection tests. All the expenditure for the above arrangements will be to the contractor's account only.

41. **INDEMNITY BOND FOR ALL MATERIALS:**

For all materials supplied by the Contractor, the Contractor shall take delivery of materials after executing an indemnity bond in favour of the Purchaser against loss, damage and any risks involved, for the full value of the materials. This indemnity bond shall be valid till the scheduled date of testing, commissioning and handing over of the line to the Purchaser.

42. **ACCOUNTING OF SURPLUS MATERIALS/ SETTLEMENT OF MATERIAL ACCOUNT**:

On completion of the works, **all surplus materials**, for which payments have been made by the APTRANSCO or materials issued by APTRANSCO for works if any, shall be handed over to the Engineer at APTRANSCO's designated stores at the cost of the contractor within a time limit of 2 month of commissioning of the project. If the materials are not returned and are in the opinion of the Executive Engineer in-charge of the work, not in a fit condition for use, they will be treated as sold to the contractor at DOUBLE the rate of issue of material and recoveries made contractor's bill accordingly.

43. **COMPLETION OF BALANCE WORKS AND HANDING OVER** :

All the balance works if any shall have to be executed by the contractor within two months of charging of substation / line. If the Contractor fails to complete the balance works with in the above time schedule the works will be executed by the other agencies and will back charge to the contractor. However the completed works shall be handed over to O&M wing by construction wing within 3 months after commissioning.

44. **FINAL ACCOUNT**:

a) Not later than one (1) month after handing over of the works complete in all respects i.e., after successful testing and commissioning, the Contractor shall submit a draft statement of 'final account' and supporting document to the Engineer/Engineer's Representative showing in detail the value of the work done in accordance with the contract.

b) Within one (1) month after receipt of the Draft Final Account and all information reasonably required for its verification, the Engineer/Engineer's representative shall determine the value of all matters to which the Contractor is entitled to under the contract. The Engineer/ Engineer's representative shall then issue to the Contractor a statement showing the final amount to which the Contractor is entitled to under the contract.

c) The Contractor shall sign the Final Account as an acknowledgement of the full and final value of the work performed under the contract and shall promptly submit a signed copy to the Engineer/Engineer's representative.

d) On receipt of Final Account, the Engineer/Engineer's representative shall promptly prepare and issue to the Contractor a "Final Payment Certificate" certifying any further amounts due to the Contractor in respect of the contract.

e) If the contractor does not finalize the material account within 2 months from the date of completion of works/ handing over of works, the field Executive Engineer will prepare the final bill duly deducting the cost of all the unaccounted / unhanded over material and will issue a notice to the contractor for signing on the bill. If the contractor do not turn-up within 15 days of that notice, the bill will be processed without signatures of the contractor. The cost of the unaccounted material will be taken as per Clause 42 above.

f.) All the completed works shall be technically audited by Quality Assurance wing within two months of the completion / commissioning of the project for the correctness of project in terms of agreements, specifications, detailed project report/technical sanction, deviation in quantities & rates.

(g) All the remarks which are pointed out by Quality assurance wing in the above technical audit shall invariably attend by TL&SS wing within one month and in case the remarks are pertaining to the execution of work which is in the scope of the agreement, the Contractor has to attend the remarks without insisting of any additional payment.

TAKING OVER:

Upon successful completion of all the commissioning tests to be performed at site on equipment furnished and erected by the Bidder, and on successful commissioning of the project, the Purchaser shall issue to the Contractor 'a taking over Certificate' as a proof of the final acceptance of the project. However such taking over certificate will be issued only after handing over of all the manuals, drawings, tower schedules as per Clause 32 above and after settlement of materials account and final bill.

46. TECHNICAL AUDIT:

i) All the completed works shall be technically audited by Quality Assurance wing within two months of the completion/ commissioning of the project for the correctness of project in terms of agreements, specifications, detailed project report/ technical sanction, deviation in quantities & rates.

ii) All the remarks which are pointed out by Quality wing in the above technical audit shall invariably attend by O&M within one month and in case the remarks are pertaining to the execution of work which is in the scope of the agreement, the Contractor has to attend the remarks without insisting of any additional payment.

TABLE OF CLAUSES - SECTION-II :: FINANCIAL

Clause. No.	Description
1.	Bid Prices
1.2	Price Variation (PV) clause
2.	Statutory Variation
2.1	Payment of taxes, levies, duties, statutes
2.2	Taxes & Cess deductible at sources
2.3	Seignorage charges
2.4	Insurance
2.5	Royalties for patents
3.	Supplemental quantities / items
4.	Performance guarantees
4.1	Performance security
4.2	Additional security
5.	Terms of payment
6.	Completion period
6.1	Penalty for late supplies / completion:
6.2	Penalties imposed for deficiencies in quality of works
6.3	Incentives for early completion of works
6.4	Extension of the completion date
6.5	Force Majeure Conditions
6.6	Delays ordered by the purchaser
7.	Responsibility of the contractor
8.	Claims due to delays beyond contractor's control
9.	Recovery of money from contractor in certain cases
10.	Way leaves & tree cutting in transmission line corridor
11.	Termination of contract
12.	Arbitration

SECTION-II: FINANCIAL

1.0 Bid Prices: The prices quoted shall be **Firm** except for Cement, Reinforcement steel and Earth Flat (GI &MS) in Schedule B&C and Transmission Line Tower Parts, Substation Structures, Cables, Isolators and Conductors in schedule –A. The estimates / Schedules are prepared with prices FADS inclusive of packing and forwarding, GST, seignorage charges, Labour welfare Cess and other legally permissible duties and levies wherever applicable, handling charges to cover the transport from destination to site / stores, unloading at destination and insurance (transit and storage at site including erection risks).

It is the responsibility of the Bidder to verify the correct rates of duties and taxes leviable on the materials and services at the time of bidding. The proforma credit available to the bidder on the purchases of inputs (raw materials) consequent to the introduction of "CENVAT" Scheme may be taken into account while quoting the prices. The Bidder shall indicate the total bid price of the Materials and services under the contract in Schedule-D.

Incase of local supplies, GST and other levies and duties solely in respect of the transaction between the APTRANSCO and the Contractor under the contract, if any shall not be included in the bid price, but those shall be indicated separately, wherever applicable. These amounts will be payable (along with subsequent variation if any), by the Employer on the supplies made by the Contractor **but limited to the tax liability on the transaction between the APTRANSCO and the contractor.** In case of bought out materials / equipment the APTRANSCO shall pay only for the taxes involving direct transaction between the bidder and APTRANSCO and not for the transaction between bidder and his sub-vendors. The C-form will not be issued for the turnkey projects.

1.2.0 **PRICE VARIATION (PV) Clause:**

The price variation is applicable for Cement and Reinforcement steel in works schedule and Transmission Line Tower Parts, Substation Structures, Earth Flat (GI &MS), Cables, Isolators and Conductors in materials schedule and the calculation of Price Variation is as below.

1.2.1 **The price variation is calculated as per G.**O.Ms.No.1 dt,25.02.2012 of finance (works & Projects –F7) department

1.2.2 The Price Variation is applicable in respect of the items Cement and Reinforcement steel as per G.O.Ms.No.94, Dt.16.04.2008 and T.O.O (CE-Civil) Ms.No.84, Dt.12.06.2012 & TOO(CE-Civil)Ms.No.260, dt.28-11-2012.

a) The price adjustment shall be applicable within original contract period or Period extended on grounds of the departmental delays and valid reasons and shall not be applicable to the extensions granted on account of the contractor's fault as envisaged in G.O.Ms.No.94 of I&CAD.
b) The price adjustment shall be applicable for actual components of works actually carried out during the period of the bill. The price variation formula is PV = (Final Rate – Basic Rate) * Quantity.

c) The adjustment scheme will be applied where the variation (increase or decrease) is more than 5% i.e. actual variation will be considered if increase or decrease is more than 5%.

d) In case variation is positive, the department will operate variation in rates only to the extent where it is above 5% over the estimated rates. Thus if the price excess is 10%, payment will be made only to the extent of 5% (10% - 5%) since the contractor factors into his original bid his risk of absorbing the first 5% increase The same shall be followed for – ve price variation.

e) For the time extension due to department fault, both +ve and –ve price variations are applicable.
f) For the time extension due to contractors fault,-ve price variation is applicable.

g) The Basic rate of reinforcement steel and cement as taken in the estimate were furnished in Project Data Sheet.

h) The final rate of reinforcement steel for the purpose of price variation shall be the rate informed by Govt. of A.P as on one month prior to the date of check measurement of work.

i) The rate communicated by Government for reinforcement steel and cement is inclusive of all taxes and duties. Hence no additional taxes and duties will be given on price variation amount.

1.2.2 The Price adjustment applicable in respect of the items Transmission Line Tower Parts, Substation Structures and Earth Flat is as per T.O.O (CE-Const)MS. No 50 and T.O.O. (CE-Const) Ms. No.87 dt.10-06-2008 and as per T.O.O. (CE-Construction-2) Ms. No.242 dt.08-11-2012 subject to the following conditions.

a) The price adjustment shall be applicable within original contract period or period extended on grounds of the departmental delays and valid reasons and shall not be applicable to the extension granted on account of the contractor's fault.

b) Price adjustment shall be applicable for actual components of items of supply actually carried out. The price variation formula is PV = (Final Rate – Basic Rate) * Quantity.

c) The adjustment scheme will be applied where the variation is more than 5% i.e. actual variation will be considered if increase or decrease is more than 5%.

d) In case variation is positive, the department will operate variation in rates only to the extent where it is above 5% over the estimated rates. Thus if the price excess is 10%, payment will be made only to the extent of 5% (10% - 5%) since the contractor factors into his original bid his risk of absorbing the first 5% increase. The same shall be followed for – ve price variation.

e) For the time extension due to department fault, both +ve and –ve price variations are applicable.

f) For the time extension due to contractors fault,-ve price variation is applicable.

g) The Basic rate of Transmission Line Tower Parts, Substation Structures and Earth Flat as taken in the estimate were furnished in the Project Data Sheet.

h) The final rate of Transmission Line Tower Parts, Substation Structures for the purpose of price variation shall be the rate communicated by the Chief Engineer / Construction for every month as on 2 months prior to the date of Dispatch and for earth flat one month prior to the date of Dispatch.

i) The rate communicated by APTRANSCO for Transmission Line Tower Parts, Substation Structures and Earth Flat is inclusive of all taxes and duties. Hence no additional taxes and duties will be given on price variation amount.

1.2.3 Price Variation is applicable in respect of Supply of Conductor and the price shall be based on and subject to adjustment due to variations in the following factors:

i. **E.C. Grade Aluminium:** The ex-works cost of indigenous E.C. grade aluminium wire rod based on average price of M/s. Nalco, Balco, Hindalco, and Malco as given in CACMAI circulars prevailing as on 30 days prior to the Bid submission closing date as per NIT.

ii. **HTGS wire:** The ex-works cost of per MT of HTGS wire exclusive of duties and taxes prevailing as on 30 days prior to the Bid opening corresponding to 3.00 to 4.09mm designationas given in CACMAI circulars prevailing as on 30 days prior to the Bid submission closing date as per NIT.

iii. The above basic prices of raw materials will remain unaltered during the execution of contract.

iv. For any variation up or down in the prices of raw materials as defined above, for every one rupee change in the rate of one MT of EC grade aluminum wire rod and HTGS wire the corresponding increase or decrease in price per KM of finished conductor allowable shall be as given below:

CONDUCTOR	Variation in Rs. Per KM of conductor	
	For Aluminium	For HTGS wire 3.00 mm to 4.09 mm
Panther ACSR Conductor	**0.588**	**0.388**
Zebra ACSR Conductor	**1.185**	**0.438**
Moose ACSR Conductor	**1.467**	**0.537**

v) For the purpose of calculation of price variation the prices of HTGS wire and EC grade aluminum shall be taken as those prevailing on the first working day of the calendar month one month prior to the date of delivery.

vi) The Base and final prices / indices shall correspond to same manufacturers as given in the CACMAI. The price variation formula is PV = (Final Rate – Basic Rate) * Quantity.

vii) The date of delivery shall be the date of receipt of materials in good condition at destination stores (i.e. check measurement date as per Form-13) for the purpose of price variation calculations.

viii) If the delivery of the material is within the scheduled delivery period, the Price Variation applicable will be based on the actual delivery.

ix) In case the purchaser advances the delivery, the price variation applicable will be based on the actual delivery.

x) Irrespective of increase in the prices of raw materials, the total Price Increase per KM of the Conductor will be limited to a maximum of **30%** over the **Unit price** mentioned in Purchase Order. However there is no ceiling for negative variation.

xi) The Price Variation amount can be claimed for each batch of supplies made from time to time (out of total quantity) and will be payable after due verification

xii) In case of conductors, where the supplier makes his own arrangements to get the Rod made out of Aluminium Ingots, the prices of Aluminium Rod as per CACMAI mentioned above will only be taken into consideration for arriving at the price variation claims irrespective of whatever expenditure the supplier might have incurred in getting the ingot converted into Rod.

xiii) If the date of delivery as defined in the P.V. formula is beyond the contracted delivery date the contracted delivery date or the actual delivery date which ever is advantageous to the Purchaser will form the basis for calculation of price variation.

xiv) Not withstanding the formula applicable for regulating the price variation, if at any time any documentary evidence proof or certificate in regard to the price variation bills is required by the Purchaser, the supplier will have to furnish the same to the Purchaser.

2. STATUTORY VARIATIONS:

It is the responsibility of bidder to consider the correct rates of duties and taxes leviable on the equipment/material/Work at the time of bidding. After considering the correct rates of duties and taxes only the bidder shall quote the percentage on ECV.

For both Schedule-A , any variation up or down in statutory levy or new levies introduced after signing of the contract under this specification will be to the account of APTRANSCO.

In cases where delivery schedule is not adhered to by the contractor and there are upward variation/ revision after the agreed delivered date the contractor will bear the impact of such levies and if there is downward variation / revision the APTRANSCO will be given credit to that effect. For the variations beyond the scheduled completion period the payment of taxes shall be limited to the tax rates applicable with in the scheduled completion period.

In case of the bought out items for which the prices shall be inclusive of Excise duty and CST, statutory variation shall not be applicable on that taxes and duties. For this purpose bought item means the material / equipment not manufactured by the bidder i.e. Statutory variation will be applicable for the taxes and duties involving direct transaction between the bidder and APTRANSCO only and not for the taxes and duties between bidder and his sub-vendors

2.1.0 PAYMENT OF TAXES, LEVIES, DUTIES, STATUTES

2.1.1 Contractor shall be entirely responsible for payment of all taxes, levies, duties, license fees, etc., incurred until delivery of the contracted goods to the purchaser.

2.1.2 In respect of supply portion, the contractor shall pay all types of fees, levies, taxes, duties etc. required to be paid by any National or State statute, ordinance or other law or any regulation or bye-law of any local or other duly constituted authority in relation to the execution of works and by the rules and regulations of all Public bodies and companies whose property or rights are affected or may be affected in any way of the works. The contractor shall in compliance with the above keep the Purchaser indemnified against all such penalties and liability of every kind for breach of any statute, ordinance or law, regulation or bye-law.

2.1.3 Nothing in the contract shall relieve the contractor from his responsibility to pay any tax that may be levied by the Government on the turnover / profits made by him in respect of the contract.

2.1.4 The present rates of taxes and duties which have been considered in the estimate are mentioned in the Project Data Sheet. However it is the responsibility of the bidder to ensure the correct rate and any change of rate of taxes and duties at the time of quoting.

2.2 TAXES & CESS DEDUCTABLE AT SOURCES: As per provision of taxation act, necessary applicable taxes will be deducted at source at all stages of payment on the bills and same will be remitted to the concerned department as per the rules / acts of the State / Central Government. The present rates of the TDS are mentioned in the Project Data Sheet. However the deductions will be as per the rates prevailing as on the date of billing.

2.3 SEIGNORAGE CHARGES:

Seignorage charges has to be paid by the bidder to the Assistant Director Mines & Geology Department other wise it will be deducted from the bills as per rules in vogue.

2.4 INSURANCE:

The prices quoted by the bidder shall include insurance charges for transit, storing and erection risks materials and insurance for labour as per the labour laws in vogue.

All the materials / equipment shall be insured by the contractor for loading, transit, unloading, storage and erection risks. Any insurance premium payable shall be borne by the contractor and shall cover the following risks also.

a) Full cover against damage to other people's property
b) Coverage against death or injury caused by the contractor's acts or omissions to:
   i. Anyone authorized to be on the site.
   ii. Third parties who are on the site.
c) Full cover against theft and damage to the works and materials during transit, storage and construction, till to the commissioning and handing over of the project in full shape.

It is the entire responsibility of the contractor for the safety of all the materials / equipment and labour till the notified project / work are handed over to APTRANSCO after commissioning. While doing the insurance for the workmen, the Contractors are advised to follow the labour department rules in vogue. Any additional insurance premiums if required during the course of project execution shall be borne by bidder.

All costs on account of insurance liabilities covered under this contract will be on contractor's account. The Contractor shall provide the purchaser with a copy of all the insurance policies

and documents taken over by him in pursuance of the contract. Such copies of documents shall be submitted to the purchaser immediately after such insurance coverage. The Contractor shall also inform the purchaser in writing at least sixty (60) days in advance, regarding expiry, change, any modification, amendments etc., without prior approval of purchaser the Contractor shall not cancel any of the Insurance policies made for the project. It is the Contractor's responsibility to ensure revalidation or renewal of the insurance policies well in advance.

The bidder shall a) Initiate and pursue insurance claim till settlement and

b) Promptly arrange for repair and/or replacement of any damaged items in full irrespective of settlement of insurance claim by the under writers.

The Goods supplied under this Contract shall be fully insured against loss or damage incidental to manufacture or acquisition, transportation, storage, delivery and erection. The insurance shall be at least for an **amount equal to 110% of the cost of the goods** from "warehouse to warehouse (final destination)" and **storage up to commissioning** thereafter on "All Risks" basis including War Risks and Strike Clauses.

In addition, the following provisions will apply to the portion of the 'Works' to be done by CONTRACTOR. Comprehensive Insurance of equipment/materials during erection and commissioning, Workmen's Compensation Insurance, Comprehensive Automobile Insurance and Comprehensive General Liability Insurance shall be the responsibility of the CONTRACTOR.

The Contractor shall also maintain an insurance policy against all claims which may be made upon Purchaser whether under the Workmen's Compensation Act or any other statue in force during the currency of the contract or at common law in respect of any employee of contractor. The Contractor shall be responsible for anything which may be excluded from the insurance policies referred to above and also for all other damage to any property or persons out of and incidental to the negligence or defective carrying out of this contract.

2.4.1 **Third party Compensation:**

In particular, the Contractor shall effect and maintain an insurance policy of at least Rs.3.00 lakhs for one person and Rs.5.00 lakhs per accidental for injury or death and at least Rs.5.00 lakhs per accident for third party property damage, to indemnify purchaser against all third party accident/damage claims which may arise in respect of the work or in consequence thereof.

2.4.2 **Workmen's Compensation Insurance:**

This insurance shall protect the CONTRACTOR against all claims applicable under the Workmen's Compensation Act, 1948 (Government of India). This policy shall also cover the CONTRACTOR against claims for injury, disability, disease or death of his or his SUB-CONTRACTOR's employees, which for any reason are not covered under the Workmen's Compensation Act, 1948. The liabilities shall not be less than:

Work Men's compensation	-	As per statutory provision
Employees liability	-	As per statutory provision.

2.4.3 **Comprehensive Automobile Insurance:**

This insurance shall be in such a form to protect the CONTRACTOR against all claims for injuries, disability disease and death to members of public including the PURCHASER's men and damage to the property of others arising from the use of motor vehicles during on or off the 'Site' operations, irrespective of the purchaser-ship of such vehicles.

The liability covered shall be as herein indicated:

Fatal injury	:	Rs. 1,00,000 each person
		Rs. 2,00,000 each occurrence
Property damage	:	Rs. 1,00,000 each occurrence

2.4.4 **Comprehensive General Liability Insurance:**

This insurance shall protect the CONTRACTOR against all claims arising from injuries, disabilities, diseases or death of members of public or damage to property of others due to any Act or omission on the part of the CONTRACTOR, his agents, his employees, his representatives and SUB-CONTRACTORs from riots, strikes and civil commotion. This insurance shall also cover all the liabilities of the CONTRACTOR arising out of the clause entitled 'Clause. 31.0' of GCC Indemnify the Purchaser under 'General Conditions of Contract'.

The above are only illustrative list of insurance covers normally required and it will be the responsibility of the CONTRACTOR to maintain all necessary insurance coverage to the extent both in time and amount to take care of all his liabilities either direct or indirect, in pursuance of the 'Contract'.

It will be the responsibility of the Contractor to lodge, pursue and settle all claims with the Insurance Company in case of any damage, loss, theft, pilferage or fire and the Purchaser shall be kept informed about it.

2.5 **ROYALTIES FOR PATENTS:**

All royalties for patents or charges for the use of infringement thereof that may be involved in the construction or use of any material shall be included in the bid prices. The bidder shall protect the APTRANSCO against any and all such claims arising out on account of the use thereof.

2.6 The bidders are requested to consider all the applicable deductions and taxes & duties while quoting and no additional payment will be made if the bidder did not consider any of the prevailing taxes and duties and deductions at the time of quoting.

3.0 SUPPLEMENTAL QUANTITES / ITEMS:

The quantities indicated in the Schedule-A (Materials / Equipment) and Schedule-B&C (Works) are only provisional and are likely to change during actual execution. When quantities of any item are likely to exceed beyond 25% over and above the scheduled quantity or any new items / supplemental items arises, the Contractor shall bring the fact to the notice of Chief Engineer/Zone well in advance and take prior orders for going ahead with the work. Without approval of this office, the Contractor shall not go ahead with the work wherever there is increase in quantities exceeds beyond 25% over and above the scheduled quantities or any new items / supplemental items arises.

The contractor is bound to execute all Increased / supplemental / new items that are found essential, incidental and inevitable during execution of the contract at the rates to be worked out as below:

3.1 **Increased Quantities:**

I) **For work items covered in Schedule-A:**

a) **For quantities increased up to 25% over and above the Scheduled quantities:** The rates accepted for individual items shall hold good up to 25% over and above the agreement quantities.

b) **For quantities increased beyond 25% over and above the Scheduled quantities:** The rates payable for the quantities beyond 25% over and above the agreement quantities will be APTRANSCO estimated rate or accepted rate which ever is less.

3.2 For Supplemental / New Items:

i) For the items relating to the Schedule-B&C where the rate can be deduced from the estimate or APTRANSCO / DISCOMS / common SSR, the rate applicable will be the estimated / SSR rate +/- tender percentage quoted/ accepted by the bidder for Schedule –B&C. The schedule of rates adopted for supplemental items / new items shall be from the current SSR applicable during the execution period.

ii) For the items relating to the Schedule-A or Schedule-B&C where the rates of new items cannot be deduced from the estimate / SSR, the rate payable will be arrived based on the prevailing market rates duly enquiring / collecting quotations and observing the reasonableness of the rates by the Purchaser.

3.3 The contractor shall plan and procure the materials indicated in the schedule 'A' duly verifying with the approved layout and profile, So that the procured quantities match with actual requirement to avoid excess supply of materials. The Executive Engineer shall give the details of the materials such as quantities of type wise towers and extensions and other line materials based on the line approved profile and the quantities of type wise structures and other materials based on the approved layout on the date of Kickoff meeting without fail.

3.4 The Schedule time required to complete the new / supplemental items including the balance work will be indicated in the approval letter issued for new / supplemental items.

4.0 PERFORMANCE GUARANTEES

i) The contractor shall warrant for the satisfactory functioning of the materials / equipment supplied and for the satisfactory operation of the workmanship performed by him, for a minimum period of **12 months** from the date of commissioning of the works/project.

ii) Equipment, sub-assemblies or spares, or parts replaced / repaired under warranty shall have further warrantee of 12 months from the date of replacement / repair.

iii) After commissioning and handing over of the project to the APTRANSCO, the contractor shall arrange for thorough patrolling of the line during the first six months of the guarantee period so that the defects if any noticed during that period may be rectified by the contractor free of cost to the APTRANSCO without having to wait for the APTRANSCO to suggest such rectifications.

iv) If during the period of guarantee, the Engineer decide and inform in writing to the contractor that any equipment, part of equipment, material or works is defective, the contractor on receiving details of such defects or deficiencies shall at his own expense, irrespective of reimbursement of insurance company, rectify/replace the defective material or works within seven (7) days of his receiving the notice or within such reasonable time as APTRANSCO may deem proper for making it good. The decision whether correction of the defects should be through repair or by replacement shall be the sole discretion of the APTRANSCO.

v) In the event of contractor not responding to the intimation of the Engineer as mentioned in above, the Engineer may arrange for a third party to correct the Defects and the extra costs for such .

vi) The Engineer is to give the contractor at least seven (7) days notice of this intention to use a third party to correct a Defect. If the contractor does not correct the Defect himself within this notice period, the Engineer may have the Defect corrected by the third party. The cost of the correction will be deducted / collected from the Retention amounts or Performance / additional / Retention securities or amounts / bills pending to the contractor either in this contract or other contracts or any other Securities.

4.1 PERFORMANCE SECURITY:

The successful bidder to whom the contract is awarded will be required to furnish performance security for a value equal to **ten percent (10%)** of the value of Schedule –A plus (+) **five percent (5%)** of the value of the Schedule-B&C for the proper fulfillment of contract with a validity of 12 months over and above the completion period and with two months claim period.

In case of Joint Venture, Performance Security shall be furnished as 75% by lead partner and 25% by associate partner respectively of the values mentioned above, which would stand surety for collective responsibility in executing the works.

In case of non execution of the works within the stipulated completion period the bidder shall extend the Bank Guarantee suitably to cover the 12 months performance + 2months claim period from the expected date of commissioning of the project.

If the successful bidder fails to furnish the performance security as specified above, the contract is liable for cancellation and forfeiture of the bid security. The performance security shall also be forfeited if the successful bidder fails to fulfill the terms of the contract.

Performance security may be made by Demand Draft on any approved Bank payable at **Visakhapatnam** in favor of Superintending Engineer/O&M Circle/Visakhapatnam u or by way of Bank Guarantee from approved Bank, in favour of tender inviting authority, in the prescribed proforma.
All Bank Guarantees, which are executed in accordance with this specification, shall be on a Stamp paper of value not less than Rs.100/- (Rupees one hundred only). Bank Guarantee executed on the stamp paper of value less than Rs.100/- will not be accepted.

The performance Bank guarantees furnished by the successful bidder shall be extended suitably as and when required to cover the Warranty period of the materials and services as per clause 4.0 above.

4.2 ADDITIONAL SECURITY:
If the rate quoted by the bidder less than 10% of the APTRANSCO's total estimated cost will be considered as abnormally low rate. In the case of abnormally low quoted rates i.e. less than 10% additional security deposit (ASD) to be paid for the less percentage below 10% i.e. if the bidder quotes 20% below the total estimated cost then he has to pay ASD equivalent to 10% of the total estimated cost.

The Additional security may be made by Demand Draft on any approved Bank payable at **Visakhapatnam** in favor of **Superintending Engineer/O&M Circle/Visakhapatnam** or by way of Bank Guarantee from a Nationalized/ Scheduled Bank as per the list enclosed, in favour of tender inviting authority covering a period of 2 months over and above the completion period.

4.3 All Bank Guarantees, which are executed in accordance with this specification, shall be on a Stamp paper of value not less than Rs.100/- (Rupees one hundred only). Bank Guarantee executed on the stamp paper of value less than Rs.100/- will not be accepted.

4.4 The Performance and Additional Securities will be forfeited in the following cases.

i) When the successful bidder does not fulfills the obligations under the contract or proven to be false submission of the information either in the bid or the information relating to the contract.
ii) When the successful bidder fails to execute the work in accordance with the clauses under this specification.

iii) When the successful bidder executing the work with in-ordinate delays or with poor quality which hampers the prime object of the contract, the securities will be en-cashed and will be adjusted towards such losses incurred by APTRANSCO or towards risk purchase.

iv) When the successful bidder pleads his inability to erect the Substation / Line and backs out of his obligations after issue of the preliminary acceptance letter or the letter of intent, the APTRANSCO will forfeit his performance/ additional/ bid securities and also recover the loss, if any, sustained by the APTRANSCO as a consequence of such backing out. In addition, the company will be blacklisted.

5. TERMS OF PAYMENT:
All the bidders who shall accept the following terms of payment are only acceptable. Bids received stipulating terms other than the following terms will be invalidated.

a) For Works:
i) Payments will be made up to 90% of the bills for the items of works completed during a month. For this purpose the contractor shall submit monthly bills to the Executive Engineer regularly to ensure payments in time. The contractor shall give full details of items of works done against each location in support of the bill. Bills submitted without supporting details will not be taken into cognizance.

(i) Out of balance 10% amount, the first 5% payment will be released after completion of all the works, provided the material account is settled. The final 5% payment will be released after completion of the Guarantee period or on submission of a Bank Guarantee for the equivalent amount, valid for the

guarantee period plus two months claim period, subject to settlement of

The Contractor shall furnish the following documents in quadruplicate for arranging initial payment as per 5.b (i) for the equipment / material supplied.

a) Copies of the invoices showing Contract No., Goods description, quantity, unit price and total amount.
b) Acknowledgement of receipt of material from consignee i.e. Form-13
c) Proof of payment of Excise duty (Not applicable for bought out items)
d) Acknowledgment of Consignee on Delivery Challan in original.
e) Detailed packing list
f) Documentary proof for the freight.
g) Copies of Insurance Certificate / policies
h) Test certificates approval and Dispatch clearance. (Not applicable for furniture and general items)
i) Certificate certifying that the defects, if any, pointed out during inspection have been rectified.
j) The payments against Schedule-A ,B&C are subject to Performance Security with a validity of **12** months as on the date of Check Measurement for proper fulfillment of performance obligations.
k) No payments will be made for the supplies made prior to scheduled delivery date or for materials which are not in full shape.

5.1 Payments will be made by cheques / by way of Electronic Fund Transfer / RTGS from APTRANSCO funds or REC/BANK funding. The contractor has to furnish requisite details for establishing RTGS in proforma as per Schedule -X. Once RTGS system is established, the bank account details submitted are final and cannot be changed till completion of the contract. An amount of Rs. 50/- will be recovered from the bill amount for each disbursement on LOA raised by unit officers towards RTGS. For payments through REC, the connected REC/Bank charges are to be borne by the contractor.

6.0 COMPLETION PERIOD:

The overall completion period for this project is as per the Project Data Sheet. For completion period for individual works, refer Schedule-E. Within one week of receipt of intimation from Superintending Engineer/Executive Engineer concerned, the contractor shall take over the approved profiles & tower schedules for the line and substation levels/ site cum Electrical Layout for the Sub-station. Otherwise the date after one week of intimation to the contractor for taking over of approved profiles, tower schedules and substation levels /site cum Electrical Layout shall be the deemed date of handing over of profiles/ substation levels /Electrical layout and will be taken as date of commencement of work. **The bidder shall execute all the works / supplies as per the program of works.**

6.1 **PENALTY FOR LATE SUPPLIES / COMPLETION**:

The completion period mentioned in Schedule-C is the essence of contract. Penalty will be levied as follows for the delay in executing the works or supply of material.

In case of delay in erection of the scheduled works, whatever be the reasons, the APTRANSCO can levy and collect the penalty @ 0.5% per week or part thereof for the portion of works not completed, subject to a maximum of 5% of the total value of works. Once the maximum is reached, APTRANSCO may consider termination of the contract.

The right of the APTRANSCO to levy penalty shall be without prejudice to its rights under the law including the right to get the balance works executed by other agencies at the risk and cost of the successful bidder. This is in addition to the right of the APTRANSCO to recover any damages from the contractor and also blacklisting.

SPECIFIC ACCEPTANCE OF THIS CLAUSE SHOULD BE IN BLACK & WHITE IN THE BID. IF SUCH ACCEPTANCE IS NOT INDICATED, THE BID IS LIABLE TO BE OVERLOOKED.

In case the successful bidder fails to execute the works as per the programme, APTRANSCO reserves its right to get the balance works executed by other agencies at the risk and cost of the successful bidder, this is in addition to the right of the APTRANSCO to recover any damage from the contractor and also blacklisting.

(i) Penalties shall be imposed for deficiencies in quality of works as per Clause No.3.30 under section – III (Technical) of the specification

6.2 **PENALTIES IMPOSED FOR DEFICIENCIES IN QUALITY OF WORKS:** During execution of works, if any deficiencies in quality of works is found in deviation to the Specification/ Agreement, a minimum penalty of Rs.5000 to Rs.30,000/- shall be levied for deficiencies as per each category as mentioned below.

Category I : Not using (i) prescribed shoring, shuttering and dewatering equipment, (ii) measurement boxes, (iii) Form boxes for different types of foundations and steel measuring boxes, (iv) not providing adequate number of chairs to the steel reinforcements, (v) not carrying out back filling and compaction of the foundation pits in layers and leveling the tower footings properly, (vi) not ensuring that the excavated earth is dumped at least 2 meters away from the pit etc., and (vii) not providing copings to the tower legs/stubs (viii) not providing water tanker, Earth rammers/Earth vibrators.

Category II : Use of reinforcement steel other than TISCO, VSP and SAIL makes, not using vibrators for effective consolidation of the concrete during foundation works, not using proper templates for firmly keeping the stubs in position when templates are supplied by the contractor & improper fixing of stubs, non ensuring of tower verticality, use of rusted stubs and tower parts if supply is by the contractor, non painting of butt joints and rusted stubs with zinc rich paint immediately after erection of the towers & stringing, non fixing of earth flat to the stub, non deployment of technical personnel for supervision of works by the contractor. Also not fixing of vibration dampers firmly, repair sleeves wherever necessary, not properly fixing of arcing horns both tower side and line side and bird guards etc., as per specification.

Category III : Use of improper grade / quality of raw material like H.B.G. metal, water and sand for concreting, using clogged and / or lump / clotted cement for concreting, not ensuring proper curing for foundation concrete, not ensuring that all the members of the tower are placed in position and firmly fixed with bolts and nuts immediately after erection of tower, not ensuring that Half round seam welding of the nuts before stringing of the line.

For the above deficiencies in the quality of works noticed by the Engineer, penalties shall be levied as given below.

Type of category	First instance (Rs.)	Second instance (Rs.)	Third instance (Rs.)
Category I	5,000/-	10,000/-	15,000/-
Category II	7,500/-	15,000/-	22,500/-
Category III	10,000/-	20,000/-	30,000/-

In the event of fourth instance of noticing the deficiency of quality of works in any of the above categories, the bidder shall be debarred from participating **in future tenders for a minimum period of one year.** Non-providing of vehicle as per the requests of the section officer shall attract a penalty of Rs. 500/- per day.

6.3 **<u>Incentives for early completion of the works:</u>** Incentives to executing agencies / contractors will be given for early completion of the transmission line works based on the formula as detailed below.

<u>Incentive Bonus Formula.</u>
$I = Z * \{Y/(X/3)\} * 4\%,$

Where as
I = Incentive Bonus,
Z = Project cost
Y = Days equivalent to the difference of the Scheduled Project commissioning Date and the Project commissioning date.
X=Time period in days for completion of a Project specified in the Specification/ bidding documents of the Project.

However Y should be > 30 days and Incentive Bonus is limited to 4% of the Project Cost
Example for a project cost of a line with 10 corers for which the completion period is 12 months (365 days.). If the agency completed the work before 60 days ahead of the schedule.

The bonus is
I = 10 Cr * {60 / (365/3)}*4% = Rs. 19, 72,602/- (However the bonus will be limited to 4% of project cost)

Incentives will not be applicable for Substation and Bulk Load, Private Power Project works and DC works. Incentives are applicable for the Transmission lines taken up under capital works. No incentives for lines less than 10KM. Length of line for incentive is the line length specified in the lot.

6.4 EXTENSION OF THE COMPLETION DATE

When a work cannot be completed within the completion period indicated for reasons beyond the control of the contractor i.e. due to Force Majeure conditions mentioned below or due to the reasons attributable to APTRANSCO, the contractor shall represent for the same and extension of time without levying penalty shall be granted only on the issue of an undertaking by the contractor that they will not put forth at a later date, any claims for extra payments towards increased overheads, material/ equipment/works costs etc. during the extended period. It is the sole discretion of the APTRANSCO to grant extension of completion period.

6.5 FORCE MAJEURE:

i) The Contractor will not be liable for forfeiture of its performance security, penalty for late delivery, or termination for default if and to the extent that its delay in performance or other failure to perform its obligations under the Contact is the result of an event of Force Majeure.

ii) For purposes of this clause, "Force Majeure" means an event beyond the control of the Supplier and not involving the Supplier's fault or negligence and not foreseeable. Such events may include, but are not restricted to, wars or revolutions, fires, floods, epidemics, quarantine restrictions, and freight embargoes.

iii) If a Force Majeure situation arises, within 15 days from the date of eventuality the Contractor shall notify the Purchaser in writing of such condition and the cause thereof. Unless otherwise directed by the Purchaser in writing, the Contractor shall continue to perform their obligations under the Contract as far as it is reasonably practice, and will seek all reasonable alternative means for performance not prevented by the Force Majeure event.

6.6 DELAYS ORDERED BY THE PURCHASER

APTRANSCO reserves the right to suspend and reinstate execution of whole or any part of the Works without invalidating the provisions of the contract. Orders for suspension or reinstatement of the Works will be issued by the Engineer to the Contractor in writing. The time for completion of the works will be extended suitably to account for duration of the suspension. Any costs incurred by the contractor due to increased overheads, idling of labour etc., as a result of such suspension will not be reimbursed to the contractor.

7.0 RESPONSIBILITY OF THE CONTRACTOR:

The successful bidder is responsible

a) For the safe delivery of the goods in good condition at destination and execution of the works ensuring quality. He should acquaint himself of the conditions obtaining in regard to supply of the materials.
b) To mobilize and plan for the labour and APTRANSCO shall not be responsible for any mismatch of work on account of whatever so the reasons and the successful bidder has to bear the costs towards ideal labour on account of mismatch of work if any.
c) To submit the work/ material bills, (preferably monthly for work bills and for materials immediately after receipt of them at site) promptly and submit all the required enclosures with out fail. APTRANSCO shall not be responsible for any delayed payments whatever so the reasons.

The bidders are requested to study all the existing conditions and all the above aspects of the project while quoting itself and no extra claim will be entertained on the above reasons.

8.0 CLAIMS DUE TO DELAYS BEYOND CONTRACTOR'S CONTROL:

The contractor shall not be entitled to claim compensation on account of delays or hindrances to the works for any cause whatsoever. Should the cause of delay or hindrance not be the responsibility of the contractor, the Chief Engineer will consider whether it is possible or not to grant extension of time to

compensate for this delay or hindrance. The contractor will not, however, be eligible for any compensation by way of increase in the rates for the works executed beyond the contract period.

9.0 RECOVERY OF MONEY FROM CONTRACTOR IN CERTAIN CASES:

In every case in which provision is made for recovery of money from the contractor, the APTRANSCO shall be entitled to retain or deduct the amount thereof from any moneys that may be due or may become due to the contractor under these present and/or under any other contract or contracts or any other account whatsoever, including Bank Guarantees, Bid Security/Performance Security etc., held up by the APTRANSCO.

10.0 WAY LEAVES & TREE CUTTING IN TRANSMISSION LINE CORRIDOR:

Payment of compensation for damaged crops or fruit bearing trees shall be initially made by the contractor, which will be reimbursed by APTRANSCO later on. The compensation so payable in respect of damaged crops shall be based on the assessment of MRO. In respect of the trees, compensation shall be payable as per the guidelines issued in T.O.O. Dir (Transmission) Ms.no.238, Dt.10.11.2006. The contractor shall take up the payment of compensation amount only after written approval by the Executive Engineer. The contractor shall pay 75% of the compensation in pursuance with the early completion of work. The payment through Account payee cheques shall be done in the presence of Assistant Engineer.

11.0 TERMINATION OF CONTRACT:

If it is found that Progress of works are not commensurate with the program of completion or if the contractor does not fulfill his obligations as per the terms of the specification APTRANSCO will be entitled to terminate the contract in part or full by giving 15 days notice and get the balance works completed through other agencies at the contractor's cost and risk.

Warning letters will be issued by **Executive Engineer /O&M** or/and **Superintending Engineer/ O&M** or/and Chief Engineer / Zone or and if the progress is not satisfactory. If the progress continues to be poor even after the second warning letter, show cause notice / final notice will be issued by the **Engineer Engineer/ O&M**. If no satisfactory reply is received or the progress of the works are not improved within 15 days of issue of the final notice / show cause notice, the **Engineer Engineer / O&M** shall terminate the agreement, which will be followed by stoppage of all payments to the contractor, encashment of the BGs, and after termination of contract the balance works / supplies shall be completed through alternate agencies at contractor's risk and cost. **The contractor so penalized shall be blacklisted for 3 years from the date of termination of contract.**

In the event of termination of contract due to delay attributable to the contractor, APTRANSCO is entitled to get back equipment/ material from the contractor for the payment already made but the material is not put to use, and then get the works done by other agency/ agencies for completion of contract by using the said recovered material. The contractor shall have no objection in this regard.

12.0 ARBITRATION

All and any disputes or differences arising out of or touching the order based on this specification shall be decided by a panel of arbitrators as detailed below:

Sl. No.	**Value of Claim**	**Panel of Arbitrators**
a)	Disputes involving amounts up to Rs.10,000/- and below.	Superintending Engineer of the APTRANSCO other than the circle to which the disputes relate.
b)	Disputes involving amounts ranging from Rs.10,001 to Rs.50,000/-	Any Chief Engineer of the APTRANSCO.

There shall not be any reference of disputes, the value of which is above Rs.50000/- to arbitration. The parties shall approach the competent Civil Courts situated in **Eluru** only, if any such disputes shall arise involving more than Rs.50,000/-.

A reference for adjudication under this clause shall be made by either party to the contract, within one year from the date of intimating the contractor of preparation of final bill or his having accepted the payment.

SECTION-III-TECHNICAL

1.0 TECHNICAL SPECIFICATION FOR COMMON BUILDING MATERIALS

The aggregates such as sand metal shall be obtained from the source as indicated in lead statements.

1.1 INDIAN STANDARDS:

The aggregates both fine and coarse shall comply with the requirements of I.S.456 for concrete and I.S.383 for sand for use in rendering, plastering and mortar except as hereinafter stated and shall be delivered to and maintained at the site, clean, washed and free from dirt.

1.2 SAMPLES:

Samples shall be submitted to the Engineer and all aggregates used in the work shall be at least equal to the approved samples.

1.3 FINE AGGREGATE:

The fine aggregate for concrete, shall consist of naturally occurring and graded in accordance with the requirements of IS.383 for grading Zone-2 or grading Zone-3 except that no particle shall exceed 3/16 inch. It shall be free from excessive sharpness. Fine aggregate cement mortar and lime mortor and rendering plastering shall comply with IS 383.

1.4 COARSE AGGREGATE:

Coarse aggregate shall be graded to produce sound concrete and for reinforced concrete shall be such that at least 5% by weight will pass a mesh of a size 1/4 per inch less than the minimum lateral distance between the reinforcing bars or 1/4 inch less than the minimum cover, whichever is smaller.

1.5 WATER FOR CONCRETE:

Clean fresh water shall be used for mixing concrete grout and mortar and curing. The water used for mixing and curing shall be free from deleterious matter and acids and alkaline substances in a solution or suspension. Potable water shall generally be used for mixing and curing concrete.

1.6 WATER FOR DRINKING:

ALL POTABLE WATER SHALL BE FREE FROM DELETORIOUS MATTER AND SHALL BE EQUAL TO OR BETTER than the minimum standard acceptable to the local and state authorities. Filtration and chlorination by approved means shall be installed as necessary to obtain the required standard. Storage and distribution of potable water shall be such as to prevent contamination.

1.7 CEMENT : Ordinary Portland cement of Grade 43 manufactured as per IS 8112 of 1989 shall only be used. Storage, certification, delivery and testing of the cement shall conform to IS:269, IS:8112 or the latest issue. IS certification mark is obligatory.

Cement procured from major cement manufacturing plants such as M/s Raasi, M/s Vishnu, M/s Andhra cement co., ACC, CCI, M/s L&T and Coramandel, Priyadarshini, Madras Cements etc will only be accepted for which prior approval shall be obtained from the employer. Copies of invoices and test certificates from the cement manufacturers shall be submitted by the contractor to the Engineer and his clearance obtained before actual use. Such clearance will be provided within a maximum period of one week.

1.8 REINFORCEMENT

Steel required for reinforcement will be procured by the contractor The contractor shall workout the requirement immediately after taking over the site and receipt of working drawings and take the approval of the site Engineer and arrange for procurement

2.0 TECHNICAL SPECIFICATION FOR EXCAVATION:

This clause covers excavation to be done for all the civil works. Excavation rates for all soils for foundations of (i) structures (ii) walls for control room shall be quoted. This rate should be inclusive of dewatering, shoring & shuttering required if any. Sides and bottoms of excavation shall be vertically true.

Blasting material for excavation in Hard rock required if any has to be procured by bidder himself. Necessary assistance in the form of approval for procurement of the material will be given by the Board to the extent possible. All rules under explosive act shall be followed by the successful Bidder. Blasting work shall be done only after approval by the site engineer during specified hours. The person incharge of blasting shall satisfy himself that all blasts have exploded before working people are permitted to re approach work site. Withdrawal of unexploded charge will not be permitted under any circumstances. Unexploded charge shall be flooded with water and the hole marked in a distinguished manner. Another hole shall be drilled at a distance of about 450 mm of old hole in the direction of unexploded charge and fired in usual way. This process shall be continued till the original blast is exploded. Excavation in decomposed or soft rock shall be carried out by crow bars or pic-axes or pneumatic drills etc. If he desires to do blasting, he shall obtain permission of site Engineer and do accordingly.

The rock obtained during excavation is the Board's property and it shall not be used under any circumstances by the contractor. Rock excavated shall be stacked by the successful tenderer at a convenient place within the Substations switchyard as directed by the site Engineer at no extra cost.

3.0 TECHNICAL SPECIFICATION FOR BACK FILLING:

After completion of foundation, footings and walls and other construction below the elevation of the final grades and prior to back filling, all forms of temporary shoring, timber, etc., shall be removed and the excavation cleaned of all trash, debris and perishable materials. Back filling shall begin only with the approval of the Engineer- in-charge. Back filling shall be done with inorganic materials obtained from the excavation of borrow pits, if suitable, and subject to the approval of the Board's Engineer-in-charge. Back fill shall not be dropped directly upon or against any structure or facility where there is danger of displacement or damage. Backfill shall be placed in horizontal layers not to exceed 15 cm. in thickness. Each layer shall be compacted with proper moisture content and with such equipment as may be required to obtain a density equal to or greater than 98% of maximum as determined by the relevant Indian Standards. Trucks or heavy equipment for depositing or compacting back-fill shall not be used within 1.5m of the foundation/structures, or other facilities which may damage by their weight or operation. The methods of compaction shall be subject to approval of the Engineer-in-charge. Backfill adjacent to pipes shall be hand placed free of stones, concrete, etc., compacted uniformly on both sides of the pipe and where practicable, to a depth of 300 mm over the top of pipes, when tampering around piping care should be taken to avoid unequal pressures.

3.1 MATERIAL & QUALITY:

All the materials used in the works shall be of the best quality of their respective kinds and specified herein. They shall be obtained from sources and suppliers approved by the Engineer or his representatives, and shall comply strictly with the tests, specified hereinafter or, where tests are not specified in this specification they should conform to the requirements of the latest issue of the relevant Indian Standards hereinafter abbreviated to as (I.S) or other approved National Standards authorized by the Engineer.

3.2 INSPECTION AND TESTING:

All the materials used in the works shall be subject to inspection and tests if required by the Engineer. Unless otherwise stated, the cost of all tests required by this specification shall be deemed to be included in the rates and prices named by the contractor in the bill of quantities. Any material which is prepared or manufactured without notice having been given in writing to the Engineer may be rejected if the Engineer considered that his inspection was necessary during the progress of manufacture of such material.

3.3 APPROVAL:

NO MATERIAL SHALL BE USED IN THE WORKS UNLESS THEY HAVE FIRST BEEN APPROVED BY THE ENGINEER OR HIS REPRESENTATIVE.

4.0 TECHNICAL SPECIFICATION FOR CEMENT CONCRETE:

4.1 CODE OF PRACTICE:

Except where otherwise specified, described or directed all concrete and reinforced concrete work shall be carried out in accordance with Indian Standard 456 Code of practice for plain and reinforced concrete.

4.2 GAUGING CONCRETE:

Aggregate shall be measured in proper gauge boxes. When measuring the fine aggregate due allowance shall be made for the moisture content and the bulk adjusted to suit the mix. The methods adopted for gauging the concrete materials shall have the approval of the Engineer.

4.3 CONCRETE PROPORTIONS: The concrete shall be of nominal mix as specified in the drawings or in the bill of quantities.

4.4 CONSISTENCY:

The quantity of water used shall be sufficient to produce a dense concrete of adequate workability for its purpose, which will surround and properly grip all the reinforcement.

4.5 CRUSHING STRENGTH:

The crushing strength of the concrete should be as specified in IS.456 for the proposed mixes.

4.6 GAUGING WATER:

The General arrangements for the supply of water for mixing concrete shall be to the satisfaction of the Engineer who will determine the quantity of water to be employed in the mix according to the degree of moisture in the aggregate. The quantity of water thus determined shall be accurately measured for each separate mixing in a suitable container.

4.7 MIXING CONCRETE:

Concrete shall be thoroughly mixed to a uniform consistency in the mixing machines of approved types. Mixing shall continue until the cement is thoroughly distributed through out the mass, and shall last atleast two minutes or for 80 turns of the mixer after the whole of the water has been added. Any concrete showing signs of initial setting before being deposited shall not be used in the works and shall be removed from the site. The concrete shall be discharged from the mixer on to a level watertight platform or floor or into a water tight receptacle. Normally hand mixing of concrete will not be allowed but where the total quantity of concrete is considerably small, the mixing may be done by hand subject to the approval and entirely at the discretion and satisfaction of the Engineer.

4.8 QUALIFIED ENGINEER:

The contractor shall employ qualified Engineer with a qualification of not less than a degree in Engineering who shall be responsible for all concreting carried out for the works. He shall be available at site at all times when concreting is being carried out.

4.9 CONCRETE IN UNSUITABLE WEATHER:

In the event of rain, storms or other severe weather conditions arising, concreting shall be stopped and appropriate temporary stop ends vee groves etc., placed as may be necessary. To meet such circumstances, the contractor shall always have in readiness on the site approved framed sheeting tarpaulins etc., for the protection of newly placed concrete. Should any concrete be damaged due to rain, streams or other weather conditions, the Engineer may order the cutting out /replacement of the damaged concrete at the expenses of the contractor.

4.10 FORM WORK:

This clause is applicable for form work/ centering etc for all the civil works under the specification.

The form work shall conform to clause 10 of I.S.456 latest. The contractor entirely responsible for the sufficiency and efficiency of the form work, which term includes moulds, and also for the safe removal of the same. Before commencing the work he shall submit for the approval of the Engineer, details of the form work he proposes to use but such approval shall in no way relieve him of any of his responsibilities for the sufficiency and efficiency of the work and that it will be resistant to the strains imposed on it in vibrating the concrete and will retain all the fines in the concrete as may be necessary to provide the desired concrete surface.

The form work shall be designed and arranged so that it will not settle under the load and can be stripped and removed without causing any blemish or jar to the concrete.

For beam soffits the contractor shall provide cambers or such other form which will ensure that undersides of beams are truly horizontal or cambered to the extent shown in the drawings.

All forms shall be securely braced and supported to prevent any sagging or bulging during construction. In no circumstances shall wire ties be used. All chamber and radius strips. Liners and cores shall be provided where necessary and shall be due to space and securely fixed. All forms shall be fixed to the proper line and trued up immediately before depositing the concrete. All joints shall be close enough to prevent leakage liquid from the concrete.

Form work for all exposed faces of mass concrete and for all faces of reinforced concrete shall consist of approved material so finished as to produce the concrete surface, finish specified without any loss of fines and without honey-combing or bulges etc., strutting shall be of such design to allow accurate adjustment and easy removal.

Strutting of form work against the sides of the structures which is subject to movement or vibration will not be permitted.

The inside faces of the forms shall be treated with mould oil or other approved preparation in either case, that will not have deteriorating effect either the strength or prevent the oil or other approved preparation coming into contact with the reinforcement.

REMOVAL OF FORM WORK:

The length of time between concreting and the removal of the formwork is the sole responsibility of the contractor. It shall, however, be competent, to the Engineer to require a minimum length of time. For structure foundation minimum time limit specified is 24 hours for removing form work from date of placing concrete.

4.11 STOP ENDS:

The position of temporary stop ends for vertical joints shall be as approved by the Engineer. Shuttering to form the stops shall be firmly fixed and secured round the reinforcing bars. Such concrete as passes through the stops shall be hacked off and removed as soon as the concrete has set.

4.12 CONSTRUCTION JOINTS :

Recesses of approved size and type shall be formed in construction joints where required by the Engineer in order to form a key with the following concrete. The cost of all shuttering to construction joints shall be deemed to be included in the rates named in the priced bill of quantities. Before Depositing any concrete resting or abutting on work previously carried out the surfaces and ends of the existing work shall be thoroughly racked to such an extent that no portion of the previous surface remains. Thus roughened surfaces shall be thoroughly cleaned off, brushed and watered immediately before the succeeding operations are commenced. The roughened surfaces shall be coated with cement/sand mortar 1/2 inch thick immediately before the concrete of the next layer is placed (special care shall be taken to put the mortor and fresh concrete thoroughly up against the hardened concrete).

4.13 CLEANLINESS OF FORM WORK :

Before concreting is commenced all formwork shall be scrupulously cleaned and wetted, and the contractor shall adopt all necessary measures to ensure that all debris, dirt, wash water and other refuse is removed. The reinforcement and form work will then be inspected by the Engineer and concreting shall not be commenced until the Engineer gives permission. Such inspection shall not however relieve the contractor any of his responsibility for the correctness of the work in every respect.

4.14 EXECUTION OF CONCRETE WORK:

Section limits: Concreting shall be carried out in Sections, not exceeding the limits specified for particular work.

4.15 CASTING PROGRAMME:

The contractor's casting programme shall be such that such section of work can be satisfactorily completed

in one operation after permission has been given to proceed. No claim for overtime working to complete a casting programme will be entertained.

4.16 CONVEYANCE OF CONCRETE:

The concrete shall be conveyed from the mixer to its place in the works as rapidly as possible and in such a manner that there shall be no separation or loss of the ingredients. In no circumstances shall more than half an hour lapse between the time when water is added to the mix and the time when the concrete is finally consolidated in position. The use of concrete distributing chutes at an angle of more than 45degrees from the horizontal will not be permitted without the prior written sanction of the Engineer. In no case shall concrete be dropped from barrows or otherwise from a height of more than one and half meters. The arrangements to be adopted by the contractor for conveying and depositing concrete shall be subject to the approval of the Engineer .

4.17 DEPOSITING CONCRETE:

Before any concrete is put in, the contractor shall carryout any filling of pockets or trimming the sides where found necessary to suit the level and line of the concrete to be laid as directed by the Engineer at site. Unless otherwise approved, concrete shall be placed in one operation to the full thickness of members.

4.18 CONSOLIDATING CONCRETE:

Concrete normally shall be consolidated by means of sufficient number of mechanical vibrators. Hand ramming and tamping will only be allowed where specified or by the prior permission of the Executive Engineer. Hand ramming and tamping where permitted shall be sufficient and efficient such as to produce uniform consolidation.

The concrete shall be thoroughly worked alround reinforcement and against shutters so that all entrained air is duly expelled and the concrete surface when stripped be found to be good and free of the formation of air pockets, honey combing or other defects.

The concrete shall be worked into position where placed and not allowed to flow, for sloping beams the work of depositing concrete shall start from the lower end and work upwards.

Except where arrangements, approved by the Engineer are made for placing concrete under water, the areas on which concrete is to be deposited shall be made and kept free from standing water during concreting operations and running water crossing or entering such areas shall be brought under control before concreting is commenced.

4.19 FINISH OF CONCRETE SURFACES:

Immediately any wrought or metal faced for work is struck the surface of the concrete will be inspected by the Engineer and after any remedial work directed or permitted by the Engineer has been completed to his satisfaction, the contractor shall remove all form marks and other imperfections in order to give uniform appearance. The cost of this work shall be included in the rates for concrete.

Floor surfaces shall be worked to a smooth even finish to correct levels of falls as indicated in the drawings or as directed.

Where so directed, concrete floor surfaces shall be treated with silicate of soda grade P.84 diluted with four times its volume of water applied to the work with a watering can or spray and afterwards spread evenly with a mop or brush. Twenty four hours later a second application shall be made and, if any surface still appears porous, further applications of the solution shall be given until the work will absorb no more. Any excess of liquid on the surface after the last coat has been absorbed shall be removed and the surface allowed to dry. When dry it shall be washed with the plain water.

4.20 CURING:

The curing period shall commence immediately after the concrete is finally trowelled or secreeded and continue for a period of 21 days. The top and side surfaces of concrete shall be kept moist and be protected from the direct rays of the sun during the period. The contractor shall submit to the Engineer, his proposals for ensuring continuous protection of the concrete during the curing period.

4.21 DEFECTIVE WORK:

CONCRETE WHICH IS DEFECTIVE FROM ANY CAUSE WHAT SO EVER SHALL, IF SO DIRECTED By THE ENGINEER, be cut out and the work reconstructed at the successful bidder's cost. No concrete thus cut out shall be reused.

The faces of the concrete work shall be sound and solid, free from honey combing. No 'Patching' of any concrete facing will be allowed without the express written permission of the Engineer.

5.0 TECHNICAL SPECIFICATION FOR REINFORCED CONCRETE:

The preceding clauses relating to concrete generally shall be read in conjunction with these following:

5.1 MIXES:

The mixes of concrete shall be as specified in the bill of quantities or shown in the drawings.

5.2 COVER TO REINFORCEMENT:

The reinforcement shall in all cases be covered with no greater and no less than the minimum thickness of concrete specified shown in the drawings. Where two bars cross, the outer should have the minimum cover and no more.

5.3 FORM BOLTS:

Any form bolts that are in the concrete shall be withdrawn when the forms are stripped . They shall not be placed within 2 inches of any steel reinforcement, so that the holes they leave do not reduce the effective cover on the steel.

5.4 BENDING REINFORCEMENT:

Bends, cranks or other labours on reinforcing bars shall be carefully formed exact in accordance with the drawings, otherwise, all bars shall be truly straight . Bends shall be made cold round a former having a diameter of atleast four times, the diameter of the bars. Heating of bars for any purpose whatsoever will not be allowed.

5.5 SPLICING REINFORCEMENT:

Where splices or overlapping in reinforcement are required the bars shall be provided with such splices or overlaps shown in the drawings. No bars may be jointed by welding unless special permission in writing has previously been given by the Engineer, in applying for such permission the contractor shall supply full details of the method he proposes to use.

5.6 FIXING REINFORCEMENT:

The number, size, form and position of all steel reinforcing bars, ties, links, stirups and other parts of the reinforcement shall be exact in accordance with the drawings, and such parts shall be kept in the correct positions in the forms without displacement during the process of working the concrete into place. Space bars, supporting stools and distance pieces to maintain the reinforcement in the correct position shall be provided by the contractors as directed by the Engineer, without any extra cost to the Board.
The use of timber blocks for welding the steel of the forms will not be permitted.

All straight bars shall be fixed parallel to each other and to the sides of the forms. Any ties, links or stirups connecting the bars shall be tied so that the bars are properly braced, the inside of their curved parts shall be in actual contact with the bars round which they are intended to fit.

5.7 BINDING WIRE:

Bars shall be bound together with black anealed steel wire No.16 S.W.G. thick and the binding shall be done tight with proper pliers or automatic binders. The free ends of the binding wire shall be sent in walls. The rate of reinforcement should be inclusive of cost of binding wire.

5.8 STEEL TO BE CLEAN:

All steel reinforcement before the concrete is deposited shall be clean and free from all loose mill scale, dust and loose rust, and coatings such as paints, cement, grout etc.,

5.9 DEPOSITING:

No concrete shall be deposited until form work and reinforcement has been inspected and approved by the Engineer. Care shall be taken that the steel reinforcement is thoroughly surrounded by the concrete and that no voids or cavities are left. Consolidation of the concrete, either by hand or mechanical vibration, shall be to the satisfaction of the Engineer, and under no circumstance shall consolidation obtained by heavy impact on the form work.

5.10 The top of each layer of concrete shall be laid parallel to the longitudinal axis of the work and shall not be at a slope to it.

6.0 DEFECTIVE WORK:

If, on the removal of the shuttering any honey combing or other defective workmanship shall be found in the face of the concrete, the Engineer will decide whether, in his opinion, the strength of the member is affected, and whether 'Patching' will be permitted in which case the defective concrete shall be cut out to the extent ordered by the Engineer, and the remaining concrete thoroughly cleaned and made good. If before or during this operation, any reinforcing bar is exposed, the cutting out shall continue right round the bar, to form a key. Where reinforcing bars are so exposed. Care shall be taken to ensure that they are not damaged by the tools used for cutting out the concrete.

If, in the opinion of the Engineer, the strength of the member is affected by the defective materials or workmanship, he may direct that the whole members shall be removed and a new member constructed by the contractor without charge. The contractor may however, elect to carry out load test at his own cost, and if the load test shows results sufficiently satisfactory to the Engineer, the member may be left in place, provided all surface defects are made good by the contractor without charge. The weight of the test load will be determined by the Engineer.

If 'Patching' is discovered which has been carried out without the permission of the Engineer, the whole member affected will be liable for rejection, in which case the member shall be removed and a new member constructed by the contractor, all without charge.

6.1 SCAFFOLDING:

The rates quoted by the tenderers for all civil works shall be inclusive of scaffolding charges required if any. Materials and labour required for scaffolding shall be arranged by the contractor himself.

SCHEDULE – I

PARTICULARS OF REGISTRATION OF THE BIDDER

Sl. No.	**Details**	**:**	**To Be filled by the Bidder**		
1.	Name &Address of the Firm / Bidder	**:**			
2.	**State whether** a) Individual b) Registered partnership firm c) Unregistered partnership firm d) Pvt. Ltd. Co. e) Public Ltd. Co. f) Others (Please specify)	**:**			
3.	Registrations under the following If Yes fill the registration No. and enclose a copy of the registration certificate	**:**	Yes / No	Registration No.	Whether copy enclosed/ not
a)	Class 'A' registration at CEIG	**:**			
b)	Company of registration Act.				
c)	Large, medium or small scale industry at Industries Dept.	**:**			

d)	Central Sales Tax act	**:**			
e)	GST	**:**			
f)	Excise Department (For manufacturers only)	**:**			
g)	Registration Under Labour Laws				
h)	Any other state sales tax law (please specify)	**:**			
4.	Name and full address of the partners (in case of partnership firm) or directors (in case of Ltd. Co.)	**:**			
5.	Name and full address of the JV partner (in case of JV)		**Not applicable**		
6.	Authority of the person signing the contract to represent the company	**:**			
7.	Whether Photostat copies of the following documents attested.	**:**			
a)	Copy of partnership deed.	**:**			
b)	Copy of Memorandum of Articles.	**:**			
c)	JV Agreement in original (Where applicable)	**:**	**Not applicable**		
d)	Authority of the partner/company to enter into business in memorandum of association.	**:**			
e)	A certified copy of the board resolution (in case the partner is a company) according permission to entering into the joint venture	**:**			
f)	A certified copy of the board resolution (in case the partner is a company) according permission for affixing the company seal on the joint venture deed.				

Signature of the Bidder/
(**along with the seal)**

SCHEDULE - II

DETAILS OF TECHNICAL PERSONNEL OF THE BIDDER

Name of the Bidder:

Sl. No.	Designation	Name & Qualifications	Professional experience and details of works carried out	Remarks
1	Construction Engineer			
2	Supervisory Staff			
i)				
ii)				
iii)				
iv)				

Signature of the Bidder/

(Along with the seal)

SCHEDULE – III A

DETAILS REGARDING FINANCIAL STANDING OF THE BIDDER

Name of the Bidder/The following particulars may be filled in.

Sl. No.	Name of the Bank	Actual balance at credit of the contractor	Permissible overdraft / credit	Total credit	Bank Guarantee Limits	Remarks

Note: Latest certificates from the Bankers in proof of cash balance and other credit facilities shall be furnished in support of the above information without fail.

Signature of the Bidder/

(Along with the seal)

SCHEDULE – III B

DETAILS OF FINANCIAL TURNOVER OF THE BIDDER

(Here the balance sheet for the last financial year, annual turn over, debt equity ratio and other relevant financial parameters and the proof of their credits standing may be furnished)

Name of the Bidder

Financial Year	Turnover in Rs.	M.F with weightage of 10% per each financial year	Updated Turnover In Rs.	State whether the required certified copies are enclosed in support of turnover
(i)	(ii)	(iii)	(iv) = (ii)*(iii)	(v)
2024-25		1		
2023-24		1.1		
2022-23		1.2		
2021-22		1.3		
2020-21		1.4		
2019-20		1.5		

The Maximum updated Turnover from the above i.e. A = Rs.
Value of existing commitments and ongoing works i.e. B = Rs.
(As per schedule V-C)
Completion period of the notified works in No. of years i.e. N=
(As per project Data Sheet)

2AN-B = Rs.

NOTE: The bidder shall furnish the Certificate from Chartered Accountant in support of their Turnover or can be certified this schedule-III B itself.

Signature of the Bidder/

(Along with the seal)

SCHEDULE – IV A

DETAILS OF WORKS/CONTRACTS EXECUTED OF SIMILAR TYPE AND MAGNITUDE CARRIED OUT BY THE BIDDER
Name of the Bidder/

Sl. No	Name of the Line with full description such as Place & State	Length of the Line in km	Order Value	Date of award of contract	Period of completion as per Agreement	Period in which actually completed	Reasons for delay	Date of actual completion & actual cost of work	Principle features
	132KV								
	Total								
	220KV								
	Total								
	400KV								
	Total								

e: Experience Certificates issued by the **an officer not below the rank of Executive Engineer or equivalent cadre of state power utility / Government organizations / PSU only"** shall be furnished by the bidder without fail, in support of the above information.

Signature of the Bidder/

(Along with the seal)

SCHEDULE - IV(B)

DETAILS OF FABRICATED AND SUPPLIED TOWERS/SUBSTATION STRUCTURES BY THE BIDDER

Name of the Bidder having physical experience

Sl. No.	Name of the work	Place and State	Quantity supplied	Period of supply	Principal features
1	2	3	4	5	6

Note: Experience Certificates issued by the **an officer not below the rank of Executive Engineer or equivalent cadre of state power utility / Government organizations / PSU only** shall be furnished by the bidder without fail, in support of the above information.

Signature of the Bidder/

(Along with the seal)

SCHEDULE - V (A)

DETAILS OF ONGOING WORKS UNDER EXECUTION IN APTRANSCO
(as on the date of submission of the bid)

Name of the Bidder/Both the Partners in case of JV:

			Works on Hand				Works bidded-for		
Sl. No.	Name of the work/ Description	Place & State	Tendered cost in Rs.	Stipulated period of completion	Cost of remaining work	Anticipated date of completion	Estimated cost in Rs.	Date when decision is expected	Stipulated date of completion
	132kV 220kV 400kV Discoms								

I / We M/s…….. here by declared that the information furnished in respect of works under execution in the above said schedule is true to the best of my / we knowledge and belief. I / We hereby undertake that in the event of any of the above information found to be false or incorrect at a later date the APTRANSCO is entitled to reject the bid or terminate the contract agreement entered into besides black listing the bidder.

Signature of the Bidder/

(Along with the seal)

SCHEDULE – V (B)

DETAILS OF ONGOING WORKS UNDER EXECUTION IN OTHER THAN APTRANSCO
(as on the date of submission of the bid)

Name of the Bidder/

Sl. No.	Name of the work	Place and State of work execu-tion	Tender-ing authority / Organisa-tion name	Work on hand			Works tendered for			
				Contrac t value in Rs.	Stipulated date of comple-tion as per contract & scheduled date of completion as per contract	Cost of remain-ing work as on date in Rs.	Anticipa-ted date of comple-tion	Estimate d cost in Rs.	Date when decision is expecte d	Stipula ted date of comple -tion
1	2	3	4	5	6	7	8	9	10	11
	132kV 220kV 400kV Other works									

Note: The bidder should submit details of all the works on hand invariably which should include the works being executed in other Departments and in other states. If any information found unfurnished or any information furnished in the bid is proven to be false at a later date, the bid will not only be rejected but the bidder will be black listed.

Signature of the Bidder/

(Along with the seal)

SCHEDULE - V (C)

DETAILS OF ALL ONGOING WORKS UNDER EXECUTION / ALREADY AWARDED AND TO BE TAKEN UP

(AWARDED BY APTRANSCO &OTHERS UTILITIES)

Name of the Bidder/

				Works on Hand				
Sl. No.	Name of the work/ Description	Place & State	Order Placing Utility /Authority	Order Value in Rs.	Value of Balance work (x)	Net Value of Balance work to be loaded	Remaining Stipulated period of completion	Whether Certificate issued by the State/Central Government Department / Undertakings enclosed
I	As an individual					x*1		
1)								
2)								
II	As Lead Partner of JV					**x*0.75**		
III	As Associate Partner of JV					**x*0.25**		
Total Value of existing commitment Rs.								

I/We, Sri./Smt./M/s..... .... do here by solemnly affirm and declare that I/we have the balance works on hand as per the details furnished above are true and also declare that in case it is found at a later date to be false fake / incorrect, I/We ..... will abide by any action such as rejection of the bid or disqualification or determination of contract under clause 60(a) PS to APSS or black listing or any action may deem fit .

Signature of the Chartered Accountant (Along with the seal)
Date:
Place:

Signature of the Bidder/ (Along with the seal)
Date:
Place:

SCHEDULE - V (D)

UNDER TAKING WITH REGARDS TO THE QUALIFICATION REQUIREMNTS

I. PHYSICAL EXPERIENCE:

The Physical Experience as per the Qualification Requirement is =

The above Qualification are meeting by considering the following works done by us=
(i)
(ii)
(ii)

Totals for each type of works. The relevant Certificates are to be attached.

II. FINANCIAL TURNOVER:

The Financial Turnover as per the Qualification Requirement is i.e. ECV = Rs.

AN-B arrived as per IIIB is = Rs.

I/We, Sri./Smt./M/s..... .... do here by solemnly affirm and declare that I/we have meeting the qualification requirements as per our knowledge and the details of Physical experience and financial turnovers furnished above are true and also declare that in case it is found at a later date to be false fake / incorrect, I/We ..... will abide by any action such as rejection of the bid or disqualification or determination of contract under clause 60(a) PS to APSS or black listing or any action may deem fit.

Signature of the Bidder
(
Name:
Designation:
Seal:
Date:
Place:

SCHEDULE - V (E)

UNDER TAKING WITH REGARDS TO THE BLACKLISTING BY OTHER UTILITIES

I/We, Sri./Smt./M/s…… …… …… …… …… do here by solemnly affirm and declare that I/we as per our knowledge are not blacklisted by any other utility and our promoters firms or companies were not blacklisted /debarred by any other utility. The information furnished above is true and also undertake that in case if it is found at a later date to be false /fake / incorrect, I/We …… will abide by any action such as rejection of the bid or disqualification or determination of contract under clause 60(a) PS to APSS or black listing or any action may deem fit.

Signature of the Bidder

Name:

Designation:

Seal:

Date:

Place:

SCHEDULE – VI

DETAILS OF GANGS AVAILABLE FOR ERECTION OF LINE

Name of the Bidder/ Partner having Physical Experience in case of JV :

Sl. No.	Name of the work under execution	Technical / supervisory staff	Skilled workers	Semi-skilled workers	Un-skilled workers
1	2	3	4	5	6

Signature of the Bidder/
Lead Partner
(Along with the seal)

SCHEDULE-VII

DETAILS OF VEHICLES AND TOOLS & PLANTS REQUIRED TO BE AVAILABLE WITH BIDDER

Name of the Bidder/ Partner having physical experience in case of Joint Venture:

Sl.No.	Tools and Plants	Quantity
1.	Theodolite	1 No.
2.	Ranging Rods	20 Nos.
3.	Chain 30 mtrs. Long	2 nos.
4.	a) Levelling instrument with	2 Nos.
	b) Levelling staves	2 Nos.
	c) No. of flags	20 Nos.
5.	Welding machine with transformer gloves, shield and handle	1 set
6.	Hand drilling machine	1 No.
7.	Steel measuring tape 15M	1 No.
8.	Steel measuring tape 30M	1 No.
9.	Column boxes for a height of 1 M	4 Nos.
10.	a) pin vibrator	1 No. + 1 no. spare
	b) Fan vibrator	1 No. + 1 no. spare
	c) Miller	1 No. + 1 no. spare
11.	Template leveling jacks	30 sets
12.	Rod bending tools (6mm to 20mm sizes)	1 set
13.	Radial drilling machine with drill bits upto 20mm diameter	1 set
14.	Concrete mixer	1 No.
15.	Water pump with hose for curing	1 No.
16.	De watering pump	1 No.
17.	Derrick pole for erection of structures	1 No.
18.	Pipe Derrick	10 Nos.
19.	Tension stringing equipment for Twin Moose Conductor (purchase proof has to be enclosed)	2 sets.
20.	100 Tonne compressors with die sets.	2 nos.
21.	Rubberised Aerial rollers	30 nos.
22.	Rubberised Ground rollers	60 nos.
23.	Come along clamps for conductor & earthwire	6 nos.
24.	Leg winches with required steel wire rope	3 nos.
25.	Poly propylene ropes of each 20M minimum length	10 Nos.
26	(a) Double end spanners of various sizes upto 22mm diameter	1 set
	(b) Ring spanners of various sizes upto 22m diameter	1 set
	(c) Box spanners of various sizes	1 set
	(d) Screw drivers and cutting pliers	1 set
27.	Equipment erection derrick pole 36' height	1 No.
28.	Tripod stand with chain pulley block of 3 Mt capaicity or arrangement to provide a crane having adequate boom length for erecting equipment	
29.	Trifor (2 Tonne) capacity	Complete set
30.	Gas welding and cutting	1 set
31.	Megger	1 set
32.	Crimping tools for cable termination	1 No.
33.	Drum lifting jack for mounting cable drums for unwinding cable	2 Nos.
34.	Safety belts for the workmen	Complete set
35.	Helmets for the workmen	Complete set
36.	Dynamo meter	Complete set
37.	Other necessary T & P required for complete	

execution of the work.

38. Vehicles :

a) Jeeps	1 No.
b) Lorries	2 Nos.
c) Tractor	1 No.

39. Templates At least 2 nos templates for each type of tower required for the notified works.

Note: Based on the studies carried out by the Engineer, minimum suggested major equipment to attain the completion of works in accordance with the prescribed construction schedule are shown in the above list.

Signature of the Bidder/
Lead Partner
(Along with the seal)

SCHEDULE - VIII

UNDERTAKING TO BE GIVEN BY THE BIDDER

I/We _____ representing the Company/ Partnership Firm/Contractor responding to the bid invitation by the APTRANSCO vide **e Proc No.04/2026-27-1st Call** hereby sincerely and solemnly affirm and state as follows:

(Strike out whichever is not applicable)

(a) That myself or any of the representative of my Company/Firm do not have any relatives as deferred in appended Annexure in the APTRANSCO/DISCOM.

OR

(b) That the following officers/employees of the APTRANSCO/DISCOM are related to me and to my representative of my Company/Firm and their status in the APTRANSCO/DISCOMS is as under:

S.No.	Name of the Officer/ Employee	Designation and Place of working	APTRANSCO/ DISCOM	Relationship
1.				
2.				
3.				
4.				
5.				
6				

It is certified that the information furnished above is true to the best of my knowledge and belief. It is hereby undertaken that in the event of any of the above information found to be false or incorrect at a later date, the APTRANSCO/DISCOM is entitled to terminate the contract agreement entered into besides recovering damages as may be found necessary with due notice.

Signature of the Bidder/

(Along with the seal)

ANNEXURE TO SCHEDULE - VIII

LIST OF RELATIVES:

1.	Father	14.	Daughter's Husband
2.	Mother (including Step .Mother)	15.	Daughter's Son
3.	Son (including Step Son)	16.	Daughter's Son's Wife.
4.	Son's Wife	17	Daughter's Daughter
5.	Daughter (including Step Daughter)	18.	Daughter's Daughter's Husband
6.	Father's Father	19.	Brother (including Step Brother)
7.	Father's Mother	20.	Brother's Wife
8.	Mother's Mother	21.	Sister (including Step Sister)
9.	Mother's Father	22.	Sister's Husband.
10.	Son's Son	23.	Son's Wife's Father
11.	Son's Son's Wife	24.	Son's Wife's Mother
12.	Daughter	25.	Daughter's Husband's Father
13.	Daughter's Husband	26.	Daughter's Husband's Mother

SCHEDULE - IX

DECLARATION TO BE GIVEN BY THE BIDDERFOR TAKING UP THE AWARDED WORKS

On award of work against Specification No. e Proc No.04/2026-27 -1st Call of Executive Engineer/O&M Division Pendurthi /APTRANSCO/Visakhapatnam I/We_____ will take up the works promptly and achieve progress as per the program communicated for each of the work. If I/We _____do not show the progress as per Schedule communicated for each work and the progress is (are) proved to be very slow which may result in delay in completion, the APTRANSCO can cancel all such works and may be got executed through alternate agencies at our cost and risk and without prejudice to the right of APTRANSCO to levy any penalties as per the terms of contract.

Signature of the Bidder/

(Along with the seal)

SCHEDULE -X

BANK ACCOUNT DETAILS FOR RTGS

1.	Name of the Bank	:
2.	Name of the Branch	:
3.	Branch Code	:
4.	City	:
5.	Account No.	:
6.	MICR No.	:
7.	IFSC No.	:
8.	Income Tax PAN No.	:
9.	GST Provisional Registration No.	:
10.	Date of GST Registration	:
11.	Place of GST Registration	:

Signature of the Bidder/

(Along with the seal)

SCHEDULE -XI
CHECK LIST

Only **Positive** Confirmations on the following will make the bid responsive. Indicate **Yes/No**

1	State whether the work / materials offered conforms to the relevant APTRANSCO's / ISS Specifications and drawings. (If not Specify the Deviations in Annexure)	
2	State whether Qualification and Performance particulars are uploaded along with necessary performance certificates in PDF formats.	
3	State whether copies of 'Tax Deducted at Source' certificates are uploaded with proper attestation in PDF formats.	
4	State whether copy of 'GST Provisional Registration Certificate' is uploaded with proper attestation in PDF formats.	
5	State whether copy of 'A-Grade license certificate' issued is uploaded with proper attestation in PDF formats.	
6	State whether copy of 'Income Tax registration / PAN registration certificate' are uploaded with proper attestation in PDF formats.	
7	State whether complete Quantity is quoted. (Bidder must quote for 100% of quantities of all items)	
8	State whether scanned copy of BG / DD for Bid Security is uploaded in PDF format.	
9	State whether Clause 1.0 of Section II Financial is accepted.	
10	State whether Bid Validity is offered as per specification.	
11	State whether APTRANSCO's Payment Terms are accepted.	
12	State whether APTRANSCO's completion period & Penalty Clause is accepted.	
13	State whether APTRANSCO's Delivery Schedule / Completion period is accepted.	
14	Are you prepared to furnish Performance Security B.G. if order is placed?	
15	State whether APTRANSCO's Guarantee Clause is accepted.	
16	State whether Bid form uploaded	
17	State whether Bank's Certificate for availability of Credit facilities are uploaded	
18	State whether all the Schedules (I to X and D& E) duly signed are uploaded.	
19	In case of J.V. firm, state whether J.V agreement as per format are uploaded	
20	In case of J.V. firm, state whether Power of attorney for J.V are uploaded	
21	State whether the Guaranteed Technical Particulars for Section-2 to Section -6 of technical Specification are uploaded	
22	State whether copies of type test certificates for equipment/materials are uploaded.	
23	State whether all the clauses are accepted.	
24	State whether price Schedule(s) – A are uploaded in PDF file. (Commercial schedules)	
25	State whether the total lump sum amount quoted in Bid form is as per schedule-A	

Signature of Bidder/

(Along with the seal)

(A) BID SECURITY FORM

Whereas. ..... (hereinafter called "the Bidder") has submitted its Bid dated (date of submission of bid) for the supply, erection, testing and commissioning of. ..... (Description of the work) (hereinafter called "the Bid").

KNOW ALL PEOPLE by these presents that WE .....(Name of bank) having our registered office at .....(address of bank)(hereinafter called "the Bank"), are bound unto .....(name of Purchaser) (hereinafter called "the Purchaser") in the sum of Rs..... for which payment well and truly to be made to the said Purchaser, the Bank binds itself, its successors, and assigns by these presents. Sealed with the Common Seal of the said Bank this day of 2024.

THE CONDITIONS of this obligation are:

1. If the Bidder
   a) withdraws its Bid during the period of bid validity specified by the Bidder on the Bid Form; or
   b) does not accept the correction of errors in accordance with the Bid Specification

OR

2. If the Bidder, having been notified of the acceptance of its bid by the Purchaser during the period of bid validity;

   (a) fails or refuses to furnish the performance security, in accordance with the Bid Specification.

   (b) fails or refuses to execute the Contract Form if required; or

   We undertake to pay the Purchaser up to the above amount upon receipt of its first written demand, without the Purchaser having to substantiate its demand, provided that in its demand the Purchaser will note that the amount claimed by it, duly mentioning the B.G. No. and date.

   We undertake APTRANSCO that once BG issued, it will not be cancelled with in its validity period with out written consent of APTRANSCO, even though bidder produces original Bank Guarantee. *(In e-procurement the bidder will upload the scanned copy of the Bid Security and shall not submit the original BG to APTRANSCO, Hence for cancellation of BG, written consent from APTRANSCO is required.)*

This guarantee will remain in force up to and including _____ ***(mention the date after the period of the bid validity + 45 days)*** and any demand in respect thereof should reach the Bank not later than the above date.

....
(Signature of the Banker)

NOTE: This will be executed on a Rs.100/- non-judicial stamp paper issued by a scheduled Bank.

(B) PERFORMANCE SECURITY FORM

To:(Name of Purchaser)

WHEREAS.....(Name of the Contractor/ Company) (hereinafter called "the Contractor") has undertaken, in pursuance of Contract No.....dated.....2024 to supply/erect ..... (Description of work) (hereinafter called "the Contract").

AND WHEREAS it has been stipulated by you in the said Contract that the contractor will furnish you with a Bank Guarantee by a recognized Bank for the sum specified therein as security for compliance with the Contractor's performance obligations in accordance with the Contract.

AND WHEREAS we ……(Name of the Bank with full address) have agreed to give the Contractor a Guarantee.

THEREFORE WE hereby affirm that we are Guarantors and responsible to you, on behalf of the Contractor, up to a total of Rs. ..... (Amount of the Guarantee in Words and Figures) and we undertake to pay you, upon your first written demand declaring the Contractor to be in default under the Contract and without cavil or argument, any sum or sums within the limit of Rs..... (Amount of Guarantee) as aforesaid, without your needing to prove or to show grounds or reasons for your demand or the sum specified therein.

This guarantee is valid until the. ... day of. ..... 2026.

Signature and Seal of Guarantors
.....
Date. .....2026.
.....
.....
Address:.....
.....
.....

NOTE: This will be executed on a Rs.100/- non-judicial stamp paper by a scheduled Bank.

(C) BID FORM

Date. .....

TO: (Name and Address of Purchaser)
Gentlemen and/or Ladies:

Having examined the Bidding documents, the receipt of which is hereby duly acknowledged, we, the undersigned, offer to execute ..... (Description of Work) in conformity with the said bidding documents for the sum quoted by us or such other sums as may be ascertained in accordance with the schedule of prices attached herewith and made part of this Bid.

We undertake, if our Bid is accepted, to complete the work in accordance with the completion period specified in the Schedule-E.

If our Bid is accepted, we will submit the Performance Bank Guarantee as per specification for the due performance of the Contract, in the form prescribed by the Purchaser.

We agree to abide by this Bid upto ……. (for the Bid Validity Period) specified in Clause ( ) and it will remain binding upon us and may be accepted at any time before the expiration of that period.

Until a formal contract is prepared and executed, this Bid, together with your written acceptance thereof and your notification of award, will constitute a binding Contract between us.

We understand that you are not bound to accept the lowest or any bid you may receive.

We certify / confirm that we comply with the eligibility requirements as per clause ( ) of the bidding documents.

Dated this. .....day of. 2026

_____

_____

[Signature]
[in the capacity of]

Duly authorized to sign Bid for and on behalf of …….

(D) FORMAT FOR BANK'S CERTIFICATE TOWARDS AVAILABILITY OF CREDIT FACILITIES

BANK CERTIFICATE

This is to certify that M/s. _____ (Full Name & Address) _____ who are submitting their bid to TRANSMISSION CORPORATION OF ANDHRA PRADESH LIMITED against their Bid specification vide ref. No._____ & dated _____.

M/s. _____ is our Customer for the past_____years. Their financial transactions with our Bank have been satisfactory. They enjoy the following fund based and non fund based limits including for guarantees, L/C and other credit facilities with us against which the extent of utilization as on date is also indicated below:

Sl.No.	**Type of Facility**	**Sanctioned Limit as on Date_____**	**Utilization as on Date_____**

This letter is issued at the request of M/s. _____.

Sd/-
Name of Bank _____
Name of Authorized Signatory _____
Designation _____
Phone No. _____
Fax No. _____
Address _____

SEAL OF THE BANK

(E) FORMAT FOR JOINT VENTURE AGREEMENT

(ON NON-JUDICIAL STAMP PAPER OF APPROPRIATE VALUE TO BE PURCHASED IN THE NAME OF JOINT VENTURE)

PROFORMA OF JOINT VENTURE AGREEMENT AMONG …… **;** ……

FOR BID SPECIFICATION NO … . …….. OF APTRANSCO.

This joint venture agreement executed on this ……day of …… two thousand and ……..among M/s ……a company incorporated under the laws of …….. and having its registered office at …….. (hereinafter called the "lead Partner" which expression shall include its successors, executors and permitted assigns) **;** M/s …… a company incorporated under the laws of ……..and having its registered office at …… (hereinafter called the "Partner" which expression shall include its successors, executors and permitted assigns) for the purpose of making a bid and entering into a contract (in case of award) against the specification No……for construction of 220 KV/132 KV SS/ transmission project on turnkey basis with Transmission Corporation of Andhra Pradesh Limited (APTRANSCO). (hereinafter called the "Purchaser"), Vidyut Soudha, Hyderabad-500 082.

WHEREAS the purchaser invited bids as per the above mentioned specification for Supply, Transport, Construction, Erection, Testing & Commissioning stipulated in the bidding documents under subject package for 220 KV/132 KV SS/transmission project on turnkey basis.

AND WHEREAS clause..……,(Qualifying requirements for bidders as Joint Ventures) and …….. forming part of the bidding documents, inter-alia stipulates that a Joint venture of two qualified firms/companies as partners, meeting the qualifying requirements, as applicable may bid, provided the Joint venture fulfills all other requirements and in such a case, the Bid Proposal Form shall be signed by all the partners so as to legally bind all the partners of the joint venture, who will be jointly and severally liable to perform the contract and all obligations hereunder.

The above clause further states that the Joint Venture agreement shall be attached to the bid and the contract Performance Guarantee will be as per the format enclosed with the bidding document without any restrictions or liability for either party.

AND WHEREAS the bid has been submitted to the Purchaser vide proposal No…….. dated…….. by Lead partner based on the Joint Venture agreement among all the partners under these presents and the bid in accordance with the requirements of clause ……, (Qualifying requirements for bidders as Joint Ventures) and (Qualifying requirements of bidder for 220 KV / 132kV SS transmission project), has been signed by all the partners.

NOW THIS INDENTURE WITNESSETH AS UNDER:
In consideration of the above premises and agreements all the partners to this Joint Venture do hereby now agree as follows:

1. In consideration of the award of the contract by the purchaser to the Joint Venture, we, the partners to the Joint Venture agreement do hereby agree that M/s ……. shall act as Lead Partner and further declare and confirm that we shall jointly and severally be bound to the purchaser for the successful performance of the contract and shall be fully, responsible for the design, manufacture, supply and successful performance of the material/equipment in accordance with the contract.

2. In case of any breach of the said contract by the Lead Partner or other partner(s) of the Joint Venture agreement, the partner(s) do hereby agree to be fully responsible for the successful performance of the contract and to carry out all the obligations and responsibilities under the contract in accordance with the requirement of the contract.

3. Further, if the purchaser suffers any loss or damage on account of any breach in the contract or any shortfall in the performance of the Equipment/material in meeting the performances guaranteed as per the specification in terms of the Contract, the partner(s) of these presents undertake to promptly make good such loss or damages caused to the Purchaser, on its demand without any demur. It shall not be necessary or obligatory for the

Purchaser to proceed against Lead partner to these presents before proceeding against or dealing with other Partner(s).

4. The financial liability of the Partners of this Joint Venture agreement to the Purchaser, with respect to any of the claims arising out of the performance or non-performance of the obligations set forth in the said Joint Venture agreement, read in conjunction with the relevant conditions of the contract shall, however, not be limited in any way so as to restrict or limit the liabilities of any of the partners of the joint venture agreement.

5. It is expressly understood and agreed to by the partners to this joint venture agreement that the responsibilities and obligations of each of the partners shall be as delineated in separate Appendix ("To be incorporated suitably by the Partners) to this agreement. It is further agreed by the Partners that the above sharing of responsibilities and obligations shall not in any way be a limitation of joint and several responsibilities of the partners under this contract.

6. This Joint Venture agreement shall be construed and interpreted in accordance with the Laws of India and the courts of Hyderabad shall have the exclusive Jurisdiction in all matters arising there under.

7. In case of an award of a contract, we the partners to the Joint Venture agreement do hereby agree that we shall be jointly and severally responsible for furnishing a Performance Security from a bank (Performance Bank Guarantee) in favour of the purchaser in the currency of the contract for the performance of the said contract.

8. It is further agreed to that this Joint Venture agreement shall be irrevocable and shall form an integral part of the contract, and shall continue to be enforceable till the purchaser discharges the same. It shall be effective from the date first mentioned above for all purposes and intents.

IN WITNESS THERE OF, the partners to this Joint Venture agreement have through their authorized representatives executed these presents and affixed common seals of their companies, on the day, month and year first mentioned above.

1. Common seal of…… has been affixed in my/our presence pursuant to the Board of Director's Resolution dated……..	For Lead Partner (Signature of authorized representative)
	Name…….. Designation…… Common Seal of the Company
Signature…… Name…….. Designation……	
2. Common seal of…… has been affixed in my/our presence pursuant to the Board of Director's Resolution dated…….	For Lead Partner (Signature of authorized representative)
	Name…….. Designation…… Common Seal of the Company
Signature…….. Name…… Designation……..	

WITNESSES

1. …… (Signature)

   Name……

   ……

   (Official address)

2. ……. (Signature)

   Name…….

   ……

   (Official address)

(F) FORMAT FOR POWER OF ATTORNEY FOR JOINT VENTURE

(On Non-Judicial Stamp Paper of Appropriate value to be Purchased in the Name of Bidder)

KNOW ALL MEN BY THESE PRESENTS THAT WE, the Partners whose details are given hereunder…….have formed a Joint Venture under the laws of ……and having our Registered Office/Head Office at …… (thereinafter called the 'Joint Venture' which expression shall unless repugnant to the context or meaning there of include its successors, administrators and assigns acting through M/s…… … ……being the partner in-charge do hereby constitute, nominate and appoint M/s…….. ……..a Company incorporated under the laws of ……and having its Registered / Head Office at ……..as our duly constituted lawful Attorney (hereinafter called "Attorney" or "Authorized Representative" or "Partner Incharge") to exercise all or any of the powers for and on behalf of the Joint Venture in regard to specification No…… Package……..the bids for which have been invited by Transmission Corporation of Andhra Pradesh Limited (APTRANSCO) (hereinafter called the "Purchaser")to undertake the following acts.

i) To submit proposal and participate in the aforesaid Bid Specification of the Purchaser on behalf of the "Joint Venture".
ii) To negotiate with the Purchaser regarding the terms and conditions for award of the Contract pursuant to the aforesaid Bid and to sign the Contract with the Purchaser for and on behalf of the "Joint Venture".
iii) To do any other act or submit any document related to the above.
iv) To receive, accept and execute the Contract for and on behalf of the "Joint Venture".

It is clearly understood that the Partner in-charge shall ensure performance of the Contract's and if one or more partner(s) fails to perform their respective portions of the Contract(s), the same shall be deemed to be a default by all the partners.

It is expressly understood that this Power of Attorney shall remain valid, binding and irrevocable till completion of the Warranty Period in terms of the contract.

The Joint Venture hereby agrees and undertakes to ratify and confirm all the acts whatsoever the said Attorney/Authorized Representative quotes in the bid, negotiates and signs the contract with the Purchaser and / or proposes to act on behalf of the Joint Venture by virtue of this Power of Attorney and the same shall bind the Joint Venture as if done by itself.

IN WITNESS THEREOF the Partners Constituting the Joint Venture as aforesaid have executed these presents on this……..day of……under the Common Seal(s) of their Companies.

For and on behalf of the
Partners of Joint Venture
……..
……..

The Common Seal of the above Partners of the joint Venture:
The Common Seal has been affixed there upto in the presence of:
WITNESS

1.	Signature	……	2.	Signature	……
	Name	……		Name	……
	Designation	……		Designation	……
	Occupation	……		Occupation	……

LIST OF APPROVED MANUFACTURERS

Sl. No	Material / Equipment		Name of the Manufacturer
1)	Conductor	i)	M/s Anam Electrical Manufacturing Co., Kadiyam
		ii)	M/s Arrow Cables Ltd, Hyderabad
		iii)	M/s Sterlite Industries India Ltd, Pune
		iv)	M/s Oswal Cables Pvt. Ltd, Hyderabad
		v)	M/s J.S.K.Industries Ltd, Mumbai
		vi)	M/s Deepak Cables (I) Ltd, Bangalore
		vii)	M/s Smitha Conductors Ltd, Mumbai
		viii)	M/s Hindustan Vidyut Products Ltd., Faridabad
		ix)	M/s APAR Industries Ltd., Vadodara
		x)	M/s KJV Alloy Conductors Pvt. Ltd, Hyderabad
		xi)	M/s. Hind Aluminium Industries Ltd., Silvasa
		xii)	M/s. Achuyuta Electricals Pvt. Ltd., Hyderabad
		xiii)	M/s. Hindustan Vidyut Products, Faridabad.
		xiv)	M/s. APAR Conductors, Vadhodara
		xv)	M/s .Cabcon India Pvt Ltd. Kolkota
		xvi)	M/s R.E. Cables and Conductors Pvt Ltd. approved by CE/Cont-2 on 01.02.2012
		xvii)	M/s Lumino Industries Ltd, Kolkatta Panther and Moose approved by CE/Cons-1 on 15.03.2013
		xviii)	M/s Diamond Power Infrastructure Ltd,Vadodara, Approved by CE/Const-1 on 18.04.2013
2)	Transmission Line Towers and Sub Station Structures fabricators	i)	M/s. Bajaj Electricals Ltd., Mumbai
		ii)	M/s. RPG Transmission Ltd., New Delhi
		iii)	M/s. Jyothi Structures Ltd., Mumbai
		iv)	M/s. BS Transcom, Hyderabad.
		v)	M/s. KEC International Ltd., Mumbai
		vi)	M/s. Kalpataru Power Transmission Ltd., Gandhinagar
		vii)	M/s. Hyundai Electrical Transmission Ltd., Gurgaon
		viii)	M/s. Sujana Towers Ltd., Hyderabad
		ix)	M/s. Deepak Galvanising & Engg. Indus. Pvt. Ltd., Hyderabad
		x)	M/s. Sangam Structures Ltd., Allahabad
		xi)	M/s. ICOMM Tele Ltd., Hyderabad
		xii)	M/s. Aster Teleservices Pvt. Ltd., Hyderabad
		xiii)	M/s. Hyderabad Steels Ltd., Hyderabad
		xiv)	M/s. R.S Engineering Co.., Hyderabad
		xv)	M/s. Venkateswara Electrical Industries., Hyderabad
		xvi)	M/s. Vertex Engineers., Hyderabad
		xvii)	M/s. Kusuma Enterprises, Hyderabad
		xviii	M/s. K.Ramachandra Rao Transmission & Projects Pvt.Ltd., Hyd.
		xix)	M/s Annapurna Constructions and Transmissions Hyderabad
		xx)	M/s Iris Infra Services Pvt Ltd Hyderabad
		xxi)	M/s Sai Tirumala Fabricators
		xxii)	M/s ECI Infra
		xxiii)	M/s MPower Infratech. Ltd., Hyd.
		xxiv)	M/s AK Power Industries ;Kolkata

Sl. No	Material / Equipment		Name of the Manufacturer
3) a	Porcelain Disc Insulators	i)	M/s. Birla NGK Industries Private Limited, Kolkata
		ii)	M/s Insulators & Electricals Company, Bhopal
		iii)	M/s W.S. Industries (I) Ltd., Chennai
		iv)	M/s BHEL, Secunderabad
		v)	M/s. Saravana Insulators, Chennai
		vi)	M/s General Power Company Ltd, New Delhi.
		vii)	M/s Bikaner Ceramics Pvt Ltd, Rajastan
3) b	Long Rod Porcelain Insulators	i)	M/s Modern Insulators Ltd. Rajasthan
4)	7/3.15mm HTGS wire	i)	M/s UIC Wires Limited, Kolkata
		ii)	M/s Geekay Wires Private Ltd., Hyderabad
		iii)	M/s. Ramsarup Industries, Kolkata
5)	Galvanised Bolts and Nuts and Washers	i)	M/s Nexo Industries, Ludhiana
		ii)	M/s HR Steel Industries, Howrah
		iii)	M/s NL Engineers Pvt. Ltd., Mohali
		iv)	M/s Himalaya Engg. Works Pvt. Ltd., Hyderabad
		v)	M/s Naveen Metal Industries, Kolkata (Washers)
		vi)	M/s Spring Lock Industries, Vadodara (Washers)
		vii)	M/s. Remax (India), Ludhiana
		viii)	M/s. BRK Industries, Ludhiana
		ix)	M/s. Maheshwari Bright and Fasteners Pvrt Ltd. Hyderabad
		x)	M/s KRSNA Transmission Hardware Mfg. Pvt Ltd. Gujrat
6)	Hardware & Accessories and Clamps and Connectors	i)	M/s. IAC Electricals Pvt. Ltd., Kolkata
		ii)	M/s. Tyco Electronics Systems India Pvt. Ltd., Bangalore
		iii)	M/s. Godavari Mechanical Industries, Hyderabad
		iv)	M/s. Durga Line Materials, Hyderabad
		v)	M/s. Kailash Enterprises, Hyderabad
		vi)	M/s. Uday Clamps Mfg. Co., Hyderabad
		vii)	M/s. Modern Malleable Ltd., Kolkata
		viii)	M/s. Star Iron Works Ltd., Kolkata
		ix)	M/s. Asbesco (I) Pvt. Ltd., Kolkata
7)	Solid Core Insulators	i)	M/s. Birla NGK Industries Private Limited, Kolkata
		ii)	M/s Insulators & Electricals Company, Bhopal
		iii)	M/s W.S. Industries (I) Ltd., Chennai
		iv)	M/s BHEL, Secunderabad
		v)	M/s Modern Insulators Ltd., Rajastan
		vi)	M/s. Saravana Insulators, Chennai.

Sl. No	Material / Equipment		Name of the Manufacturer
8)	Circuit Breakers	i)	M/s ABB Ltd, Baroda
		ii)	M/s Siemens Ltd., Aurangabad
		iii)	M/s Crompton Greaves Ltd., Nasik
		iv)	M/s. Areva T & D India Ltd., Chennai
		v)	M/s. Megawin Switchgear P Ltd, Secunderabad (up to 33 kV only)
9)	Current Transformers & Potential Transformers	i)	M/s ABB Ltd., Baroda
		ii)	M/s. Areva T & D India Ltd, Banglore
		iii)	M/s BHEL, Secunderabad
		iv)	M/s Crompton Greaves Ltd., Nasik
		v)	M/s. Vidyut Controls Systems (P)Ltd, Hyderabad (up to 132kV)
		vi)	M/s. Vijai Electricals Ltd, Hyderabad(PTs up to 132 kV only)
		vii)	M/s. LAMCO, Hyderabad (up to 132 kV only)
10)	Capacitor Voltage Transformers	i)	M/s ABB Ltd., Baroda
		ii)	M/s. Areva T & D India Ltd, Banglore
		iii)	M/s BHEL, Secunderabad
		iv)	M/s Crompton Greaves Ltd., Nasik
11)	Lightning Arrestors	i)	M/s. Areva T & D India Ltd, Bangalore
		ii)	M/s Crompton Greaves Ltd., Nasik
		iii)	Ms/. Lamco Lightning Arrestors Pvt., Ltd., Hyderabad
		iv)	M/s. Oblum Electricals Industries P Ltd., Hyderabad
		v)	M/s Elektrolites power ltd.;Jaipur
12)	Isolators	i)	M/s G.R.Power Switchgear Ltd., Hyderabad
		ii)	M/s Versatek Switchgears Pvt. Ltd., Secunderabad
		iii)	M/s Southern Switchgear Industries, 145/B, Hyderabad
		iv)	M/s Switchgears & Structurals (I) Private Ltd, Hyderabad
		v)	M/s Universal Isolators, Indore, MP
		vi)	M/s Siemens Ltd., Hyderabad
		vii)	M/s Crompton Greaves Ltd., Nasik
		viii)	M/s Switches and Switchgears Hyderabad
		ix)	M/s Elekrolites (Power) Pvt Ltd,Jaipur
		x)	M/s S.L.Electricals
13)	Control and Relay Panels	i)	M/s G.E. Power Controls, Banglore
		ii)	M/s Easun Reyrolle ltd., Hossur
		iii)	M/s. Areva T & D India Ltd, Chennai
		iv)	M/s ABB Ltd., Baroda
		v)	M/s. Siemens Ltd.
		vi)	M/s Schneider Electricals
14)	LTAC Panels & Marshalling Kiosks	i)	M/s G.E.Power Controls, Banglore
		ii)	M/s Easun Reyrolle ltd., Hossur
		iii)	M/s. Areva T & D India Ltd, Chennai
		iv)	M/s ABB Ltd., Bparoda
		v)	M/s Popular Switchgears Pvt Ltd., Nasik
		vi)	M/s. Jain Enterprises, Rajahmundry
		vii)	M/s Bose Corpration, Kolkata
		viii)	M/s Bodapati Controls

Sl. No	Material / Equipment		Name of the Manufacturer
16)	Chargers & DCDB	i)	M/s Dubas Engineering Pvt. Ltd., Bangalore
		ii)	M/s Amara Raja Batteries Ltd., Renigunta
		iii)	M/s Kejriwal Auto Electric & Engineering Works, Kolkata
		iv)	M/s. Automatic Pvt., Ltd., Kolkata

		v)	M/s. Sabnife, Hyderabad
		vi)	M/s. Statcon Power Controls, Noida
		vii)	M/s. Agarwal Salt Co., Bikaner
		viii)	M/s Chhabi Electricals Pvt. Ltd., Jalgoan , Maharastra
		ix)	M/s Servilink Engineers Pvt Ltd, Vadodara
		x)	M/s Caldyne Automatics Ltd., Kolkata
		xi)	M/s Swati Electronics Pvt Ltd.
17)	Batteries (Sealed Maintenance free Batteries)	i)	M/s Amara Raja Batteries Ltd., Renigunta
		ii)	M/s. HBL Nife, Hyderabad
		iii)	M/s. Exide, Hyderabad
		iv)	M/s AAJ Accumulators Pvt Ltd., Kolhapur
18)	Control Cables & LT power Cables	i)	M/s. Arrow Cables Ltd, Hyderabad
		ii)	M/s. Arun manufacturing Company, Delhi
		iii)	M/s. Vikas Cable Company, New Delhi
		iv)	M/s. Polycab Wires Pvt., Ltd., Daman
		v)	M/s. Finecab Ltd., Hyderabad
		vi)	M/s. Goyolene Fibres (India) Private Limited, Mumbai
		vii)	M/s. Deco Industries (India) Pvt., Ltd., Delhi
		viii)	M/s. Grid India Power Cables Pvt., Ltd Gurgao
		ix)	M/s. Pavan Power Cables, Hyderabad
		x)	M/s. SPM Power and Telecom Pvt Ltd Hyderabad
		xi)	M/s .Thermo Cables Limited , Hyderabad
		xii)	M/s. KEI Industries Limited, Hyderabad
		xiii)	M/s. GEMSCAB Industries Ltd,Rajastan
		xiv)	M/s. Diamond Power Infrastructure Ltd,Vadodara
19)	XLPE U/G Power Cables	i)	M/s. Universal Cables Ltd., Satna
		ii)	M/s. RPG Cables Ltd, Hyderabad
		iii)	M/s. Hindustan Vidyut Products Ltd., New Delhi-110001
		iv)	M/s. Cable Corporation of India Ltd, Mumbai
		v)	M/s. Torrent Cables Gujrat.
20)	Sub-Station Lighting	i)	M/s. Bajaj Electricals
		ii)	M/s. Philips
		iii)	M/s. Crompton Greaves Ltd
21)	Wave Traps	i)	M/s. Areva T & D India Ltd, Bangalore
		ii)	M/s. BPL, Palakkad, Kerala
22)	PLCC and Telecom Equipment	i)	M/s. ABB, Mumbai
		ii)	M/s. BPL, Palakkad, Kerala
		iii)	M/s. Puncom, Mohali

Sl. No	Material / Equipment		Name of the Manufacturer
23)	Energy Meters	i)	M/s. Secure meters Ltd., Udaipur
		ii)	M/s. L & G,
		iii)	M/s. L & T, Mysore
24)	a) 33kV Station Transformers	i)	M/s. Vijay Elec. Ltd, Hyderabad
		ii)	M/s. Crompton Greaves Ltd, Nasik
		iii)	M/s. Esennar Trasformers (P) Ltd., Hyderabad
		iv)	M/s. EMCO Ltd., Jalgoan
		v)	M/s. Inra Vidyut Ltd., Banglore
		vi)	M/s. ECE, Hyderabad
		vii)	M/s. HIT, Hyderabad
		viii)	M/s. PETE, Hyderabad.
		ix)	M/s HINT
	b) 11kV Station Transformer		As per DISCOMs approved list of Companies.
25)	Diesel Generator Set	i)	M/s. Powerica

		ii)	M/s. Jakson Ltd., Noida
		iii)	M/s. Escorts Ltd.,
		iv)	M/s. Genset India (P) Ltd., Kolkata
		v)	M/s. Greaves Cotton Ltd.
26)a)	Steel Brands for use in Reinforcement		**1**) VSP **2**) SAIL **3**) TISCO
26)b)	Steel Brands for use for Structures & Towers		**1**) VSP **2**) SAIL **3**) TISCO and 4) ISCO
27)	Cement Brands		**1**) ACC Limited **2**) Kesoram Cements **3**) Orient Cements **4**) Zuari Cements **5**) CCI Limited **6**) Andhra Cements **7**) Coramandel Cement **8**) Raasi Cement **9**) Sri Vishnu Cements **10**) Madras Cements **11**) Ultra Tech Cement Limited – APCW **12**) KCP Limited **13**) Penna Cements **14**) Panayam Cement **15**) Grasim 1**6**) Rajashree Cement **17**) Mysore Cement **18**) Century Cement **19**) Ambhuja Cement and **20**) Priya Cement21) Deccan Cement 22) Dalmia Cement 23) Maha Cement 24) Prajashakti Cement 25) JSW Cements 26) M/s Bharati Cements 27)M/s Sagar Cements 28) Nagarjuna Cement
28)	Furniture		Godrej
29)	T & P		Taparia, Indore.
30)	EOT Crane		RANK, ACCESS, TRANS RAIL, SIVPRA, ATLAS CRANES
31)	AIR CONDITIONING SYSTEM		Voltas, Bluestar, Carrier
32)	PLCC Terminal & Coupling units		M/s. ABB Limited., Andheri (west) Mumbai
33)	PLCC Terminals, wave traps & Exchanges		M/s. BPL Telecom PVt.Ltd., Kasturba Road, Banglore.

34)	Exchanges	1	M/s. Intellicon Pvt.Ltd., Karkhana, Secunderabad
		2	M/s. Intellicon Pvt.Ltd., Gandhinagar.
		3	M/s. Siemens Ltd., Banglore
35)	Wave Traps	1	M/s. Areva T&D PVt Ltd., Hosur, Tamilnadu
		2	M/s. Techno Labs, Hyderabad
36)	Telephone Cables		M/s. Maharaja Cables, New Delhi
37)	Batteries & Chargers		M/s. Amar Raja Power Systems Ltd., Karakamb, Tirupathi.
38)	Chargers	1	M/s. Swati Electronics & Automation, Balanagar, Hyderabad
		2	M/s. Exide Batteries Ltd., Hosur, Banglore.
		3	M/s. SPM Cables, Secunderabad
39)	OLT equipment		M/s. Comtel Networks Pvt Ltd., Trombay Road, Chember, Mumbai.
40)	Chargers		M/s. S. Creative Electronics, Gurgaon, Haryana
41)	A/C units		M/s. Varun Techno Services, Ameerpet, Hyderabad
42)	Cables		M/s. Kothari Brothers, Hyderbasti, Secunderabad
43)	A/C Units		M/s. Globo Tech Services, Vijayanagar Colony, Hyderbad
44)	Electric Testing Instrument		M/s. World Traders & Company, Veernariman road, Mumbai
45)	PLCC terminals		M/s. Punjab communications Ltd., S.A.S.Nagar, Chandigarh.
46)	Batteries		M/s. Exide Industries LTd., Chowranghee road, Kolkata.
47)	Chargers	1	M/s. Dubas Engineerin g PVt.Ltd., BTM Layout, Banglore.
		2	M/s. SM Creative Electronics Ltd., Gurgaon, Haryana
		3	M/s. Exide Industries Ltd., Krishnagiri Dist, Tamilnadu

The following Power Grid approved steel manufacturers are also approved by APTRANSCO for procuring steel and angles which are not manufactured/rerolled by SAIL, VSP. TISCO & ISCO subject to stage inspection as per clause 24.1 section- I General.

a. M/s Southern iron & steel company Ltd, Pattaneri, Salem (Tamil Nadu)

b. M/s Lloyds Steel (only for steel plates)

c. M/s Jindal Steel & Power Ltd, Raigarh (only slabs, booms & billets).

d. M/s Jindal Vijaya Nagar Steel Limited, Bellary, Karnataka, (only for plates 3mm to 10mm thick)

e. M/s Jindal Iron & Steel Company Ltd, Thane.

f. M/s. Hazira works (for MS & HT plates upto 20mm thick)

The following re-rollers are approved by APTRANSCO subject to procurement of the input Ingots from SAIL, VSP. TISCO & ISCO

a. M/s Sanvijay Rolling & Engineering Limited, Nagpur.
b. M/s Shri Bhajarang Alloys Ltd, Nagpur.
c. M/s Vandana Ispat Ltd, Raipur.
d. M/s Shilpa Re-Rollers Ltd, Butibori, Nagpur.
e. M/s Bhuwalka Steel Industries Limited, Thane.
f. M/s P.R. Rolling Pvt Ltd, Jaipur.
g. M/s Vinar Ispat Ltd, Chandrapur.
h. M/s Eagle Steels, Taloja.
i. M/s Shilpa Re-Rollers Ltd, Kamptee Road, Nagpur.
j. M/s Mahavir Steel Industries Ltd, Pune.

k. M/s Unique Structures & Tower Ltd, Raipur.
l. M/s Rita Steel Industries, Nagpur.
m. M/s United Metal Industries, Chennai.
n. M/s Hamma Steel Industries, Chennai.
o. M/s Sharda Ispat, Nagpur.
p. M/s Maharashtra Steel Rolling & Engineering works, Mumbai.
q. M/s SRMB Udyog Limited Paharpur, Kolkata.
r. M/s Shyam Steel Industries Limited, Kolkata.
s. M/s Prakash Ispat Udyog, Nagpur.
t. M/s Swastika Steel & Allied products, Kolkata.
u. M/s Sujana Metal products, Hyderabad.
v. M/s Kadevi Engineering Co. Pvt Ltd, Hyderabad.
w. M/s Sawaria Pipes Pvt Ltd, Hyderabad.
x. M/s Abhishek Steels Pvt. Ltd, Hyderabad.

The above re-rollers are approved with the following guide lines into consideration.

i) The steel angles re-rolled shall be as per the approved manufacturing quality assurance plan.

ii) The steel supplied to APTRANSCO shall be embossed with company name ending with APT

LIST OF THE APPROVED BANKS FOR ISSUING BG /DD

S.No.	NAME OF THE BANK
1	Allahabad Bank
2	Andhra Bank
3	APCO. Bank, Main Branch, Hyderabad.
4	Bank of Baroda
5	Bank of India
6	Bank of Maharashtra
7	Bank of Punjab Ltd
8	Central Bank of India
9	Centurion Bank Ltd
10	City Bank
11	Corporation Bank
12	Federal Bank
13	HDFC Bank,
14	HSBC BANK
15	I.C.I.C.I Bank
16	I.D.B.I. Bank
17	Indian Bank
18	Indian Overseas Bank
19	ING Vysya Bank Ltd
20	Karnataka Bank
21	Karur Vysya Bank
22	Kotak Mahindra Bank
23	Lakshmi Vilas Bank
24	Oriental Bank of Commerce
25	Punjab National Bank
26	Standard Chartered Bank
27	State Bank of India and other State Bank Group Banks / Associate Banks.
28	Syndicate Bank
29	U.T.I. Bank Ltd
30	Union Bank Of India
31	Vijaya Bank
32	YES Bank Ltd
33	Development Credit Bank Ltd.,
34	UCO Bank

PROJECT DATA

1	Name of the Project / work	:	Replacement of existing ABB make 6Nos 132 kV feeder C&R panels of 132 kV Kothavalasa & 132 kV Goldstar,Kasimkota-1,Simhachalam 1&2,132kv Anandapuram -1 feeders under RMI Scheme 2023-24 at 220/132/33 kV Pendurthi SS.
2	Location	:	Located at Pinagadi, Pendurthi,Visakhapatnam.
3	District, Mandal	:	Visakhapatnam
4	O&M Circle and Division	:	SE/O&M/ Visakhapatnam and EE/O&M/ PDT
5	Bid Inviting Authority	:	The Executive Engineer /O&M Division/Pendurthy/ A.P.TRANSCO/ Visakhapatnam
6	Master Copy of the Bid Specifications available at	:	O/o The Executive Engineer /O&M Division/Pendurthy/ A.P.TRANSCO/ Visakhapatnam
7	Rates of Taxes adopted	:	Applicable taxes adopted
8	Present Rates of Tax Deductions in the Bills		Applicable Taxes will be deducted.
9	Completion Period of the Project (Overall).		3 months
10	Financial Tie up / Funding	:	APTRANSCO
11	Any amendments / deviations given to the Standard Bidding Documents.		Will be furnished in Special Conditions of the Contract.

TRANSMISSION CORPORATION OF ANDHRA PRADESH LIMITED
O&M CIRCLE/VISAKHAPATNAM

SCHEDULE – D

SUMMARY OF PRICE

NAME OF THE WORK: Replacement of Old/Damaged 220 KV & 132 KV CTMBs, CVTMBs, PTMBs at 220/132/33 KV Pendurthi SS Under O&M.

S.No.	Item	Total Price
01	Total value of scheduled items (Quantities and prices are as per the Schedule appended to the specification)	Rs.6,04,120.00 (Total Estimate Value)
02	Percentage rate quoted below/at par/above the total value vide item.1 above	
	a. in Figures	%
	b. in words	
03	Total value of schedule after loading the percentage rate quoted against S.No.2 above	
	i) in Figures	
	ii) in words	

Note:-

1. Percentage is to be rounded off up to two decimals.
2. Amounts are to be rounded off to nearest rupee
3. The total estimate value is inclusive of taxes
4. Copy of GST Provisional registration certificates are to be enclosed to the tenders, failing which the tender will be treated as non-responsive.

Signature of the Bidder/ Lead Partner
(Along the seal)

COMPLETION PERIOD - SCHEDULE – E

Program of execution:

The successful bidder shall execute the works as per the program given in below work.

The overall completion period of the project is as per the Project Data Sheet. The commencement date of the Project will be reckoned from the date of issue of detailed Contract award letter or from the date of issue of the approved layout for Sub-Station whichever is later.

1	Replacement of Old/Damaged 220 KV & 132 KV CTMBs, CVTMBs, PTMBs at 220/132/33 KV Pendurthi SS Under O&M	**2 months**

I/We agree to the above mentioned completion period.

Signature of the Bidder/
Lead Partner
(Along the seal)

ABSTRACT

Name of the Work: Replacement of Old/Damaged 220 KV & 132 KV CTMBs, CVTMBs, PTMBs at 220/132/33 KVPendurthi SS Under O&M

Sl. No.	Description			Amount
1	Material Portion	:	Rs	316902.16
2	Labour Portion	:	Rs	195064.17
	Total	**:**	**Rs**	**511966.33**
	Add: GST @ 18%			**92153.94**
	Total Estimation Cost			**604120.27**
	or say Rs.			**₹ 6,04,120.00**
(Rupees Six Lakhs Four Thousand One Hundred and Twenty only)				

EXECUTIVE ENGINEER

O & M DIVISION PENDURTHY

(MATERIAL PORTION)								
S.No	**SSR 26-27 SI.NO/ PG.NO**	**DESCRIPTION OF THE MATERIALS**	**QTY**	**UNIT**	**RATE**	**Freight @ 3%**	**AMOUNT**	**Total amount**
1	1.B/7	220 KV CT Marshalling Box suitable for 5 core made with 3mm thickness Aluminium sheet.	10	No	5,869.00	176.07	6,045.07	60,450.70
2	1.B/7	220 KV PT Marshalling boxes suitable for 3 secondary windings with 3 sets of fuses and links made with 3mm thickness Aluminium sheet. Wiring from Terminal Blocks to Fuses/Links and back to Terminal Blocks in the Marshalling Box shall be done at Factory	2	No	7,822.00	234.66	8,056.66	16,113.32
3	1.B/7	220 KV CVT Marshalling Box suitable for 3 secondary windings with 3 sets of fuses and links made with 3mm thickness Aluminium sheet. Wiring from Terminal Blocks to Fuses/Links and back to Terminal Blocks in the Marshalling Box shall be done at Factory	8	No	7,822.00	234.66	8,056.66	64,453.28
4	1.B/7	132 KV CT Marshalling Box suitable for 3 core made with 3mm thickness Aluminium sheet.	12	No	4,695.00	140.85	4,835.85	58,030.20
5	1.B/7	132 KV PT Marshalling Box suitable for 3 secondary windings with 3 sets of fuses and links made with 3mm thickness Aluminium sheet. Wiring from Terminal Blocks to Fuses/Links and back to Terminal Blocks in the Marshalling Box shall be done at Factory	1	No	7,822.00	234.66	8,056.66	8,056.66

6	1.B/7	132 KV CVT Marshalling Box suitable for 3 secondary windings with 3 sets of fuses and links made with 3mm thickness Aluminium sheet. Wiring from Terminal Blocks to Fuses/Links and back to Terminal Blocks in the Marshalling Box shall be done at Factory	10	No	7,822.00	234.66	8,056.66	80,566.60
7	71/15	GI Flexible Earth Bonds(43*2)	86	No	330.00	9.90	339.90	29,231.40
							Total Material Portion Amount in Rs.	**3,16,902.16**

EXECUTIVE ENGINEER
O & M DIVISION PENDURTHY

Name of the Work: Replacement of Old/Damaged 220 KV & 132 KV CTMBs,CVTMBs,PTMBs at 220/132/33 KVPendurthi SS Under O&M

LABOUR PORTION

S.No.	SSR 26-27 SI.NO/PG.NO	DESCRIPTION OF THE WORK	QTY	UNIT	RATE	COP 13.615%	Area allowance@40%	AMOUNT	Total amount
1	S20/8	Erection of Marshalling Boxes(ALL 220KV&132 KV CT,CVT,PT Boxes)	43	Each	690.00	93.94	276.00	1,059.94	45,577.57
2	S20/8	Dismantling of Marshalling Boxes(CT,CVT,PT)(75% of Erection Charges)	43	Each	517.50	70.46	207.00	794.96	34,183.18
3	S13/8	Cable terminations to the switchgear, Marshallingboxes /Panel terminal blocks/ control & relay panels, LTAC panel including providing suitable ferrules and lugs as per specification (including cost of ferrules and lugs)							
		2.5 sqmm Copper with copper lugs each core at both ends.	1668	Each	45.00	6.13	18.00	69.13	1,15,303.42
								Total Labour Portion Amount in Rs.	**1,95,064.17**

EXECUTIVE ENGINEER
O & M DIVISION PENDURTHI